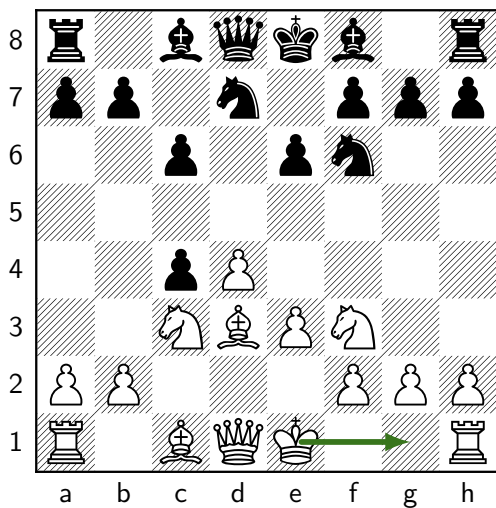


# Melanie-MASTERS (human masters) — Semi-Slav

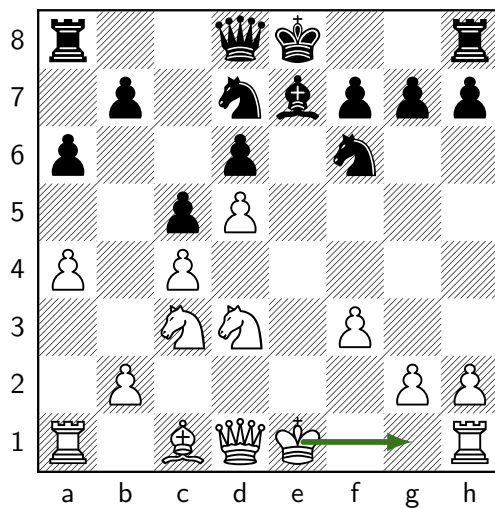
Melanie (Deep\_Vision)

## Abstract

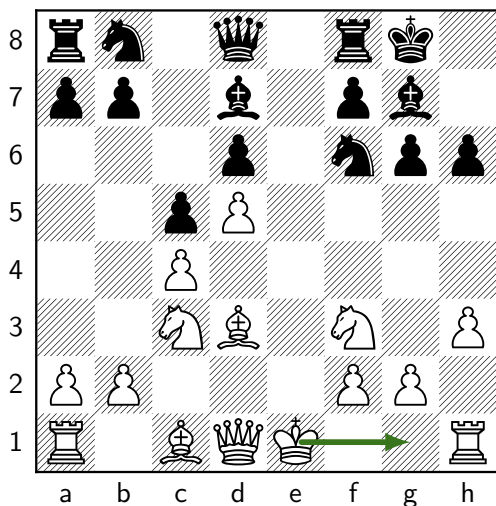
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



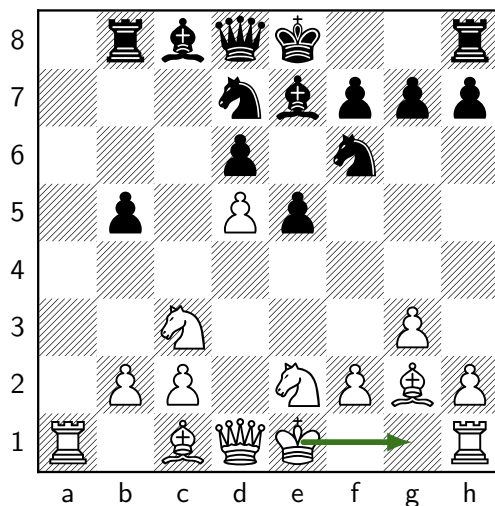
(a)



(b)

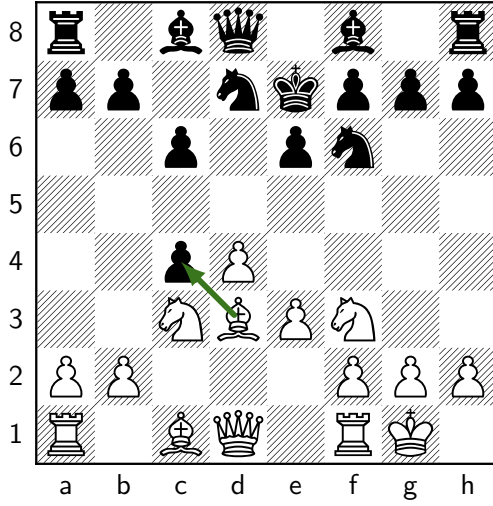


(c)

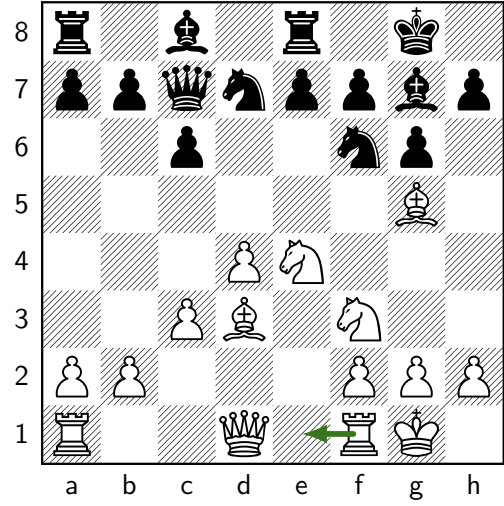


(d)

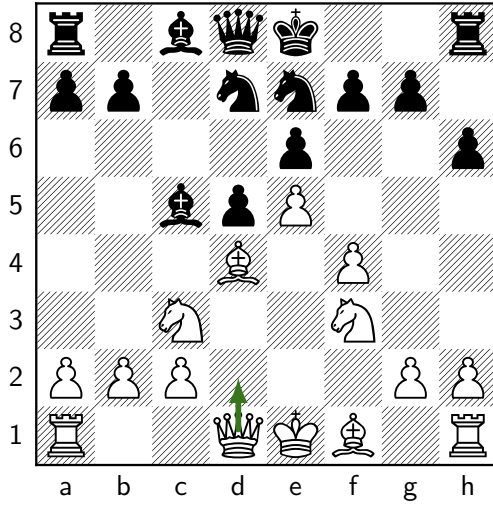
Figure 1: **Move 7, White: 0-0**. (a) Our position; Melanie plays 0-0 (e1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#1393` (ply 24) [structural distance  $d=75$ , QAP obj=14]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf3* as their *Ra1*, our *ra8* as their *Bc1*; transferred move e1g1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#1793` (ply 22) [structural distance  $d=76$ , QAP obj=11]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *nd7* as their *Ph3*, our *qd8* as their *nf6*; transferred move e1g1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0018#2182` (ply 22) [structural distance  $d=79$ , QAP obj=10]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *nf6* as their *Ra1*, our *ra8* as their *Bc1*; transferred move e1g1 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to 0-0 (summed QAP objective 81, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



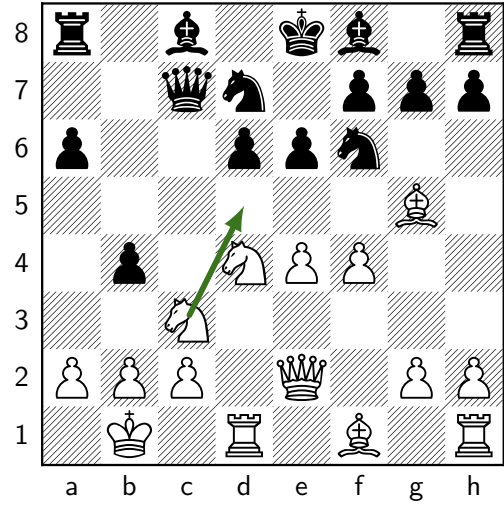
(a)



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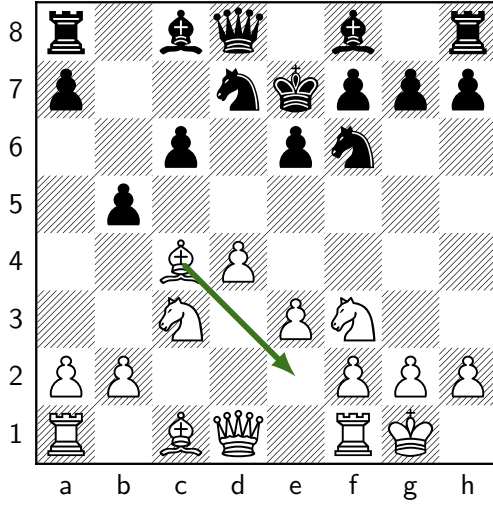


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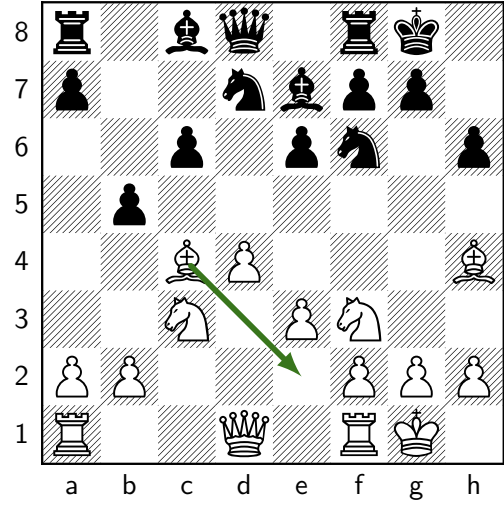


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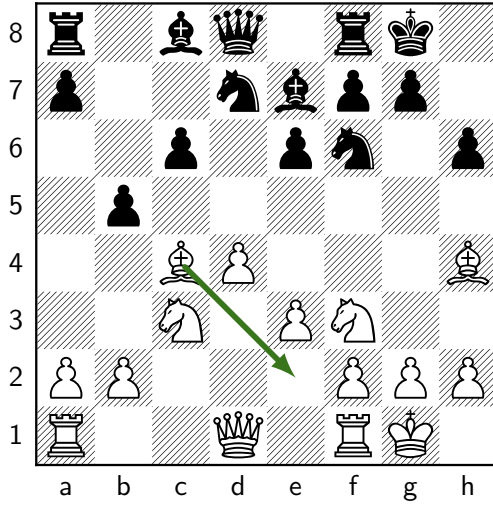
Figure 2: **Move 8, White: Bxc4.** (a) Our position; Melanie plays Bxc4 (d3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#223` (ply 20) [structural distance  $d=94$ , QAP obj=38]: the master played Re1; under  $\mu$  our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pd4 as their Ra1, our nf6 as their Qd1; transferred move f1e1. (c) Retrieved from `lichess_elite_2024-09_p0025#3594` (ply 22) [structural distance  $d=82$ , QAP obj=34]: the master played Qd2; under  $\mu$  our Nf3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pg7 as their Ra1, our nf6 as their Bf1; transferred move f3e1. (d) Retrieved from `lichess_elite_2024-09_p0020#3490` (ply 20) [structural distance  $d=96$ , QAP obj=33]: the master played Nd5; under  $\mu$  our Bd3 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Bf1, our nf6 as their Rh1; transferred move d3e4. Synthesis: no retrieved analog transferred to Bxc4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



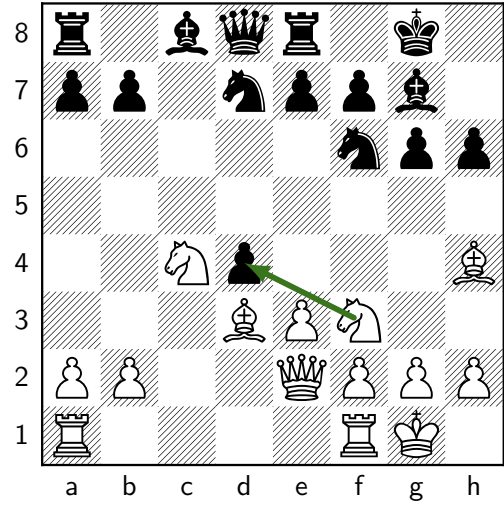
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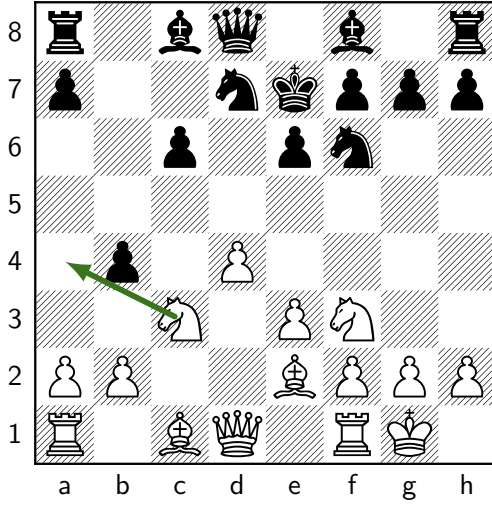


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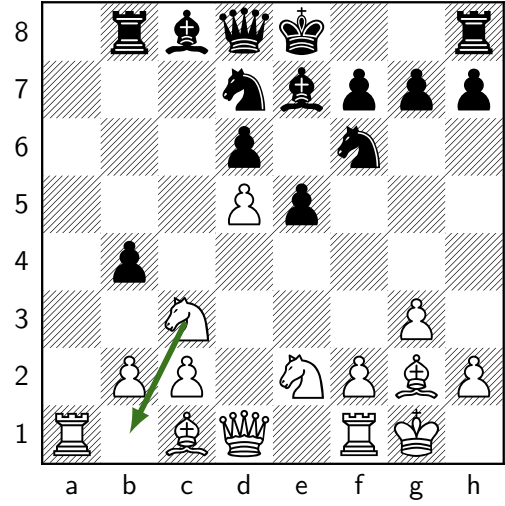


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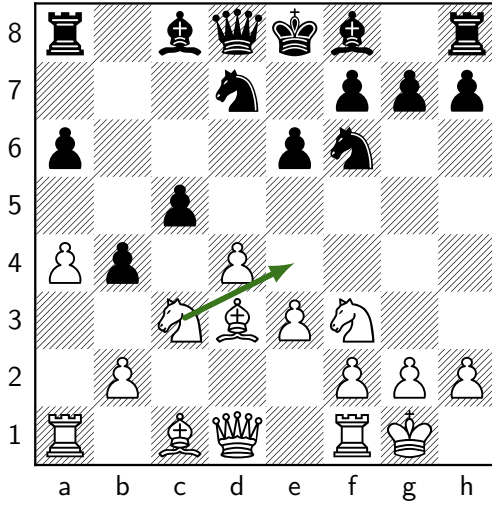
Figure 3: **Move 9, White: Be2.** (a) Our position; Melanie plays Be2 (c4e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#217` (ply 20) [structural distance  $d=103$ , QAP obj=68]: the master played Be2; under  $\mu$  our *Bc4* plays the role of that game’s *Bc4*; cross-type roles: our *pe6* as their *Ra1*, our *Nf3* as their *Pe3*; transferred move c4e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#241` (ply 20) [structural distance  $d=103$ , QAP obj=68]: the master played Be2; under  $\mu$  our *Bc4* plays the role of that game’s *Bc4*; cross-type roles: our *pe6* as their *Ra1*, our *Nf3* as their *Pe3*; transferred move c4e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#40` (ply 22) [structural distance  $d=101$ , QAP obj=35]: the master played *Nxd4*; under  $\mu$  our *Nc3* plays the role of that game’s *Nf3*; cross-type roles: our *Bc1* as their *Ra1*, our *Ra1* as their *Pa2*; transferred move c3e4. Synthesis: 2 retrieved analogs transferred to Be2 (summed QAP objective 136, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



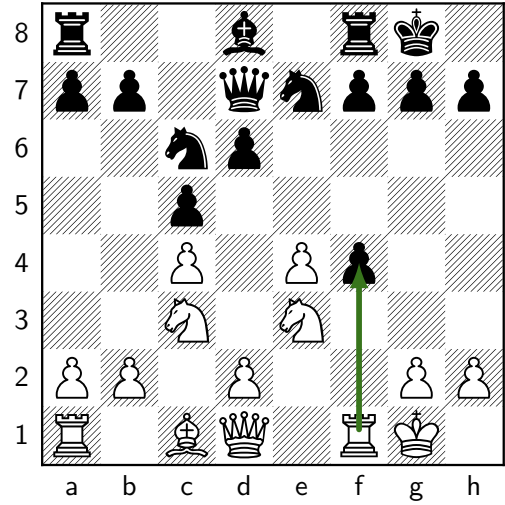
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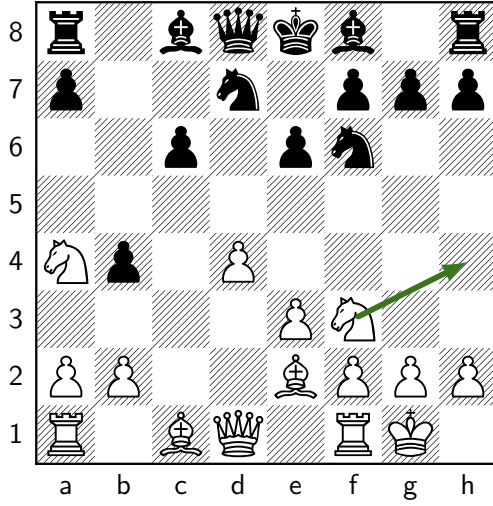


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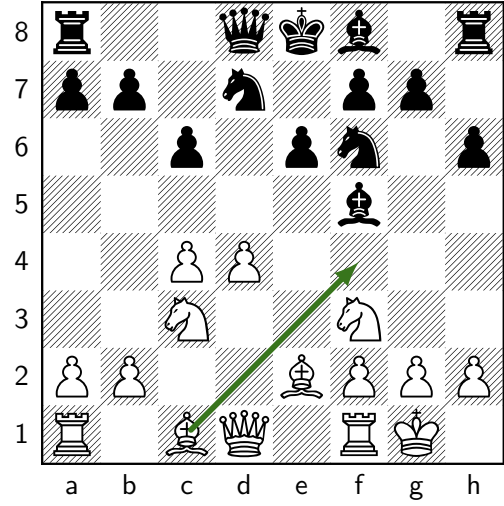


(d)

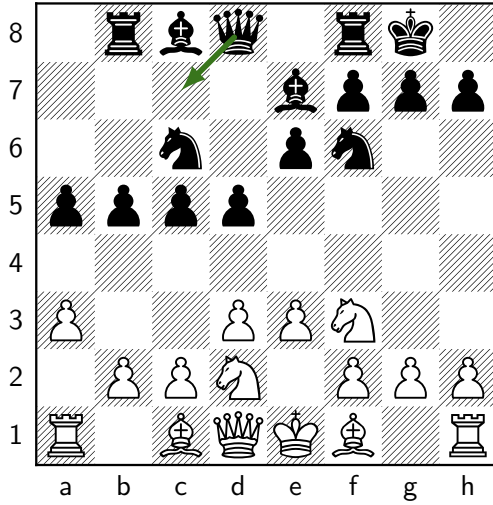
Figure 4: **Move 10, White: Na4.** (a) Our position; Melanie plays Na4 (c3a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#2182` (ply 24) [structural distance  $d=83$ , QAP obj=30]: the master played Nb1; under  $\mu$  our Nc3 plays the role of that game’s Nc3; cross-type roles: our Qd1 as their Ra1, our Be2 as their Qd1; transferred move c3a4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#1480` (ply 20) [structural distance  $d=99$ , QAP obj=30]: the master played Ne4; under  $\mu$  our Nc3 plays the role of that game’s Nc3; cross-type roles: our ph7 as their Ra1, our nf6 as their Bd3; transferred move c3e4. (d) Retrieved from `lichess_elite_2024-09_p0026#300` (ply 22) [structural distance  $d=116$ , QAP obj=27]: the master played Rxf4; under  $\mu$  our Rf1 plays the role of that game’s Rf1; cross-type roles: our nf6 as their Bc1, our Bc1 as their Pa2; transferred move f1e1. Synthesis: 1 retrieved analog transferred to Na4 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



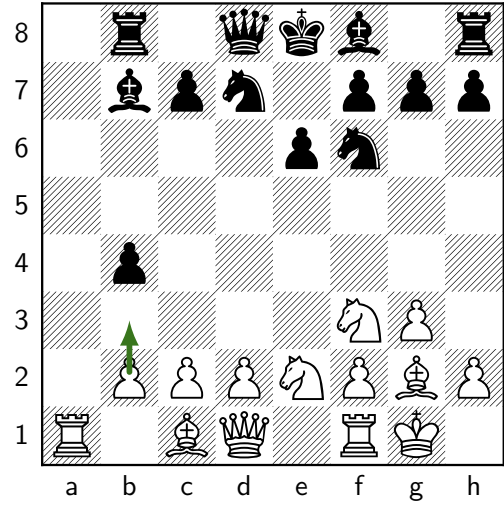
(a)



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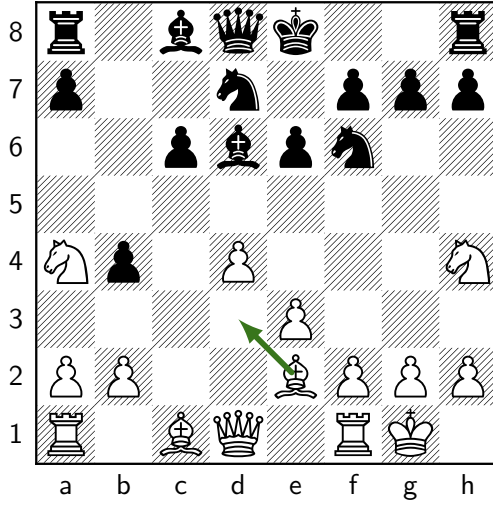


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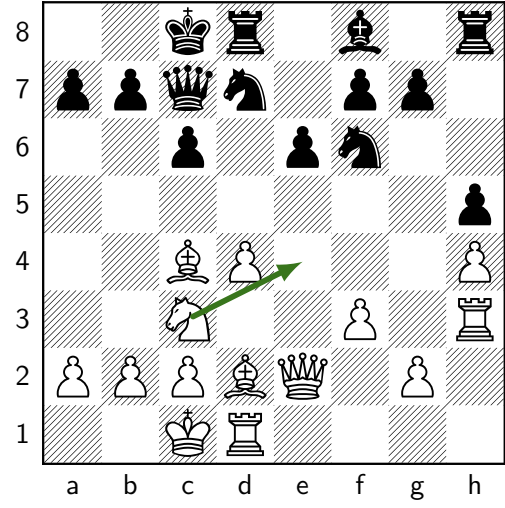


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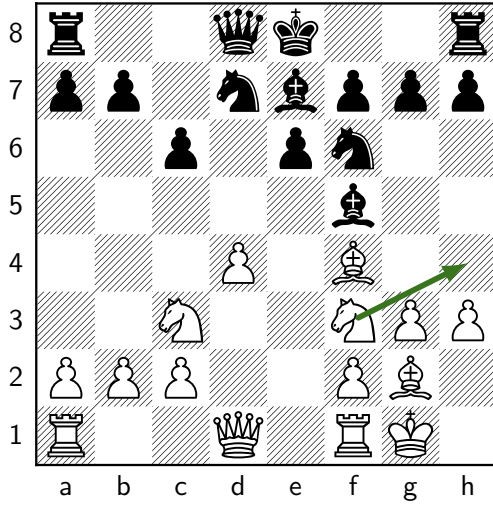
Figure 5: **Move 11, White: Nh4.** (a) Our position; Melanie plays Nh4 (f3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#3185` (ply 20) [structural distance  $d=67$ , QAP obj=9]: the master played Bf4; under  $\mu$  our Nf3 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Ra1, our pb4 as their Qd1; transferred move f3h4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#3757` (ply 21) [structural distance  $d=87$ , QAP obj=7]: the master played Qc7; under  $\mu$  our Nf3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our nf6 as their Rb8, our Ra1 as their Bc8; transferred move f3h4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#507` (ply 22) [structural distance  $d=71$ , QAP obj=11]: the master played b3; under  $\mu$  our Be2 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Na4 as their Ra1, our pe6 as their Bc1; transferred move e2b5. Synthesis: 2 retrieved analogs transferred to Nh4 (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



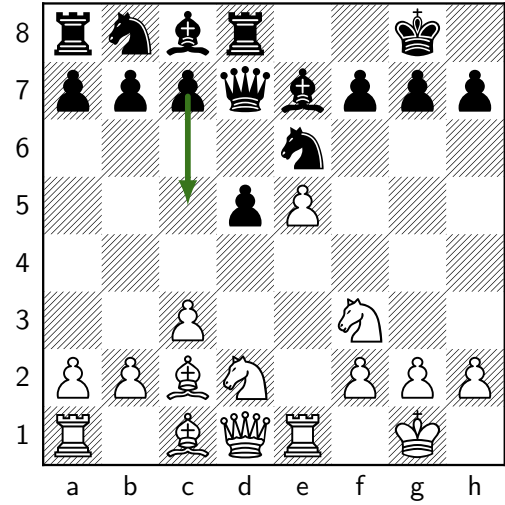
(a)



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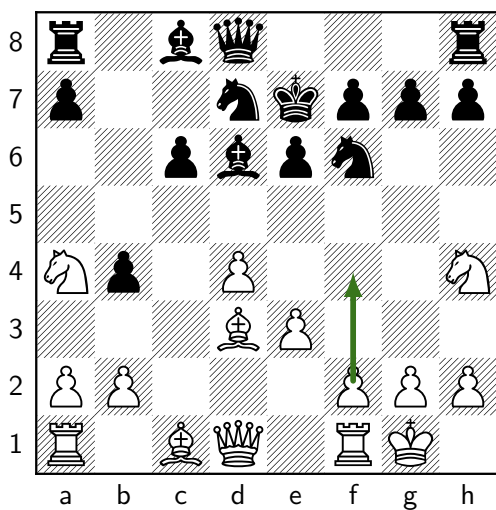


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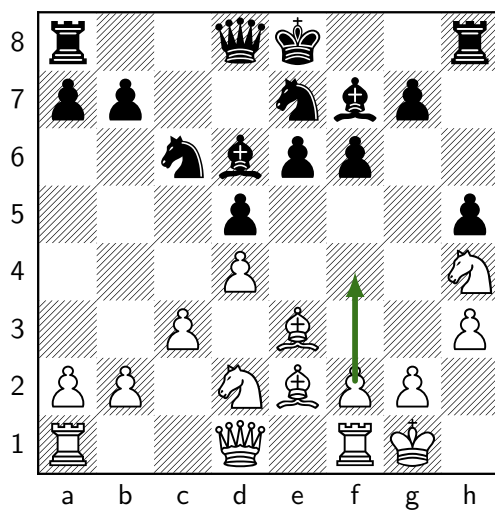


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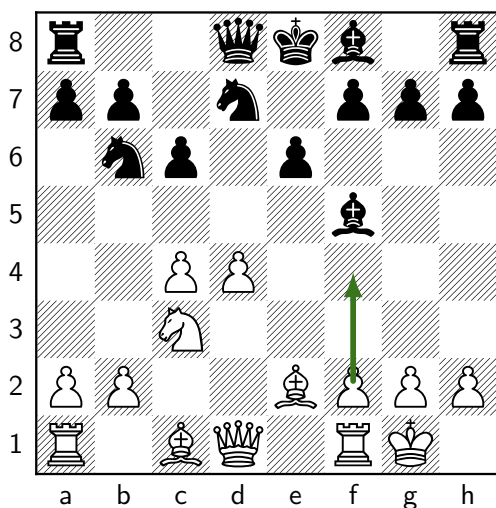
Figure 6: **Move 12, White: Bd3.** (a) Our position; Melanie plays Bd3 (e2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#102` (ply 24) [structural distance  $d=89$ , QAP obj=25]: the master played Ne4; under  $\mu$  our Be2 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pa2, our Pg2 as their Bd2; transferred move e2d3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#1386` (ply 20) [structural distance  $d=80$ , QAP obj=23]: the master played Nh4; under  $\mu$  our Qd1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our nd7 as their Ra1, our Pd4 as their Qd1; transferred move d1e1. (d) Retrieved from `lichess_elite_2024-09_p0013#1703` (ply 21) [structural distance  $d=99$ , QAP obj=22]: the master played c5; under  $\mu$  our Pf2 plays the role of that game’s Pc2; cross-type roles: our Qd1 as their Ra8, our Bc1 as their Pa7; transferred move f2f3. Synthesis: 1 retrieved analog transferred to Bd3 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



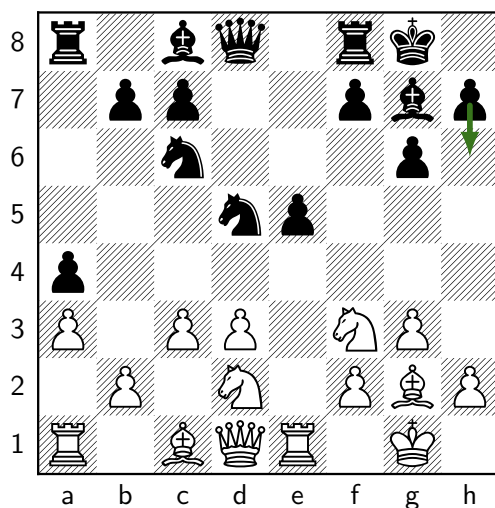
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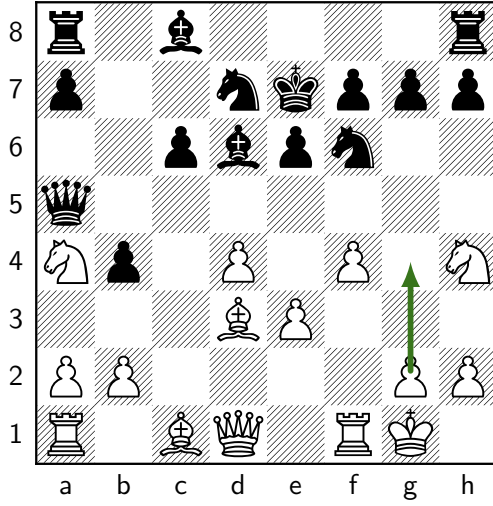
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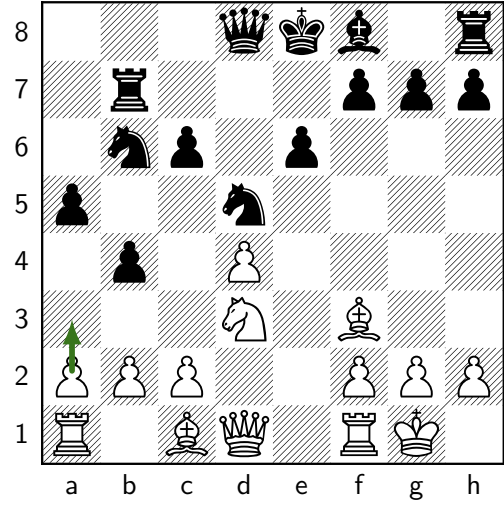
(d)

Figure 7: **Move 13, White: f4.** (a) Our position; Melanie plays f4 (f2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#1438` (ply 30) [structural distance  $d=87$ , QAP obj=28]: the master played f4; under  $\mu$  our *Pf2* plays the role of that game’s *Pf2*; cross-type roles: our *Pd4* as their *Ra1*, our *Bc1* as their *Pb2*; transferred move f2f4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#241` (ply 22) [structural distance  $d=83$ , QAP obj=20]: the master played f4; under  $\mu$  our *Pf2* plays the role of that game’s *Pf2*; cross-type roles: our *Na4* as their *Bc1*, our *Pg2* as their *Qd1*; transferred move f2f4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#1946` (ply 21) [structural distance  $d=82$ , QAP obj=34]: the master played h6; under  $\mu$  our *Pa2* plays the role of that game’s *Ph2*; cross-type roles: our *pe6* as their *Ra8*, our *ra8* as their *Bc8*; transferred move a2a3. Synthesis: 2 retrieved analogs transferred to f4 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

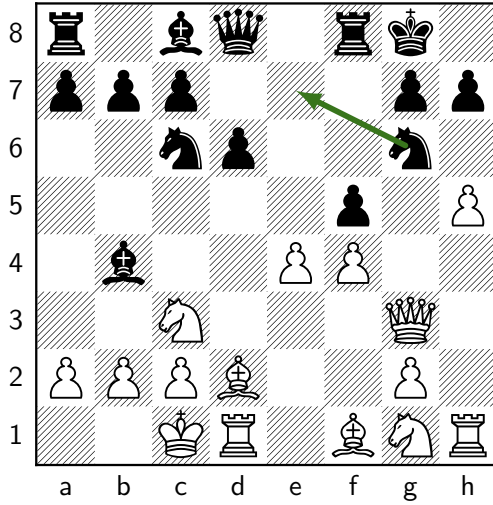




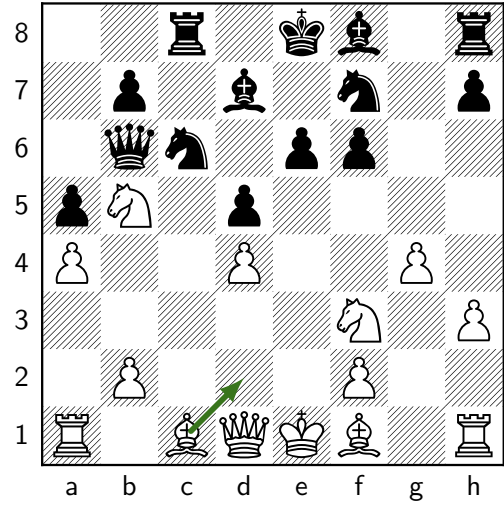
(a)



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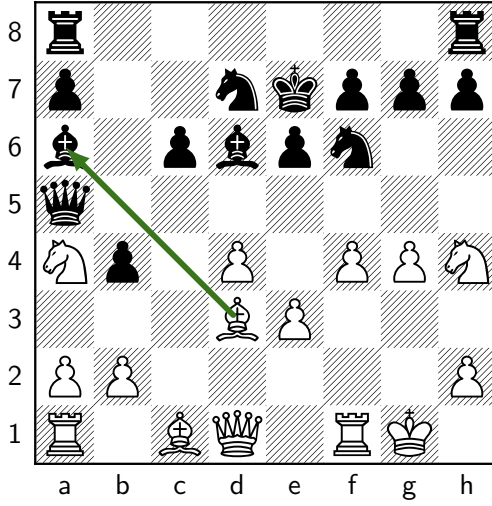


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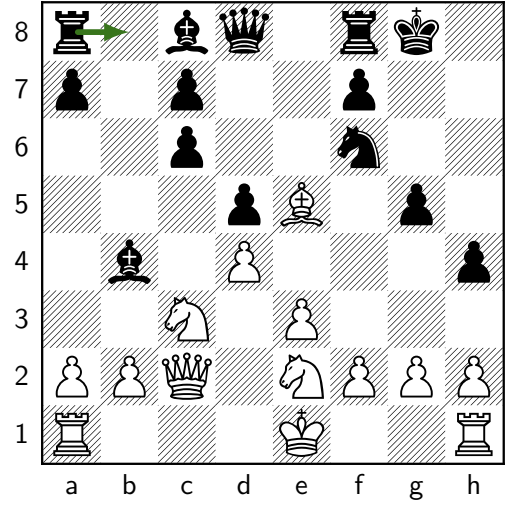


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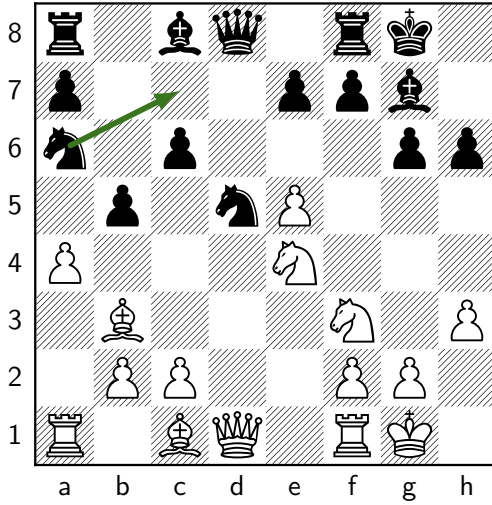
Figure 8: **Move 14, White: g4.** (a) Our position; Melanie plays **g4** (g2g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#3160` (ply 28) [structural distance  $d=102$ , QAP obj=15]: the master played **a3**; under  $\mu$  our  $Pg2$  plays the role of that game’s  $Pa2$ ; cross-type roles: our  $Bc1$  as their  $Ra1$ , our  $Pa2$  as their  $Bc1$ ; transferred move **g2g4** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#3862` (ply 23) [structural distance  $d=79$ , QAP obj=33]: the master played **Nge7**; under  $\mu$  our  $Na4$  plays the role of that game’s  $Ng3$ ; cross-type roles: our  $Bc1$  as their  $Ra8$ , our  $Pd4$  as their  $Bc8$ ; transferred move **a4c3**. (d) Retrieved from `lichess_elite_2024-09_p0030#3563` (ply 26) [structural distance  $d=103$ , QAP obj=25]: the master played **Bd2**; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Bc1$  (a **cross-type** correspondence); cross-type roles: our  $Pg2$  as their  $Ra1$ , our  $Rf1$  as their  $Qd1$ ; transferred move **d4d5**. Synthesis: 1 retrieved analog transferred to **g4** (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



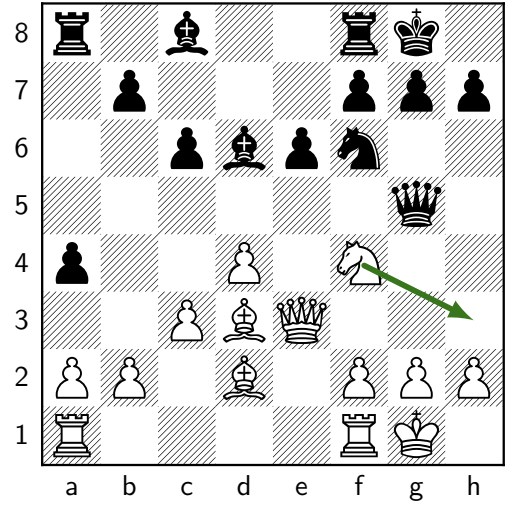
(a)



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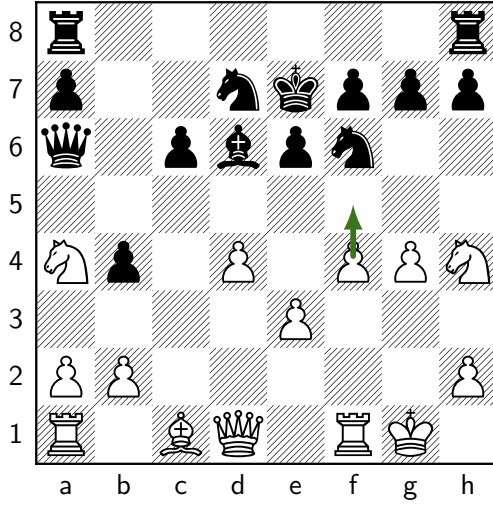


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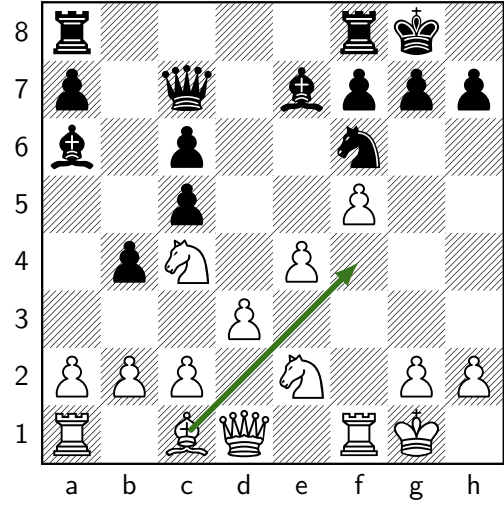


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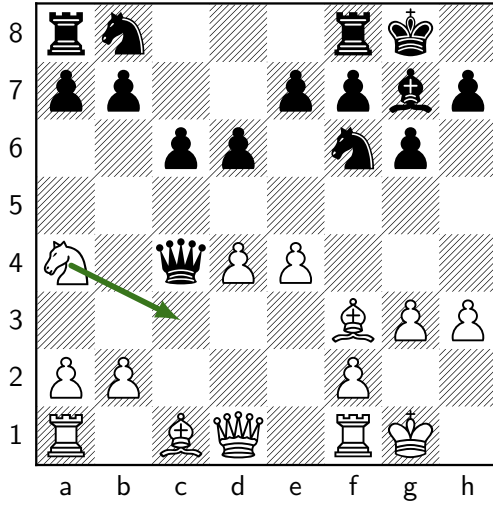
Figure 9: **Move 15, White: Bxa6.** (a) Our position; Melanie plays Bxa6 (d3a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#527` (ply 25) [structural distance  $d=117$ , QAP obj=37]: the master played Rb8; under  $\mu$  our Bc1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Nh4 as their Bc8, our rh8 as their Qd8; transferred move c1d2. (c) Retrieved from `lichess_elite_2024-09_p0012#3249` (ply 23) [structural distance  $d=116$ , QAP obj=32]: the master played Nac7; under  $\mu$  our Pd4 plays the role of that game’s Na3 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Bc8, our Pf4 as their Rf8; transferred move d4d5. (d) Retrieved from `lichess_elite_2024-09_p0031#1648` (ply 30) [structural distance  $d=113$ , QAP obj=31]: the master played Nh3; under  $\mu$  our Na4 plays the role of that game’s Nf4; cross-type roles: our Nh4 as their Pa2, our Pd4 as their Bd2; transferred move a4c3. Synthesis: no retrieved analog transferred to Bxa6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



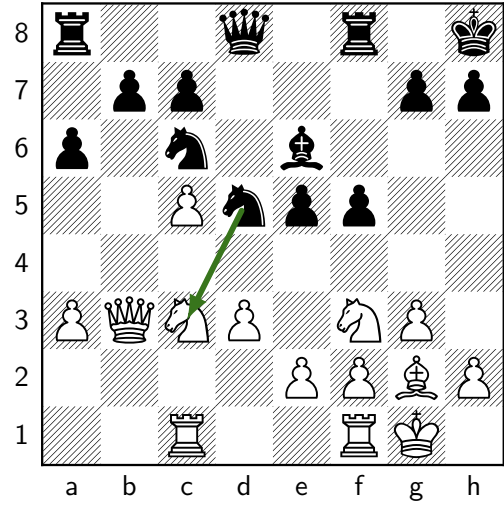
(a)



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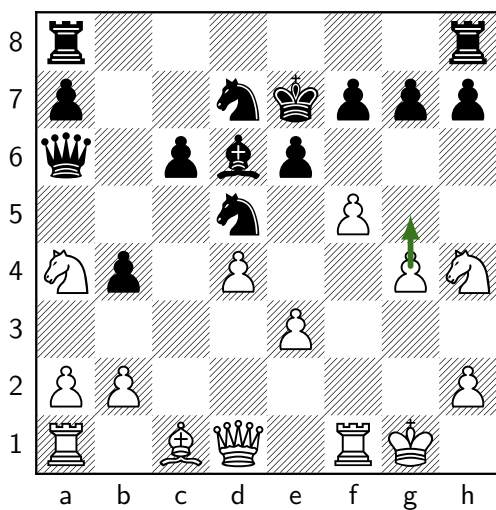


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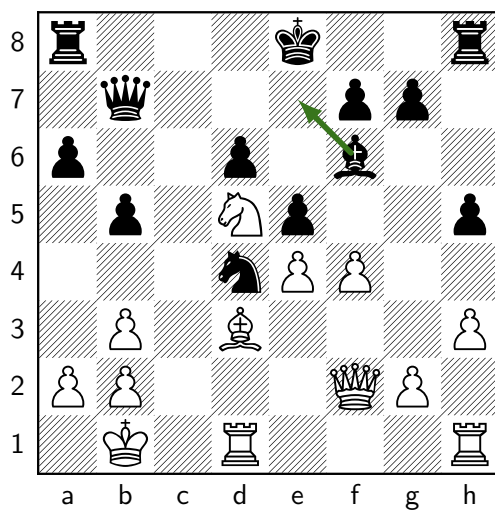


(d)

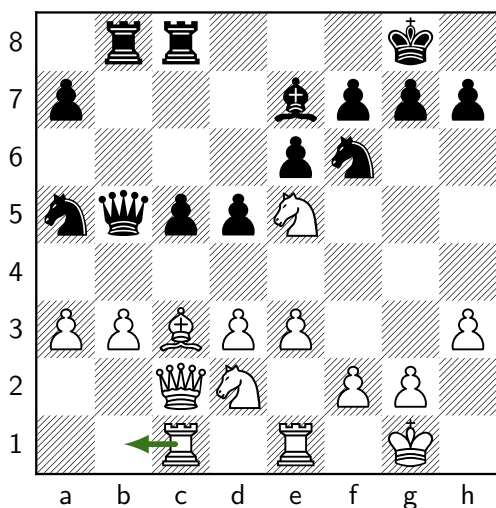
Figure 10: **Move 16, White: f5**. (a) Our position; Melanie plays f5 (f4f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#1221` (ply 28) [structural distance  $d=95$ , QAP obj=30]: the master played Bf4; under  $\mu$  our Pf4 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our pc6 as their Ra1, our Ra1 as their Pa2; transferred move f4f5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#2897` (ply 22) [structural distance  $d=87$ , QAP obj=64]: the master played Nc3; under  $\mu$  our Na4 plays the role of that game’s Na4; cross-type roles: our Qd1 as their Ra1, our Rf1 as their Bc1; transferred move a4c3. (d) Retrieved from `lichess_elite_2024-09_p0016#596` (ply 29) [structural distance  $d=78$ , QAP obj=47]: the master played Nxc3; under  $\mu$  our Na4 plays the role of that game’s Nd4; cross-type roles: our Pb2 as their Ra8, our nf6 as their Qd8; transferred move a4c5. Synthesis: 1 retrieved analog transferred to f5 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



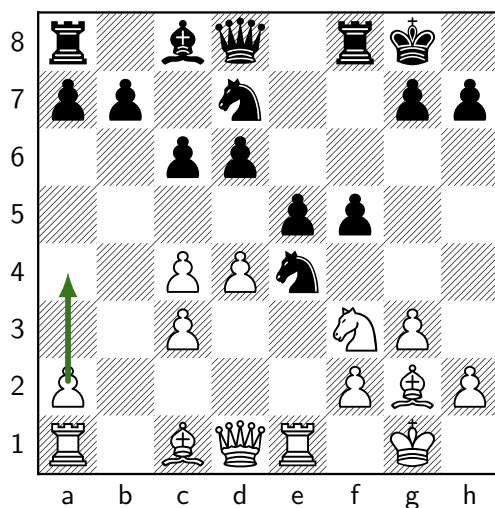
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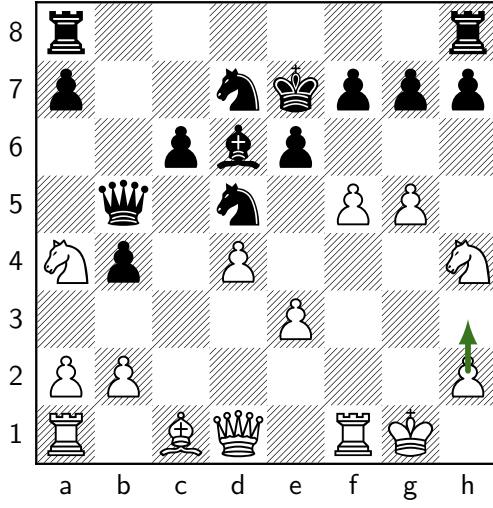


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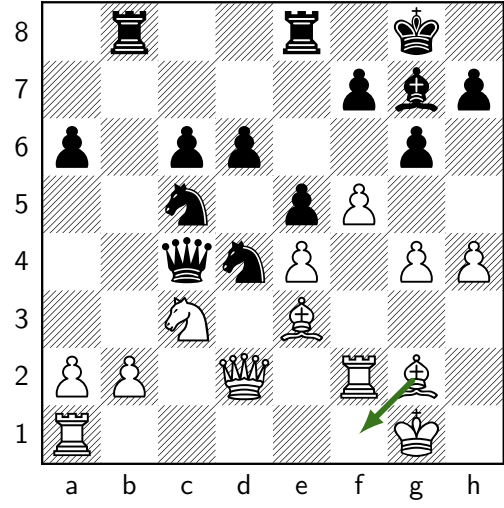


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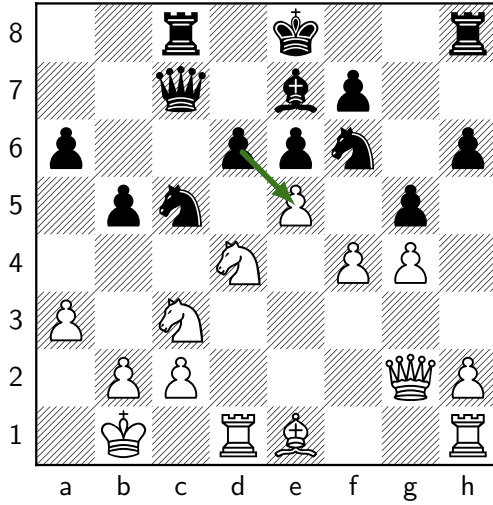
Figure 11: **Move 17, White: g5.** (a) Our position; Melanie plays g5 (g4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#1963` (ply 33) [structural distance  $d=91$ , QAP obj=67]: the master played Be7; under  $\mu$  our  $Pg4$  plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our nd7 as their Pf7, our Qd1 as their Pa6; transferred move g4g5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#137` (ply 38) [structural distance  $d=83$ , QAP obj=37]: the master played Rb1; under  $\mu$  our  $Pg4$  plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Re1, our pa7 as their Qc2; transferred move g4g5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#2715` (ply 20) [structural distance  $d=95$ , QAP obj=47]: the master played a4; under  $\mu$  our Ra1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra1, our nd5 as their Bc1; transferred move a1b1. Synthesis: 2 retrieved analogs transferred to g5 (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



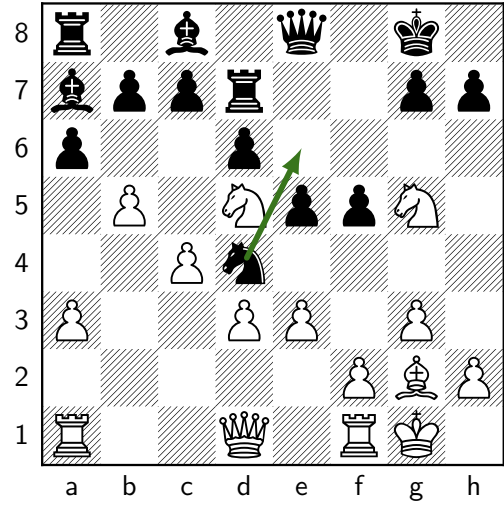
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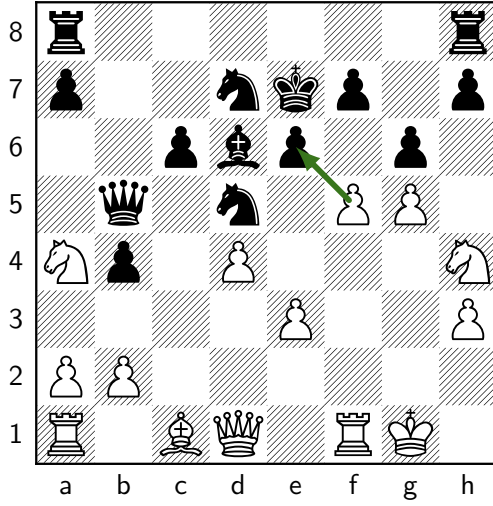


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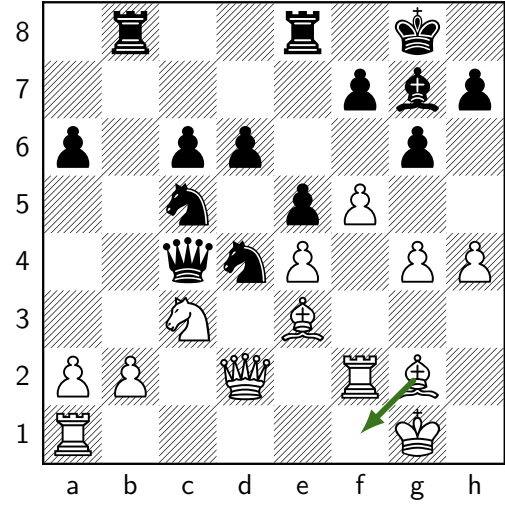


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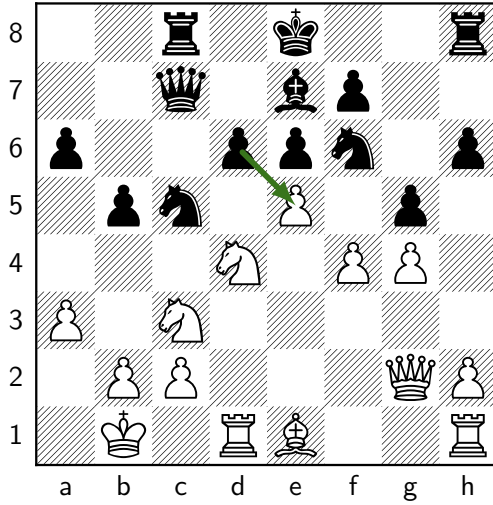
Figure 12: **Move 18, White: h3.** (a) Our position; Melanie plays h3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#501` (ply 40) [structural distance  $d=101$ , QAP obj=54]: the master played Bf1; under  $\mu$  our *Ph2* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Pa2*, our *nd5* as their *Pb2*; transferred move h2h3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#748` (ply 33) [structural distance  $d=99$ , QAP obj=43]: the master played dxe5; under  $\mu$  our *Na4* plays the role of that game’s *Pd3* (a **cross-type** correspondence); cross-type roles: our *Ph2* as their *Be7*, our *nd5* as their *Pe6*; transferred move a4c3. (d) Retrieved from `lichess_elite_2024-09_p0012#3651` (ply 31) [structural distance  $d=101$ , QAP obj=34]: the master played Ne6; under  $\mu$  our *Na4* plays the role of that game’s *Nd5*; cross-type roles: our *pb4* as their *Ra8*, our *nd7* as their *Qe8*; transferred move a4c3. Synthesis: 1 retrieved analog transferred to h3 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



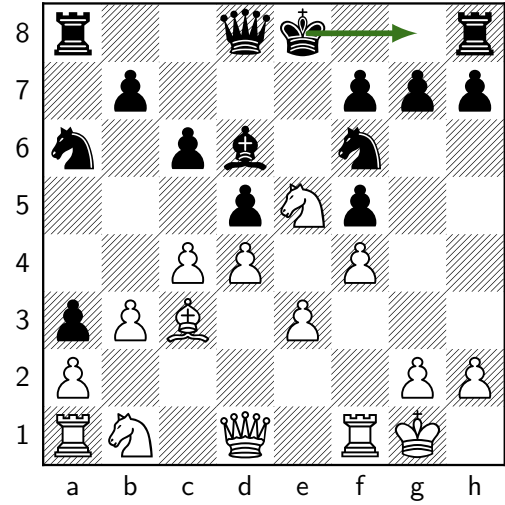
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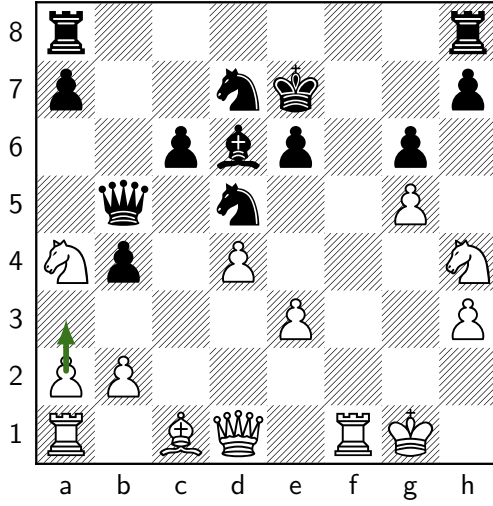


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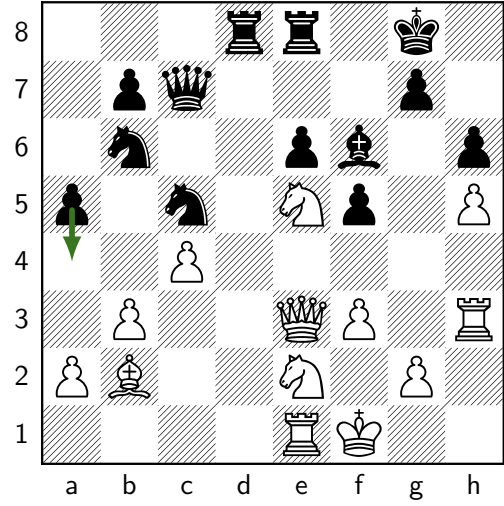


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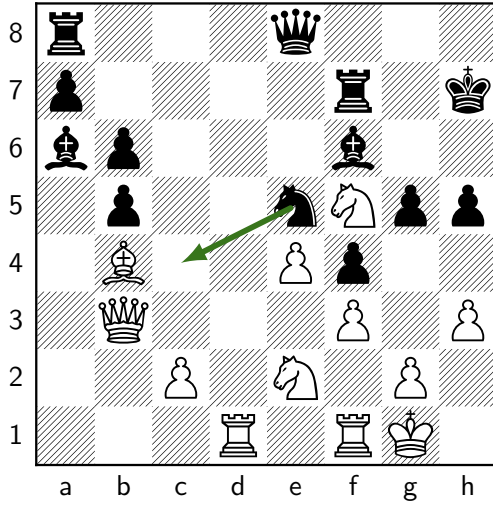
Figure 13: **Move 19, White: fxe6.** (a) Our position; Melanie plays fxe6 (f5e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#501` (ply 40) [structural distance  $d=95$ , QAP obj=57]: the master played Bf1; under  $\mu$  our *Pa2* plays the role of that game’s *Bg2* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Pa2*, our *pe6* as their *Qd2*; transferred move a2a3. (c) Retrieved from `lichess_elite_2024-09_p0024#748` (ply 33) [structural distance  $d=99$ , QAP obj=41]: the master played dxe5; under  $\mu$  our *Na4* plays the role of that game’s *Pd3* (a **cross-type** correspondence); cross-type roles: our *nd5* as their *Qc7*, our *Pd4* as their *Be7*; transferred move a4c3. (d) Retrieved from `lichess_elite_2024-09_p0013#3379` (ply 23) [structural distance  $d=120$ , QAP obj=38]: the master played O-O; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Rh8*, our *Nh4* as their *Pg7*; transferred move g1g2. Synthesis: no retrieved analog transferred to fxe6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



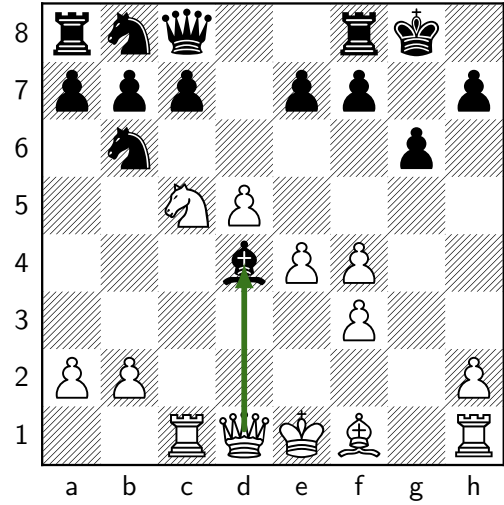
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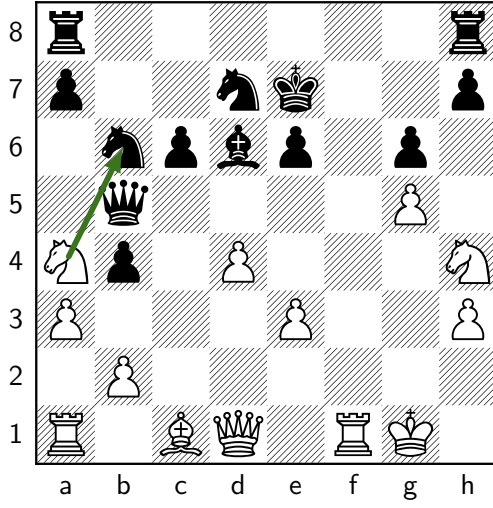
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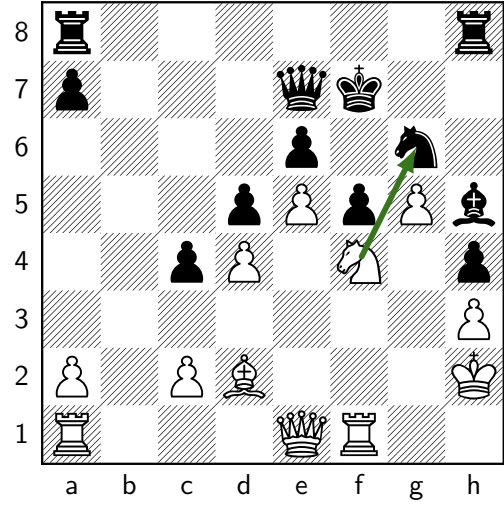
(d)

Figure 14: **Move 20, White: a3.** (a) Our position; Melanie plays a3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#530` (ply 51) [structural distance  $d=92$ , QAP obj=35]: the master played a4; under  $\mu$  our *Pa2* plays the role of that game’s *Pa4*; cross-type roles: our *pc6* as their *Rd8*, our *Qd1* as their *Re8*; transferred move a2a3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#2924` (ply 47) [structural distance  $d=84$ , QAP obj=16]: the master played Nc4; under  $\mu$  our *Pa2* plays the role of that game’s *Ne4* (a **cross-type** correspondence); cross-type roles: our *pa7* as their *Ra8*, our *ra8* as their *Pa7*; transferred move a2a3 (**endorses this move**). (d) Retrieved from `twic1500#183` (ply 30) [structural distance  $d=89$ , QAP obj=26]: the master played Qxd4; under  $\mu$  our *Rf1* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Rc1*, our *qb5* as their *bd4*; transferred move f1f3. Synthesis: 2 retrieved analogs transferred to a3 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

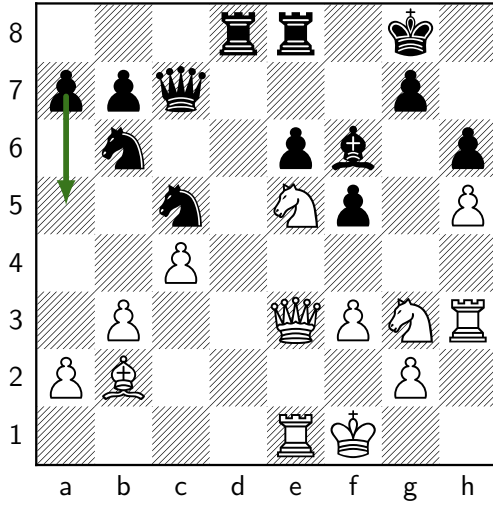




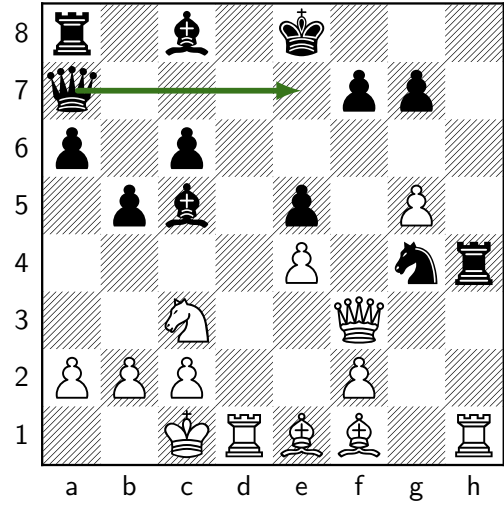
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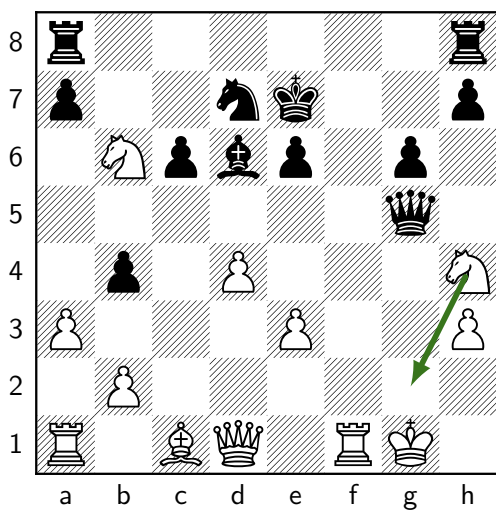
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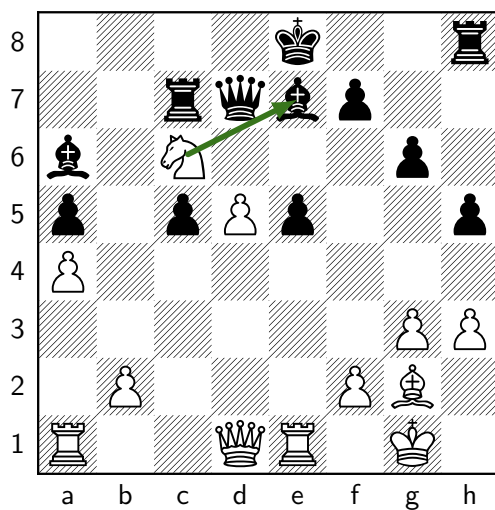
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Figure 15: **Move 21, White: Nxb6.** (a) Our position; Melanie plays Nxb6 (a4b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `twic1500#327` (ply 52) [structural distance  $d=105$ , QAP obj=41]: the master played Nxg6; under  $\mu$  our Na4 plays the role of that game’s Nf4; cross-type roles: our Qd1 as their Ra1, our Pg5 as their Qe1; transferred move a4b6 (**endorses this move**). (c) Retrieved from `lichess.elite_2024-09_p0003#530` (ply 49) [structural distance  $d=92$ , QAP obj=42]: the master played a5; under  $\mu$  our Pd4 plays the role of that game’s Pa2; cross-type roles: our nd7 as their Rd8, our Ra1 as their Pb7; transferred move d4d5. (d) Retrieved from `lichess.elite_2024-09_p0019#3939` (ply 33) [structural distance  $d=100$ , QAP obj=42]: the master played Qe7; under  $\mu$  our Qd1 plays the role of that game’s Qa2; cross-type roles: our ra8 as their Bc8, our pg6 as their Bc5; transferred move d1d2. Synthesis: 1 retrieved analog transferred to Nxb6 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

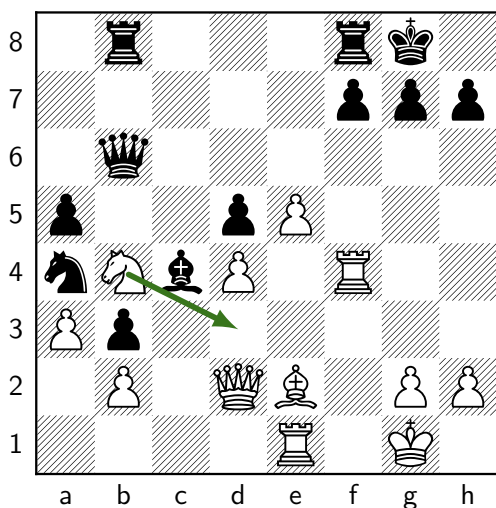




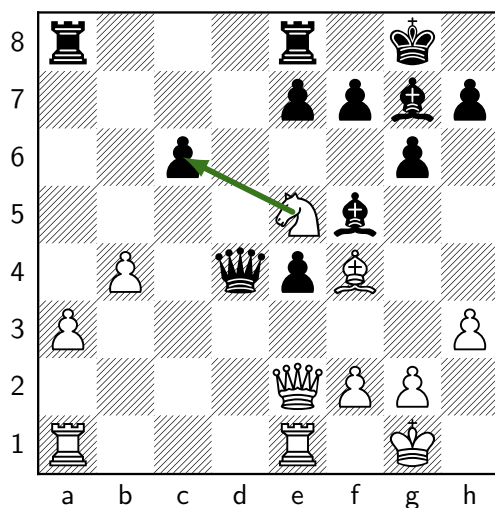
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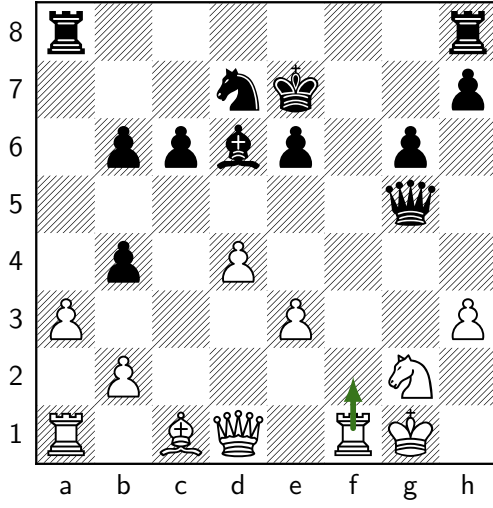


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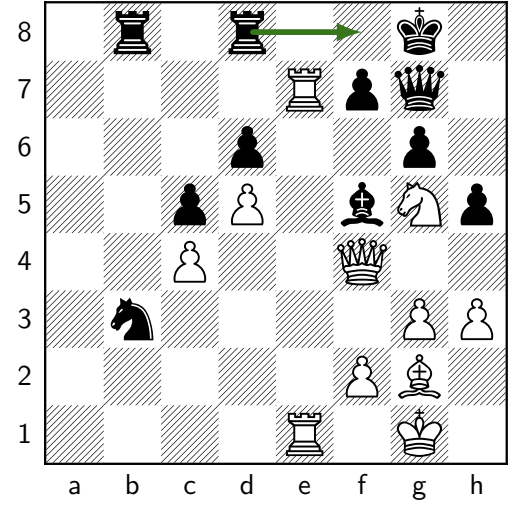


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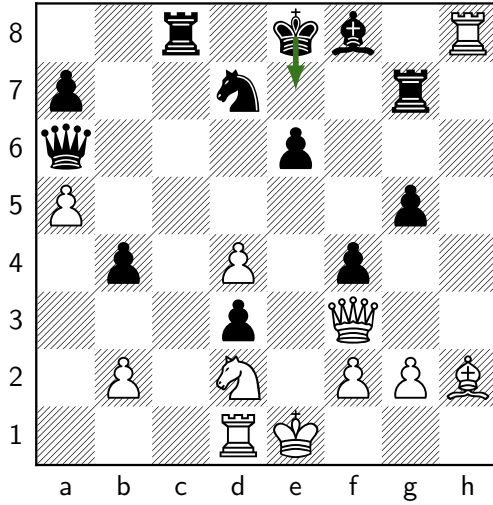
Figure 16: **Move 22, White: Ng2.** (a) Our position; Melanie plays Ng2 (**h4g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#753` (ply 44) [structural distance  $d=81$ , QAP obj=34]: the master played Nxe7; under  $\mu$  our Nh4 plays the role of that game’s Nc6; cross-type roles: our rh8 as their Qd1, our bd6 as their Re1; transferred move h4g2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#753` (ply 52) [structural distance  $d=104$ , QAP obj=29]: the master played Nd3; under  $\mu$  our Nh4 plays the role of that game’s Nb4; cross-type roles: our Bc1 as their Qd2, our rh8 as their Be2; transferred move h4g2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#3312` (ply 36) [structural distance  $d=94$ , QAP obj=29]: the master played Nxc6; under  $\mu$  our Nh4 plays the role of that game’s Ne5; cross-type roles: our Pd4 as their Ra1, our Bc1 as their Qe2; transferred move h4g2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ng2 (summed QAP objective 111, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



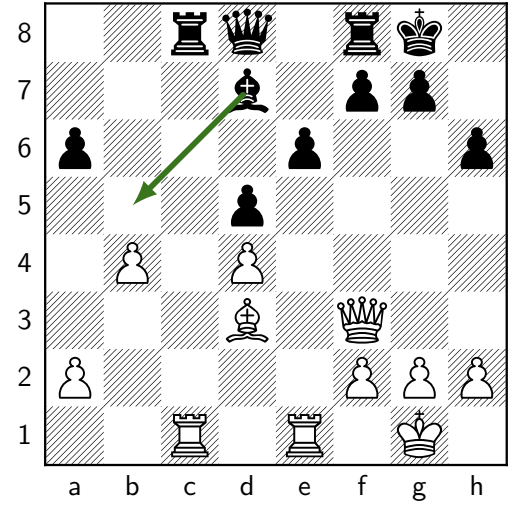
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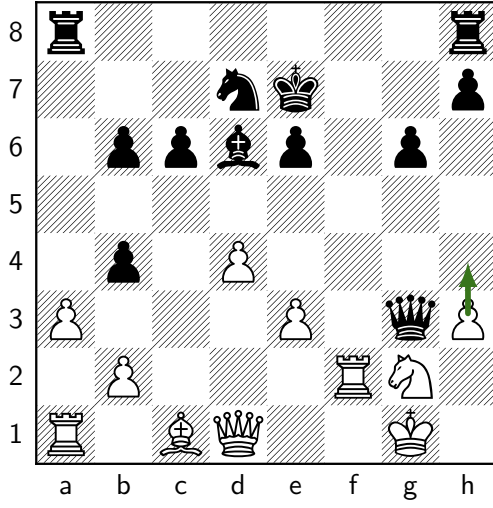


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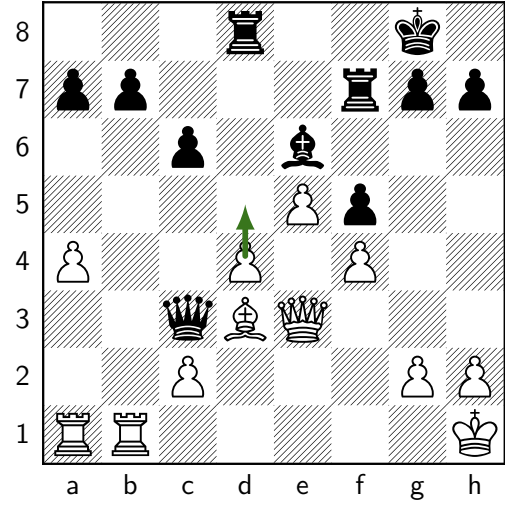


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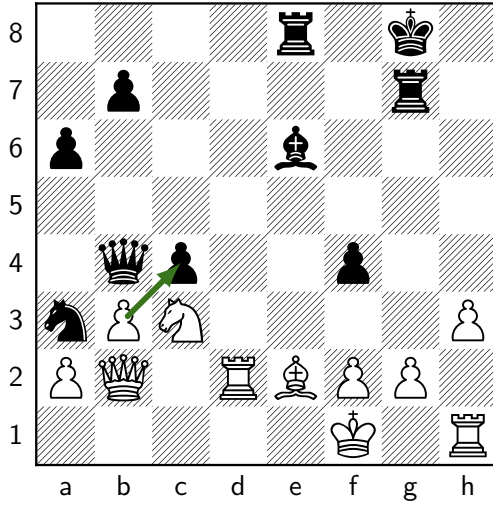
Figure 17: **Move 23, White: Rf2.** (a) Our position; Melanie plays Rf2 (f1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#3645` (ply 55) [structural distance  $d=76$ , QAP obj=50]: the master played Rf8; under  $\mu$  our Rf1 plays the role of that game’s Rd1; cross-type roles: our pb4 as their Rb8, our qg5 as their re7; transferred move f1f2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#2849` (ply 47) [structural distance  $d=72$ , QAP obj=68]: the master played Ke7; under  $\mu$  our Kg1 plays the role of that game’s Ke1; cross-type roles: our pg6 as their Rc8, our Ng2 as their Bf8; transferred move g1f2. (d) Retrieved from `lichess_elite_2024-09_p0024#2301` (ply 43) [structural distance  $d=71$ , QAP obj=56]: the master played Bb5; under  $\mu$  our Bc1 plays the role of that game’s Bd2; cross-type roles: our Pa3 as their Rc8, our Ra1 as their Qd8; transferred move c1d2. Synthesis: 1 retrieved analog transferred to Rf2 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



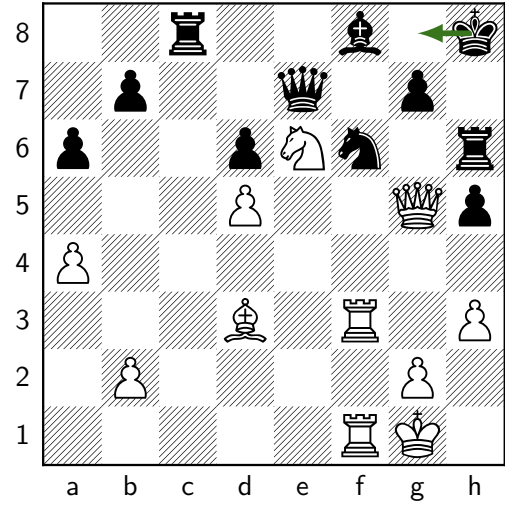
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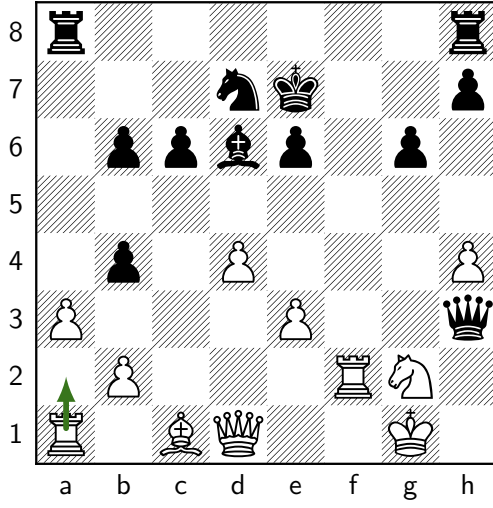


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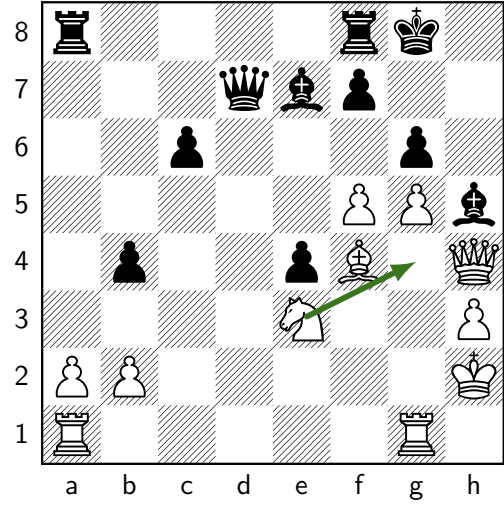


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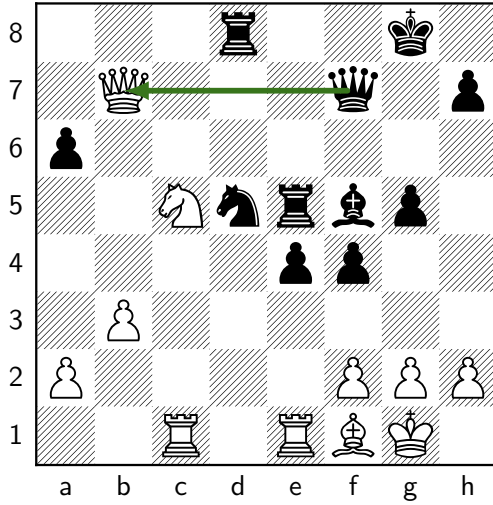
Figure 18: **Move 24, White: h4.** (a) Our position; Melanie plays h4 (h3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `twic1500#580` (ply 38) [structural distance  $d=81$ , QAP obj=86]: the master played d5; under  $\mu$  our  $Ph3$  plays the role of that game’s  $Pd4$ ; cross-type roles: our  $pb4$  as their  $Rb1$ , our  $ra8$  as their  $Pg2$ ; transferred move h3h4 (**endorses this move**). (c) Retrieved from `lichess_elite.2024-09_p0018#3426` (ply 52) [structural distance  $d=80$ , QAP obj=57]: the master played bxc4; under  $\mu$  our  $Ph3$  plays the role of that game’s  $Pb3$ ; cross-type roles: our  $Qd1$  as their  $Rh1$ , our  $nd7$  as their  $Rd2$ ; transferred move h3h4 (**endorses this move**). (d) Retrieved from `lichess_elite.2024-09_p0028#2662` (ply 51) [structural distance  $d=85$ , QAP obj=111]: the master played Kg8; under  $\mu$  our  $Kg1$  plays the role of that game’s  $Kh1$ ; cross-type roles: our  $Qd1$  as their  $Rc8$ , our  $Ng2$  as their  $Bf8$ ; transferred move g1h1. Synthesis: 2 retrieved analogs transferred to h4 (summed QAP objective 143, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



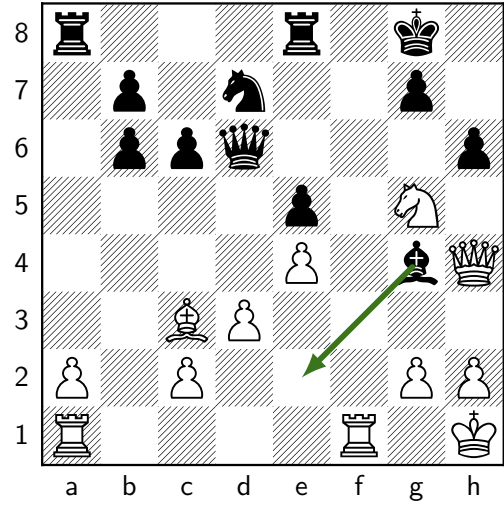
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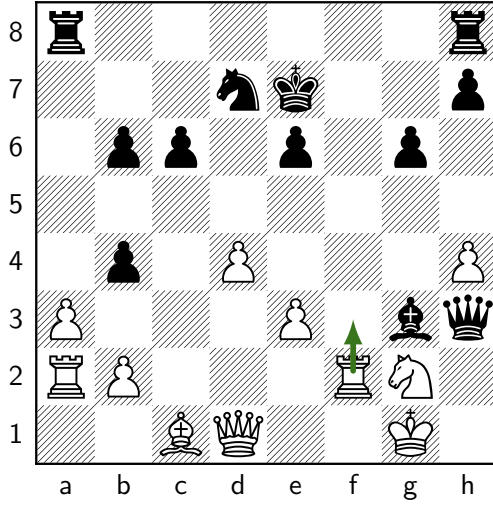


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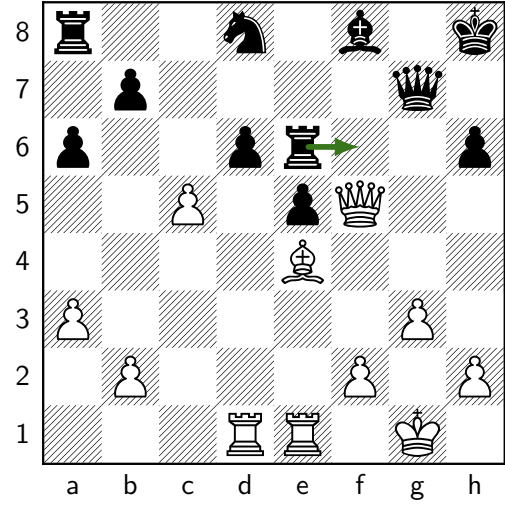


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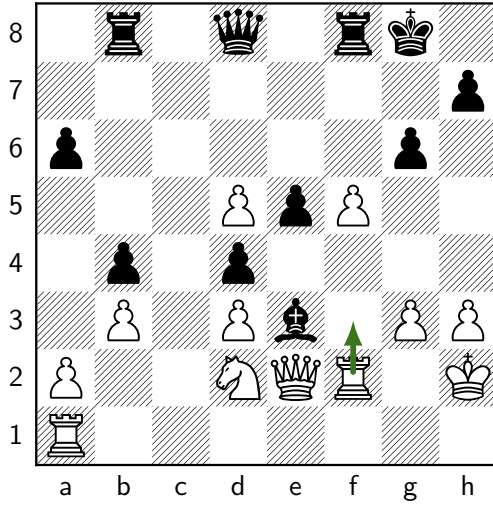
Figure 19: **Move 25, White: Ra2.** (a) Our position; Melanie plays Ra2 (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#1216` (ply 52) [structural distance  $d=76$ , QAP obj=40]: the master played Ng4; under  $\mu$  our *Ra1* plays the role of that game’s *Ne3* (a **cross-type** correspondence); cross-type roles: our *nd7* as their *Ra1*, our *bd6* as their *Rg1*; transferred move a1a2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#1576` (ply 47) [structural distance  $d=70$ , QAP obj=50]: the master played Qxb7; under  $\mu$  our *Ng2* plays the role of that game’s *Qf2* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Rd8*, our *Rf2* as their *Ph7*; transferred move g2e1. (d) Retrieved from `lichess_elite_2024-09_p0017#788` (ply 39) [structural distance  $d=77$ , QAP obj=50]: the master played Be2; under  $\mu$  our *Ng2* plays the role of that game’s *Bg5* (a **cross-type** correspondence); cross-type roles: our *pb4* as their *Nd7*, our *Rf2* as their *Pg7*; transferred move g2e1. Synthesis: 1 retrieved analog transferred to Ra2 (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



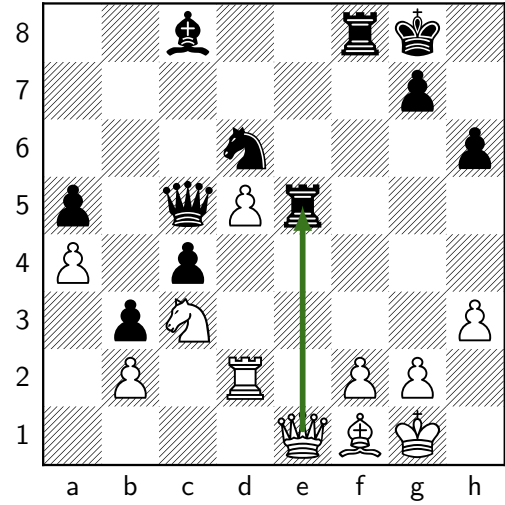
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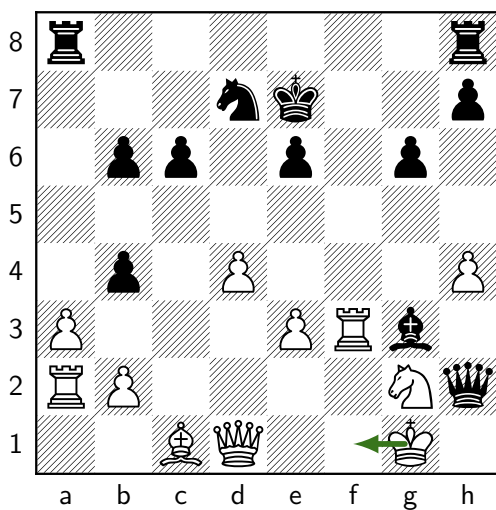


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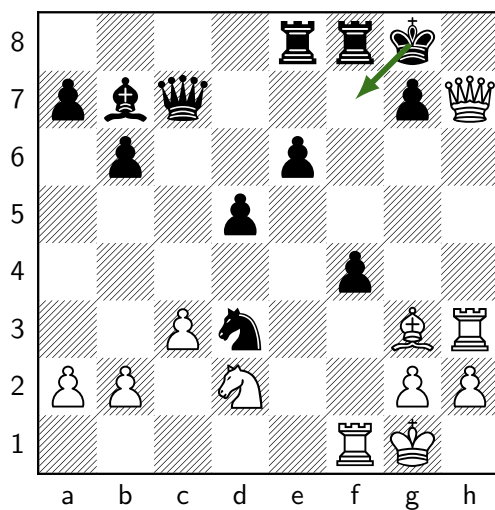


(d)

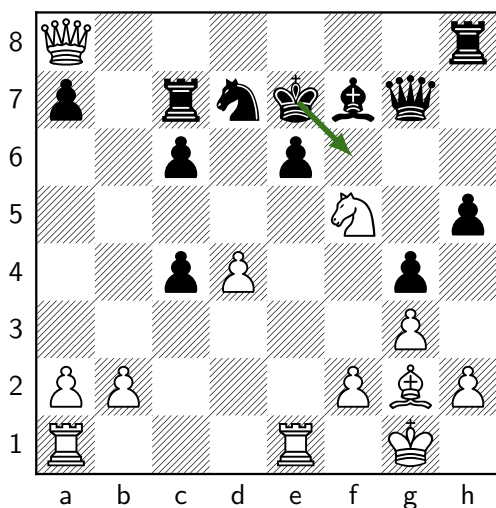
Figure 20: **Move 26, White: Rf3**. (a) Our position; Melanie plays Rf3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#1271` (ply 49) [structural distance  $d=74$ , QAP obj=64]: the master played Rf6; under  $\mu$  our Rf2 plays the role of that game’s Re3; cross-type roles: our Bc1 as their Ra8, our pg6 as their Nd8; transferred move f2f3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#2665` (ply 54) [structural distance  $d=83$ , QAP obj=53]: the master played Rf3; under  $\mu$  our Rf2 plays the role of that game’s Rf2; cross-type roles: our pb6 as their Ra1, our Rf2 as their Nd2; transferred move f2f3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0012#3603` (ply 66) [structural distance  $d=85$ , QAP obj=32]: the master played Qxe5; under  $\mu$  our Rf2 plays the role of that game’s Qe1 (a **cross-type** correspondence); cross-type roles: our Ng2 as their Bf1, our Pd4 as their Nc3; transferred move f2f3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rf3 (summed QAP objective 167, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



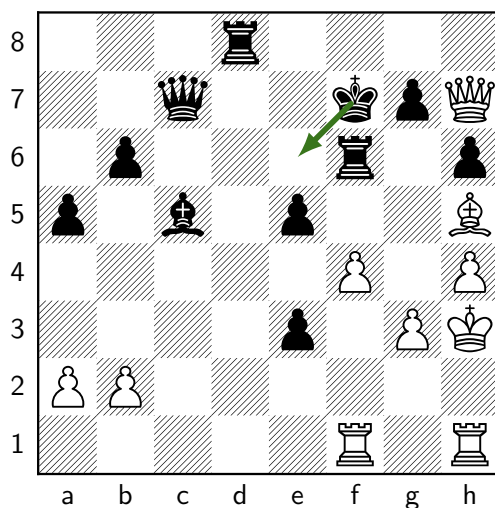
(a)



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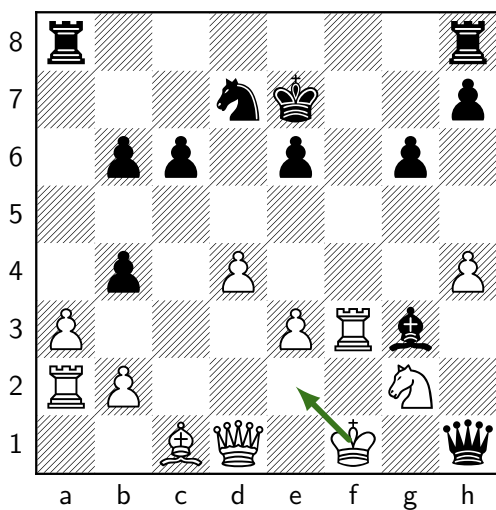


(c)

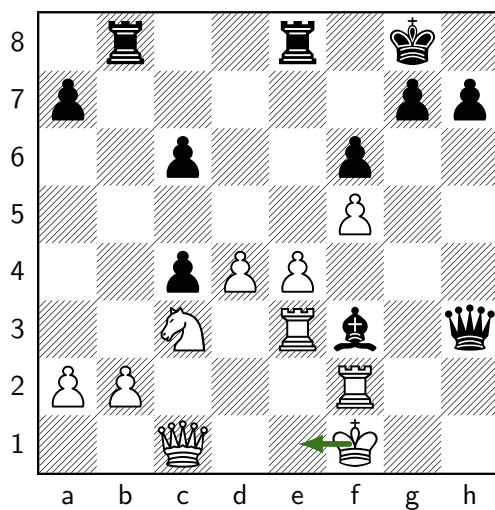


(d)

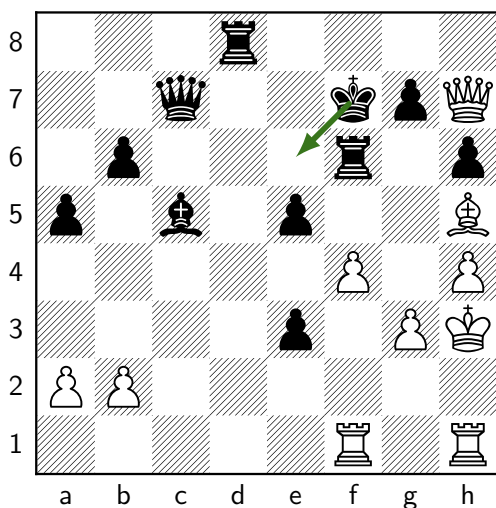
Figure 21: **Move 27, White: Kf1.** (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0012#2701` (ply 43) [structural distance  $d=80$ , QAP obj=79]: the master played Kf7; under  $\mu$  our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Qd1* as their *Rf8*, our *Pe3* as their *Bb7*; transferred move g1f1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#503` (ply 45) [structural distance  $d=91$ , QAP obj=60]: the master played Kf6; under  $\mu$  our *Kg1* plays the role of that game’s *Ke2*; cross-type roles: our *bg3* as their *qa8*, our *Ng2* as their *Rh8*; transferred move g1f1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#2445` (ply 67) [structural distance  $d=91$ , QAP obj=44]: the master played Ke6; under  $\mu$  our *Kg1* plays the role of that game’s *Kf2*; cross-type roles: our *Pe3* as their *Rd8*, our *Ng2* as their *Pg7*; transferred move g1f1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kf1 (summed QAP objective 227, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



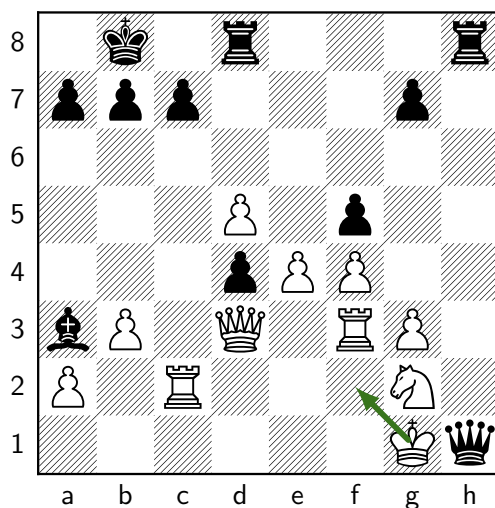
(a)



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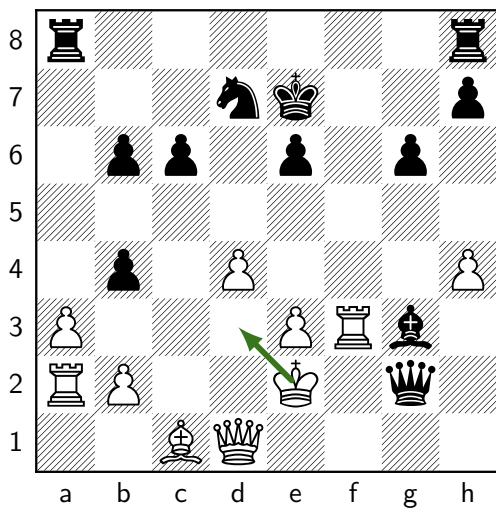
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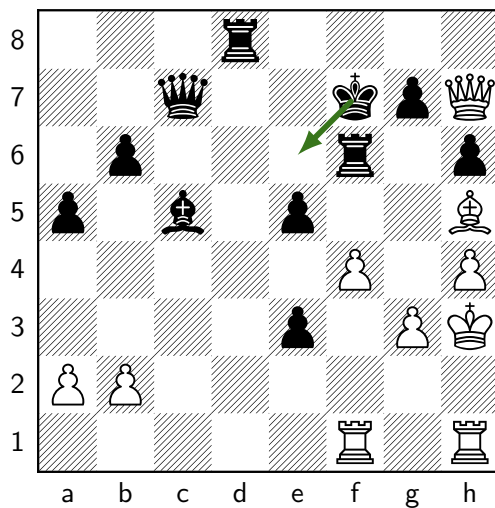
(d)

Figure 22: **Move 28, White: Ke2.** (a) Our position; Melanie plays Ke2 (f1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2099` (ply 68) [structural distance  $d=91$ , QAP obj=97]: the master played Ke1; under  $\mu$  our Kf1 plays the role of that game’s Kf1; cross-type roles: our nd7 as their Pb2, our bg3 as their Nc3; transferred move f1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#2445` (ply 67) [structural distance  $d=78$ , QAP obj=88]: the master played Ke6; under  $\mu$  our Kf1 plays the role of that game’s Kf2; cross-type roles: our bg3 as their Rd8, our Ng2 as their Pg7; transferred move f1e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#1572` (ply 46) [structural distance  $d=89$ , QAP obj=66]: the master played Kf2; under  $\mu$  our Kf1 plays the role of that game’s Kg1; cross-type roles: our ra8 as their Pa2, our pg6 as their ba3; transferred move f1e2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ke2 (summed QAP objective 251, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

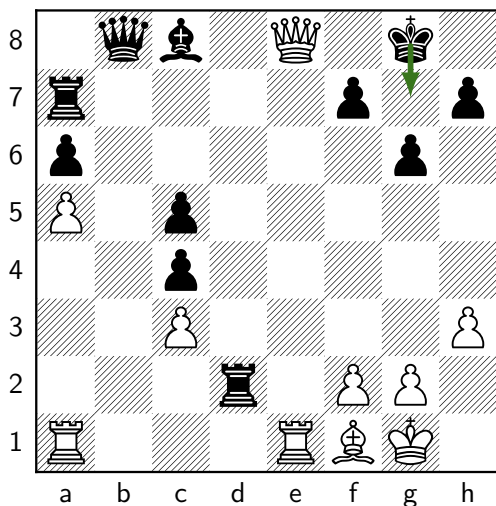




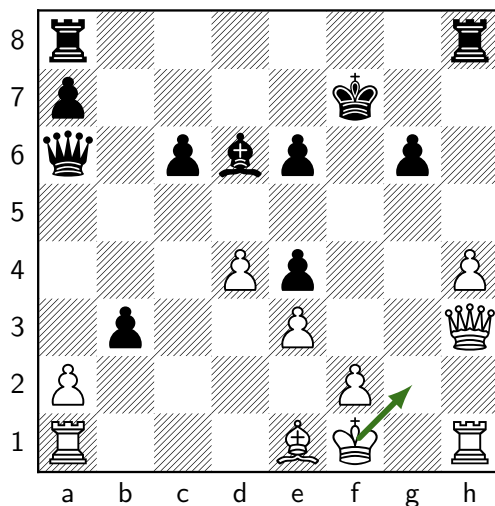
(a)



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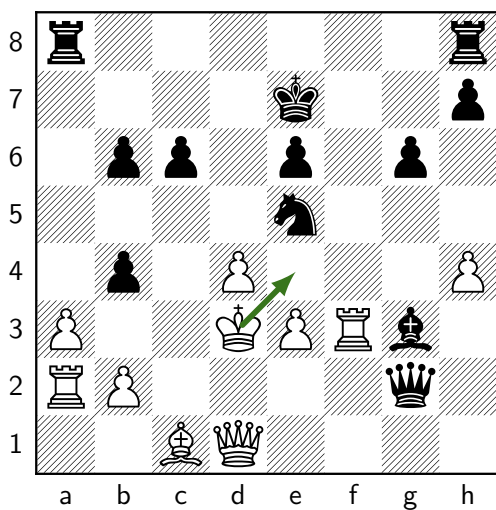


(c)

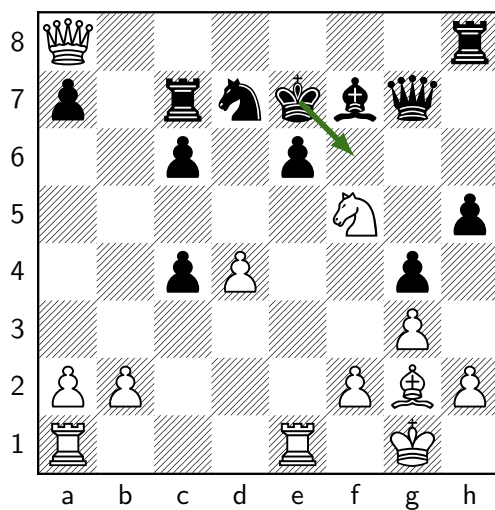


(d)

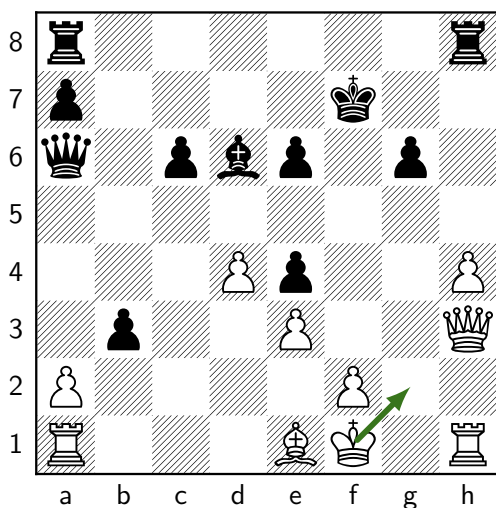
Figure 23: **Move 29, White: Kd3.** (a) Our position; Melanie plays Kd3 (e2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#2445` (ply 67) [structural distance  $d=66$ , QAP obj=76]: the master played Ke6; under  $\mu$  our Ke2 plays the role of that game’s Kf2; cross-type roles: our pe6 as their Qc7, our Rf3 as their Pg7; transferred move e2d3 (**endorses this move**). (c) Retrieved from `twic1500#123` (ply 51) [structural distance  $d=77$ , QAP obj=75]: the master played Kg7; under  $\mu$  our Ke2 plays the role of that game’s Kg1; cross-type roles: our Pb2 as their Qb8, our Pa3 as their Bc8; transferred move e2d3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#2387` (ply 50) [structural distance  $d=55$ , QAP obj=53]: the master played Kg2; under  $\mu$  our Ke2 plays the role of that game’s Kf1; cross-type roles: our Pe3 as their Ra1, our Qd1 as their Rh1; transferred move e2d3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kd3 (summed QAP objective 245, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



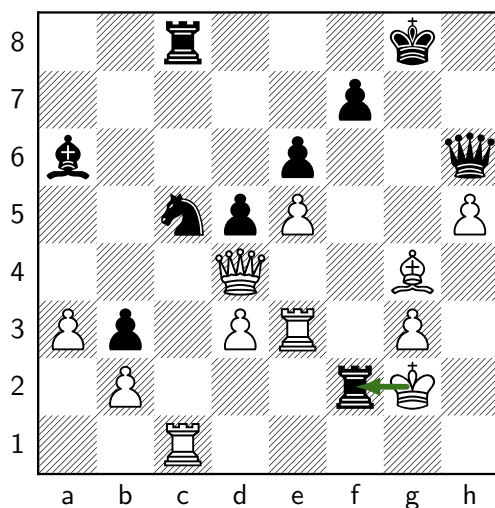
(a)



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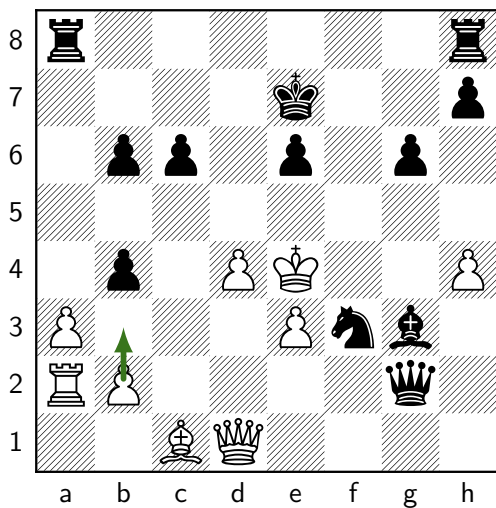


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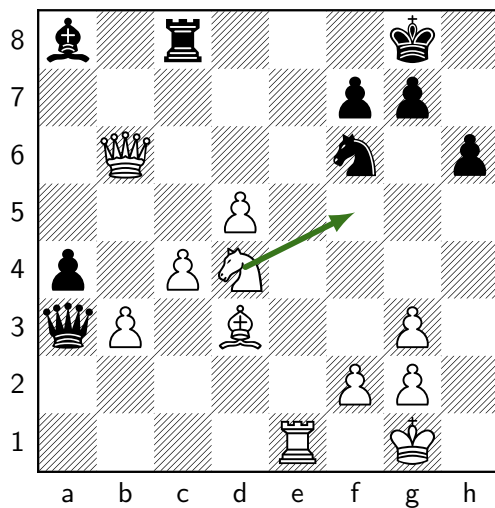


(d)

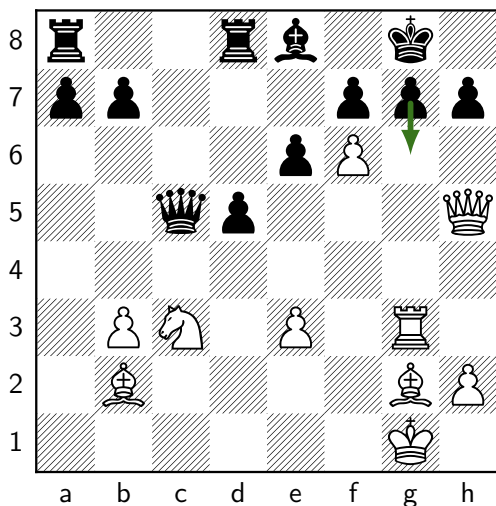
Figure 24: **Move 30, White: Ke4.** (a) Our position; Melanie plays Ke4 (d3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#503` (ply 45) [structural distance  $d=70$ , QAP obj=90]: the master played Kf6; under  $\mu$  our Kd3 plays the role of that game’s Ke2; cross-type roles: our pb4 as their Rh8, our Pe3 as their Rc7; transferred move d3e4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#2387` (ply 50) [structural distance  $d=87$ , QAP obj=57]: the master played Kg2; under  $\mu$  our Kd3 plays the role of that game’s Kf1; cross-type roles: our Ph4 as their Be1, our Qd1 as their Rh1; transferred move d3e4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#1182` (ply 62) [structural distance  $d=94$ , QAP obj=57]: the master played Kxf2; under  $\mu$  our Kd3 plays the role of that game’s Kg2; cross-type roles: our ne5 as their rf2, our Qd1 as their Pa3; transferred move d3e4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ke4 (summed QAP objective 203, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



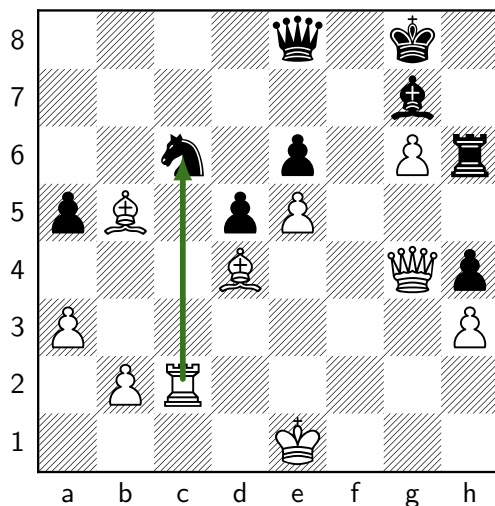
(a)



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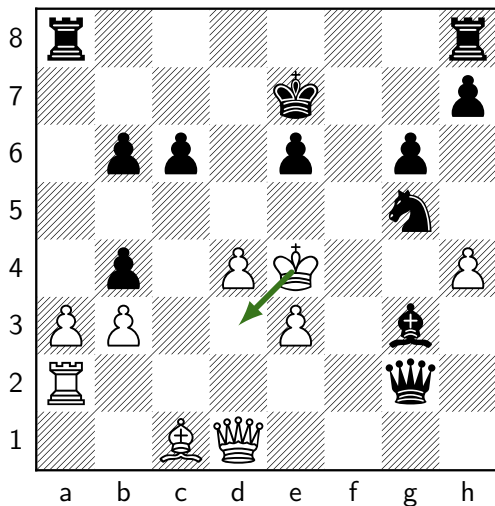


(c)

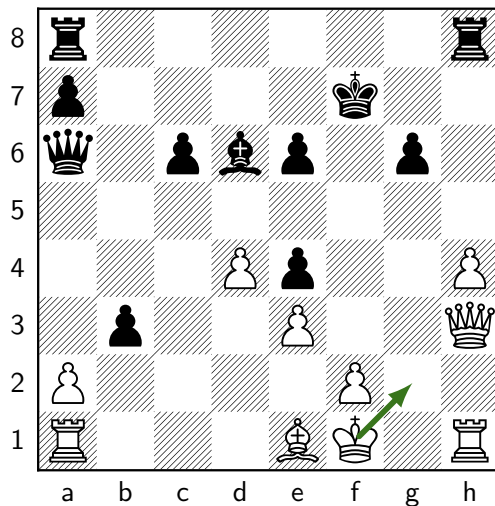


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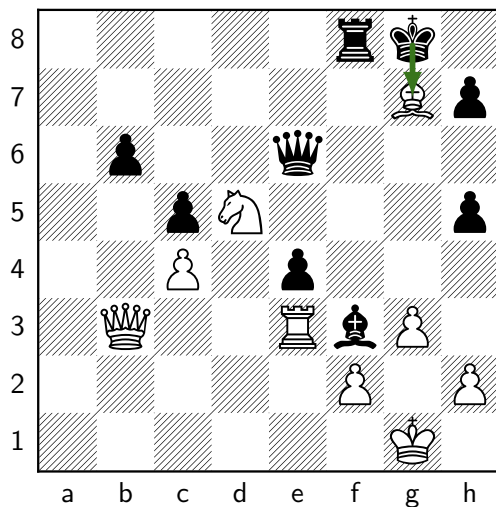
Figure 25: **Move 31, White: b3.** (a) Our position; Melanie plays **b3** (**b2b3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#1212` (ply 48) [structural distance  $d=88$ , QAP obj=34]: the master played **Nf5**; under  $\mu$  our **Pb2** plays the role of that game’s **Nd4** (a **cross-type** correspondence); cross-type roles: our **Pe3** as their **Re1**, our **ra8** as their **qa3**; transferred move **b2b3** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#227` (ply 45) [structural distance  $d=99$ , QAP obj=71]: the master played **g6**; under  $\mu$  our **Pa3** plays the role of that game’s **Pg2**; cross-type roles: our **Bc1** as their **Ra8**, our **Qd1** as their **Rd8**; transferred move **a3a4**. (d) Retrieved from `lichess_elite_2024-09_p0031#1204` (ply 66) [structural distance  $d=102$ , QAP obj=54]: the master played **Rxc6**; under  $\mu$  our **Ke4** plays the role of that game’s **Rc2** (a **cross-type** correspondence); cross-type roles: our **Bc1** as their **Pb2**, our **bg3** as their **Pa3**; transferred move **e4d3**. Synthesis: 1 retrieved analog transferred to **b3** (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)



(b)



(c)

Figure 26: **Move 32, White: Kd3.** (a) Our position; Melanie plays Kd3 (e4d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#2387` (ply 50) [structural distance  $d=95$ , QAP obj=51]: the master played Kg2; under  $\mu$  our *Ke4* plays the role of that game’s *Kf1*; cross-type roles: our *pc6* as their *Ra1*, our *pe6* as their *Be1*; transferred move e4d3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#1184` (ply 63) [structural distance  $d=96$ , QAP obj=5]: the master played Kxg7; under  $\mu$  our *Ke4* plays the role of that game’s *Kg1*; cross-type roles: our *Pe3* as their *Rf8*, our *ng5* as their *bg7*; transferred move e4d3 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Kd3 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

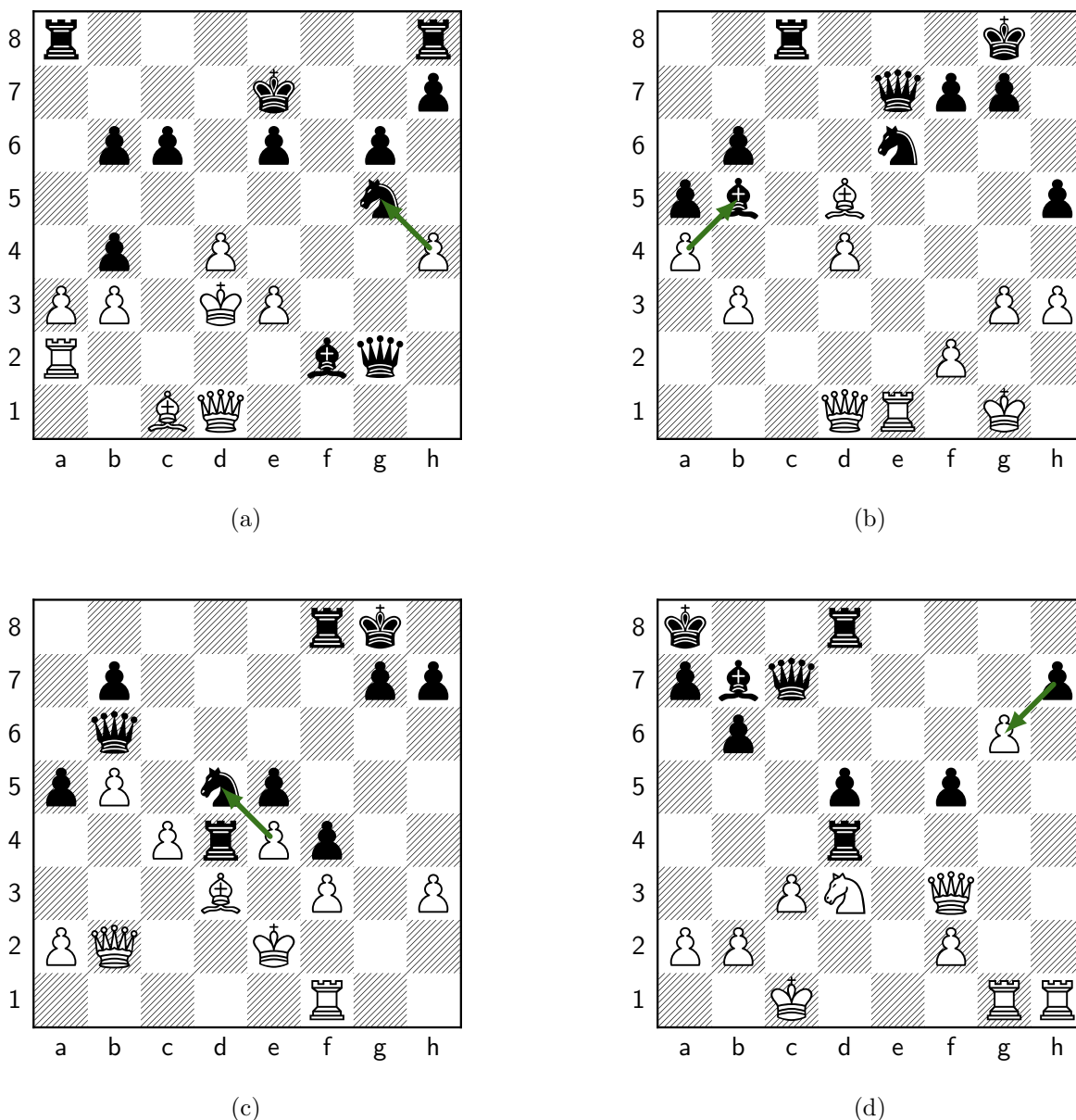


Figure 27: **Move 33, White: hxg5**. (a) Our position; Melanie plays **hxg5** (**h4g5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#1120` (ply 48) [structural distance  $d=75$ , QAP obj=61]: the master played **axb5**; under  $\mu$  our *Ph4* plays the role of that game’s *Pa4*; cross-type roles: our *Pa3* as their *Qd1*, our *Bc1* as their *Pf2*; transferred move **h4g5** (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#571` (ply 54) [structural distance  $d=67$ , QAP obj=56]: the master played **exd5**; under  $\mu$  our *Ph4* plays the role of that game’s *Pe4*; cross-type roles: our *Qd1* as their *Rf1*, our *bf2* as their *Qb2*; transferred move **h4g5** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#1433` (ply 53) [structural distance  $d=81$ , QAP obj=37]: the master played **hxg6**; under  $\mu$  our *Bc1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *gg2* as their *Bb7*, our *ra8* as their *Qc7*; transferred move **c1b2**. Synthesis: 2 retrieved analogs transferred to **hxg5** (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

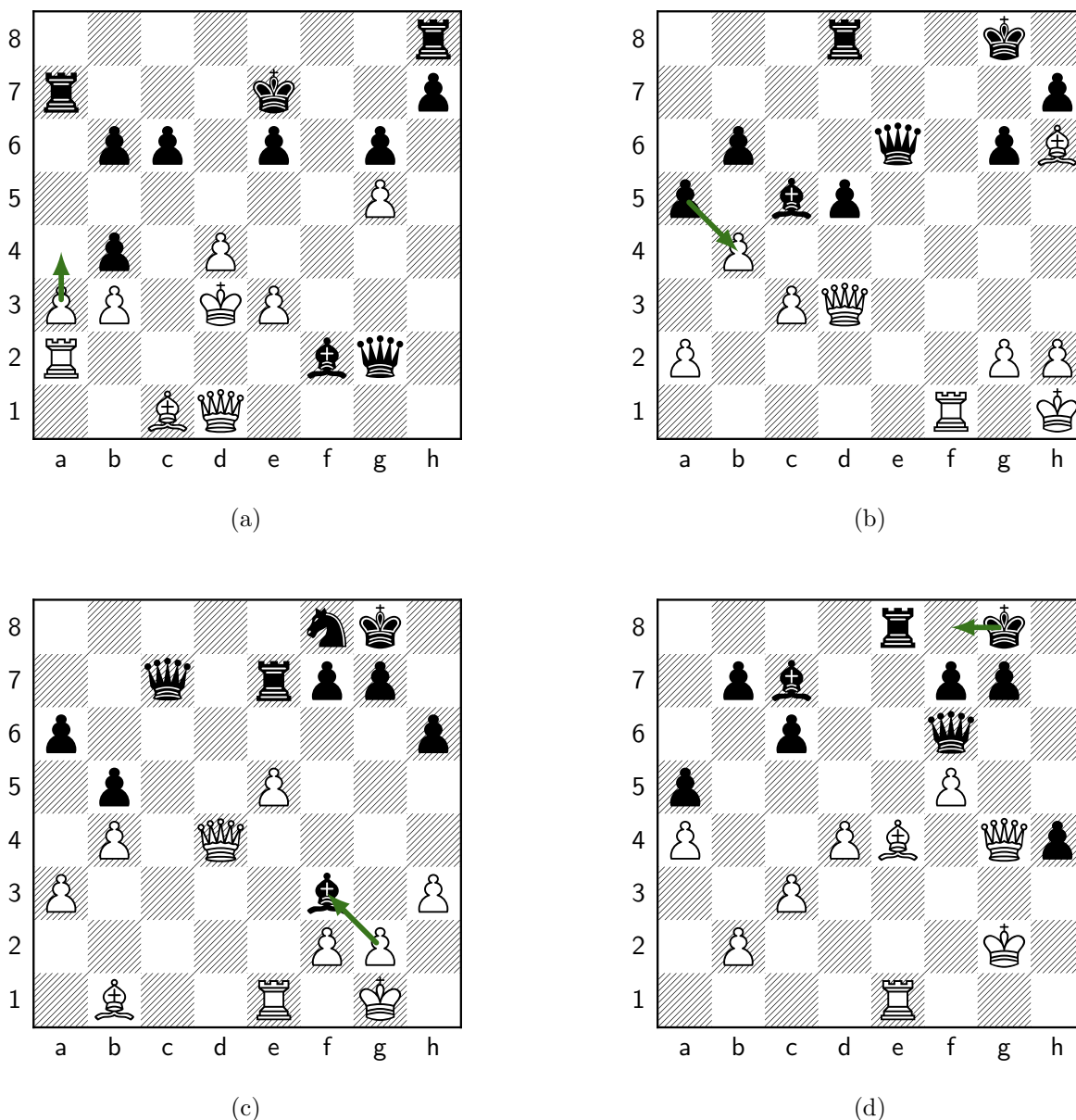
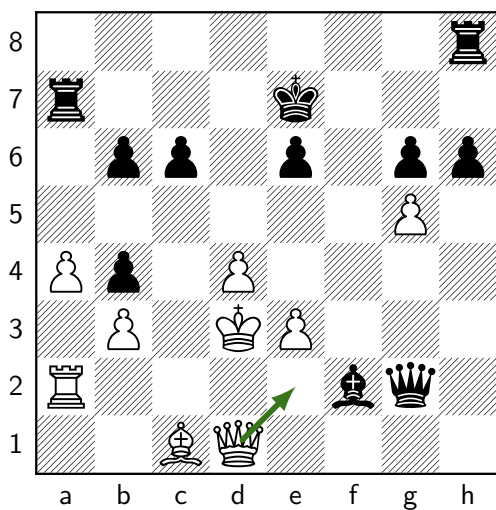
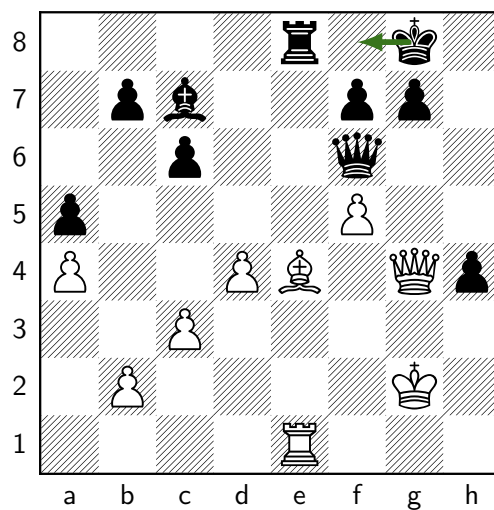


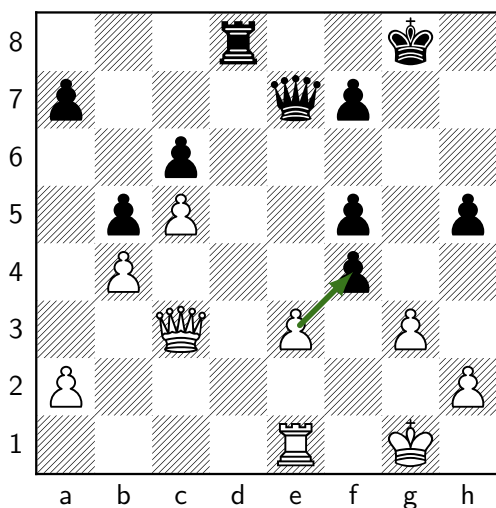
Figure 28: **Move 34, White: a4.** (a) Our position; Melanie plays **a4** (a3a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#115` (ply 59) [structural distance  $d=58$ , QAP obj=42]: the master played **axb4**; under  $\mu$  our  $Pa3$  plays the role of that game's  $Pa4$ ; cross-type roles: our  $Qd1$  as their  $Rd8$ , our  $pg6$  as their  $bh6$ ; transferred move **a3a4** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#316` (ply 56) [structural distance  $d=63$ , QAP obj=35]: the master played **gxf3**; under  $\mu$  our  $Pa3$  plays the role of that game's  $Pg2$ ; cross-type roles: our  $Pd4$  as their  $Bb1$ , our  $Qd1$  as their  $Re1$ ; transferred move **a3a4** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0030#1368` (ply 69) [structural distance  $d=65$ , QAP obj=89]: the master played **Kf8**; under  $\mu$  our  $Kd3$  plays the role of that game's  $Kg1$ ; cross-type roles: our  $Bc1$  as their  $Pb7$ , our  $Qd1$  as their  $Bc7$ ; transferred move **d3e2**. Synthesis: 2 retrieved analogs transferred to **a4** (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



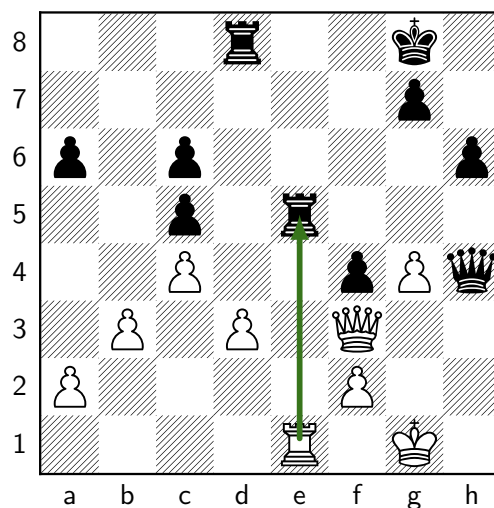
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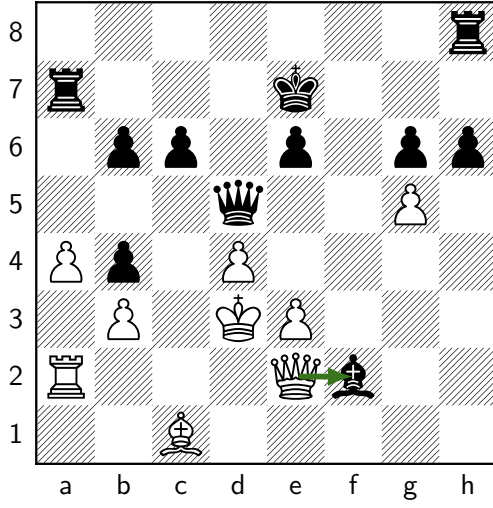
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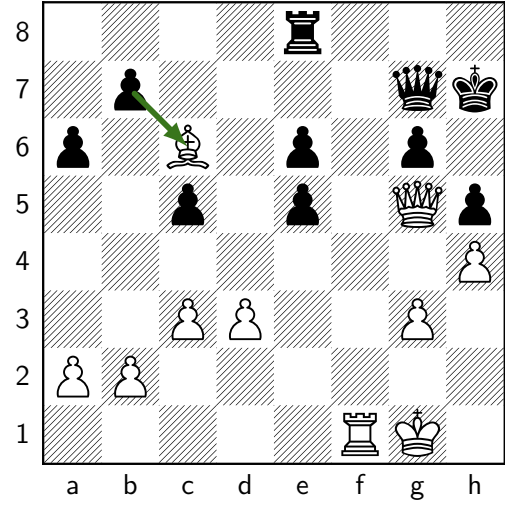
(d)

Figure 29: **Move 35, White: Qe2.** (a) Our position; Melanie plays Qe2 (d1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#1368` (ply 69) [structural distance  $d=63$ , QAP obj=94]: the master played Kf8; under  $\mu$  our Kd3 plays the role of that game’s Kg1; cross-type roles: our Qd1 as their Pb7, our pg6 as their Bc7; transferred move d3e2. (c) Retrieved from `lichess_elite_2024-09_p0019#1501` (ply 52) [structural distance  $d=57$ , QAP obj=89]: the master played exf4; under  $\mu$  our Pg5 plays the role of that game’s Pe3; cross-type roles: our Qd1 as their Re1, our Bc1 as their Pa2; transferred move g5h6. (d) Retrieved from `lichess_elite_2024-09_p0023#957` (ply 60) [structural distance  $d=59$ , QAP obj=69]: the master played Rxe5; under  $\mu$  our Pg5 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our bf2 as their Pf2, our bf2 as their Qf3; transferred move g5h6. Synthesis: no retrieved analog transferred to Qe2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

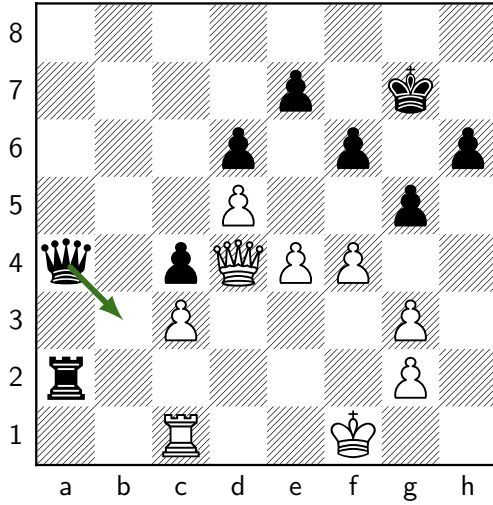




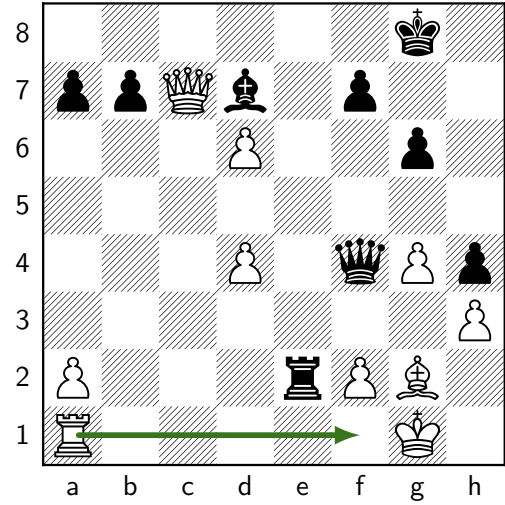
(a)



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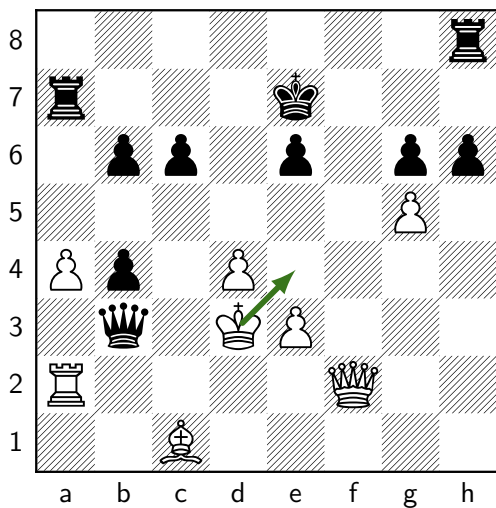


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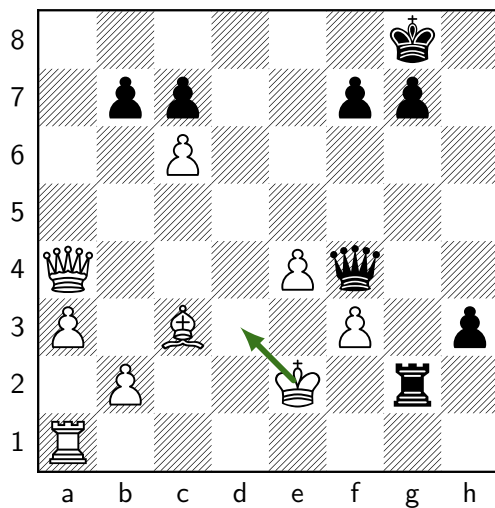


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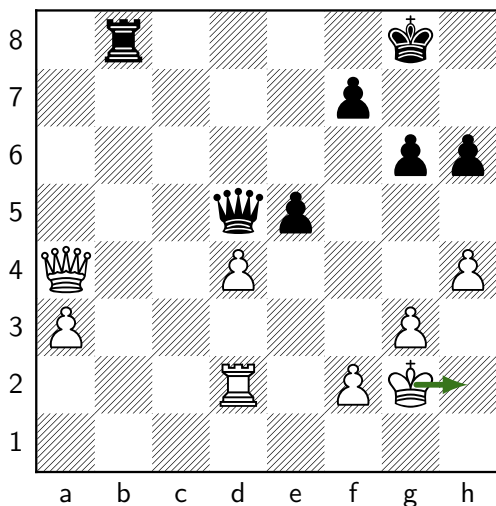
Figure 30: **Move 36, White: Qxf2**. (a) Our position; Melanie plays Qxf2 (e2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#252` (ply 61) [structural distance  $d=61$ , QAP obj=79]: the master played **bxc6**; under  $\mu$  our *Qe2* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *Pe3* as their *Re8*, our *Ra2* as their *Qg7*; transferred move **e2f2** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#1306` (ply 75) [structural distance  $d=81$ , QAP obj=76]: the master played **Qb3**; under  $\mu$  our *Ra2* plays the role of that game’s *Qa5* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Ph6*, our *Qe2* as their *Ra2*; transferred move **a2a3**. (d) Retrieved from `twic1500#7135` (ply 58) [structural distance  $d=74$ , QAP obj=75]: the master played **Rf1**; under  $\mu$  our *Bc1* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Ra2* as their *Pa2*, our *qd5* as their *re2*; transferred move **c1d2**. Synthesis: 1 retrieved analog transferred to **Qxf2** (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



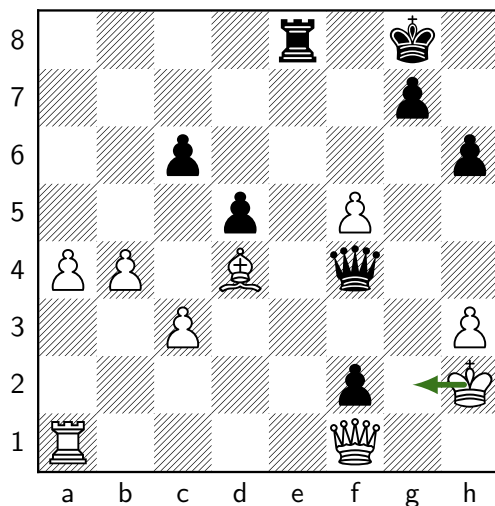
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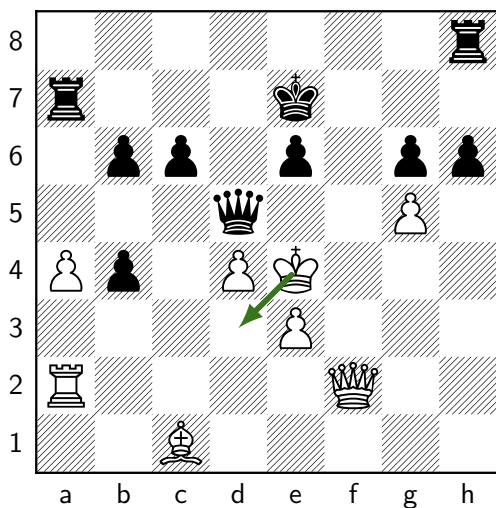


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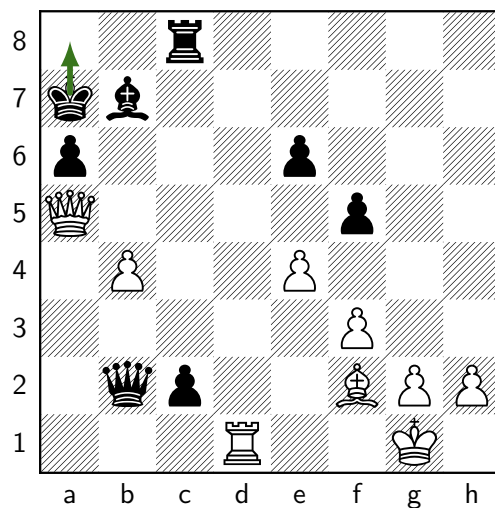


(d)

Figure 31: **Move 37, White: Ke4**. (a) Our position; Melanie plays Ke4 (d3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#1832` (ply 78) [structural distance  $d=57$ , QAP obj=75]: the master played Kd3; under  $\mu$  our Kd3 plays the role of that game’s Ke2; cross-type roles: our Bc1 as their Ra1, our qb3 as their rg2; transferred move d3e4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#2721` (ply 80) [structural distance  $d=55$ , QAP obj=49]: the master played Kh2; under  $\mu$  our Kd3 plays the role of that game’s Kg2; cross-type roles: our pb4 as their Rd2, our Ra2 as their Qa4; transferred move d3e4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#1586` (ply 66) [structural distance  $d=54$ , QAP obj=75]: the master played Kg2; under  $\mu$  our Kd3 plays the role of that game’s Kh2; cross-type roles: our pb6 as their Ra1, our ra7 as their Qf1; transferred move d3e2. Synthesis: 2 retrieved analogs transferred to Ke4 (summed QAP objective 123, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



(a)



(b)

Figure 32: **Move 38, White: Kd3.** (a) Our position; Melanie plays Kd3 (e4d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#563` (ply 71) [structural distance  $d=72$ , QAP obj=23]: the master played Ka8; under  $\mu$  our *Ke4* plays the role of that game’s *Ka2*; cross-type roles: our *Pg5* as their *Bb7*, our *ra7* as their *qa5*; transferred move e4f4. Synthesis: no retrieved analog transferred to Kd3 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

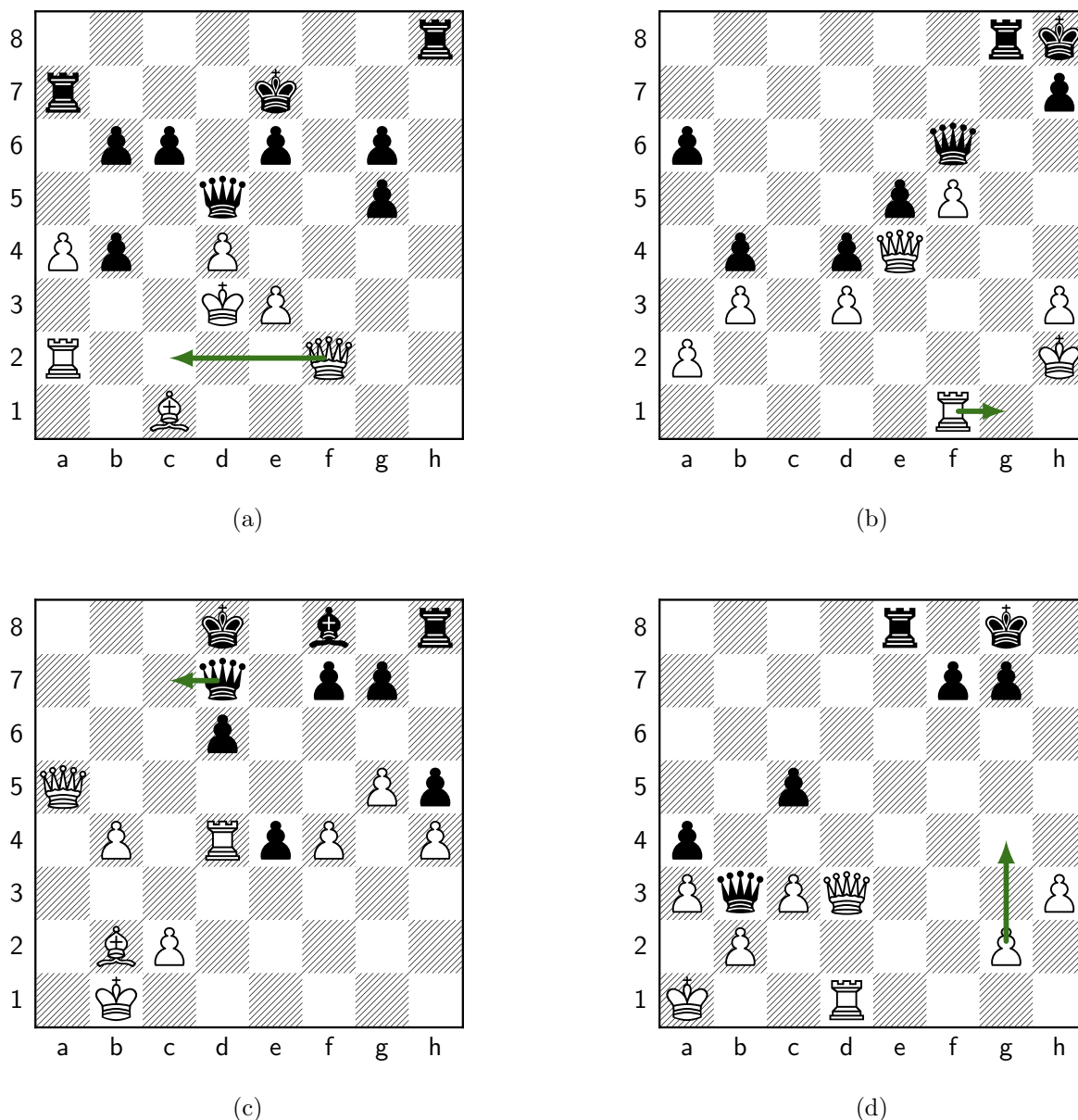
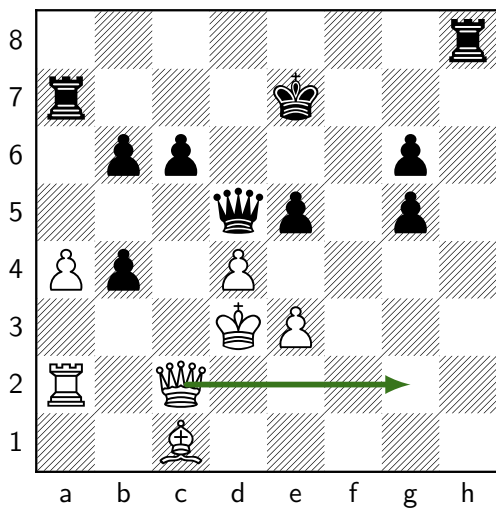
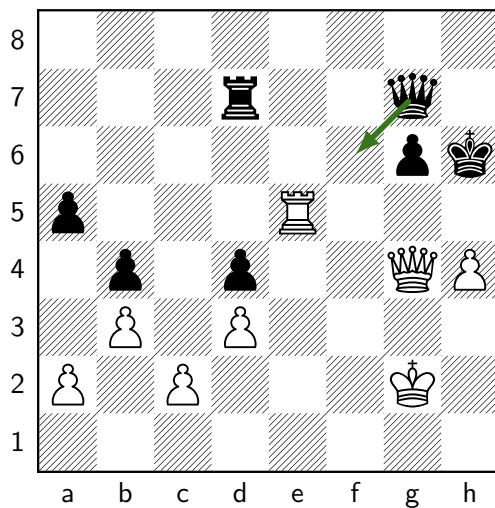


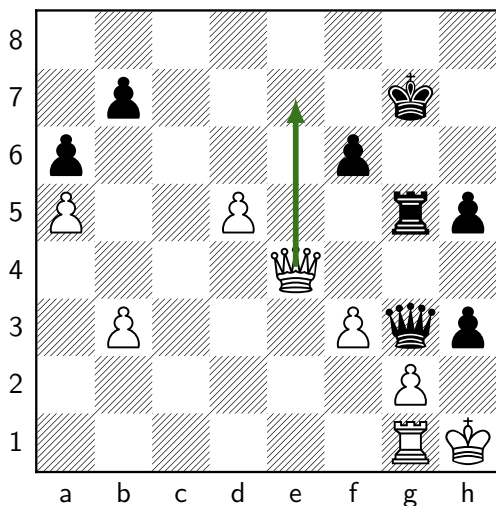
Figure 33: **Move 39, White: Qc2.** (a) Our position; Melanie plays Qc2 (f2c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#2665` (ply 78) [structural distance  $d=58$ , QAP obj=49]: the master played Rg1; under  $\mu$  our Qf2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Ra2 as their pd4, our Bc1 as their pa6; transferred move f2c2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#240` (ply 67) [structural distance  $d=57$ , QAP obj=20]: the master played Qc7; under  $\mu$  our Qf2 plays the role of that game’s Qd2; cross-type roles: our Pa4 as their Bf8, our pb6 as their qa5; transferred move f2c2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#807` (ply 68) [structural distance  $d=54$ , QAP obj=77]: the master played g4; under  $\mu$  our Pa4 plays the role of that game’s Pg2; cross-type roles: our Ra2 as their Pb2; transferred move a4a5. Synthesis: 2 retrieved analogs transferred to Qc2 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



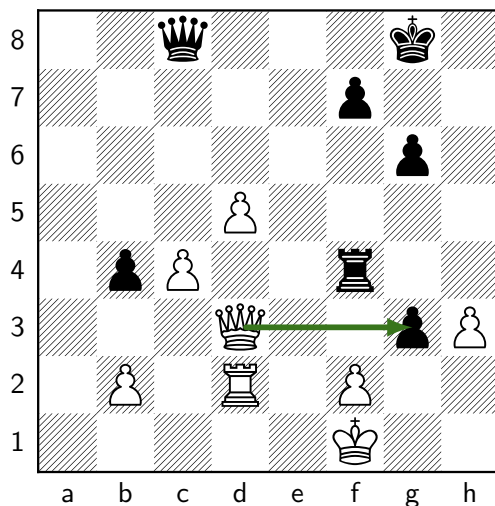
(a)



(b)

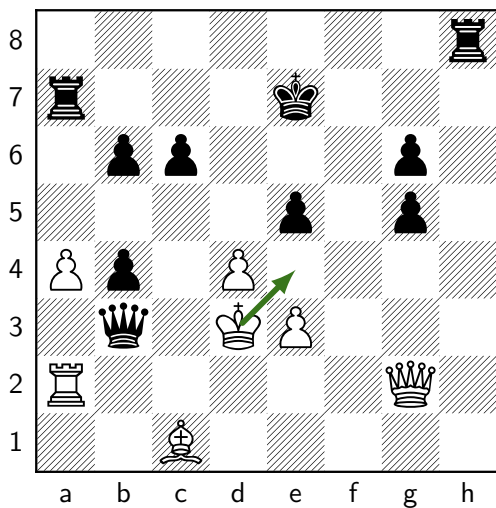


(c)

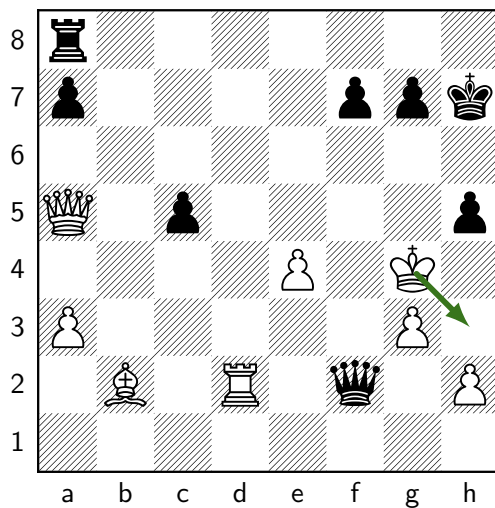


(d)

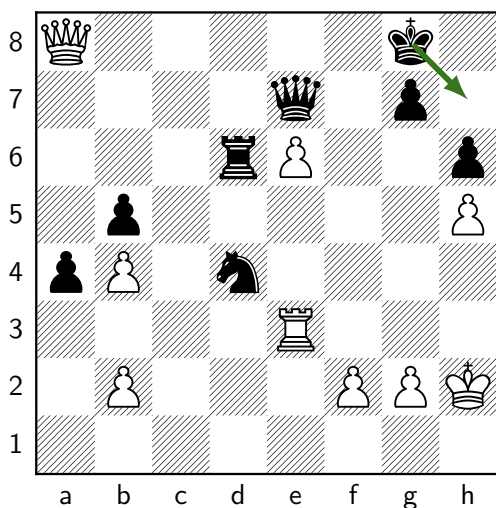
Figure 34: **Move 40, White: Qg2.** (a) Our position; Melanie plays Qg2 (c2g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#1391` (ply 83) [structural distance  $d=62$ , QAP obj=49]: the master played Qf6; under  $\mu$  our Qc2 plays the role of that game’s Qg2; cross-type roles: our pc6 as their re5, our ra7 as their pd3; transferred move c2g2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#203` (ply 84) [structural distance  $d=66$ , QAP obj=137]: the master played Qe7+; under  $\mu$  our Pe3 plays the role of that game’s Qe4 (a **cross-type** correspondence); cross-type roles: our Qc2 as their Rg1, our Ra2 as their Pf3; transferred move e3e4. (d) Retrieved from `lichess_elite_2024-09_p0005#1108` (ply 84) [structural distance  $d=71$ , QAP obj=93]: the master played Qxg3; under  $\mu$  our Qc2 plays the role of that game’s Qd3; cross-type roles: our Bc1 as their Pb2, our Pe3 as their Rd2; transferred move c2f2. Synthesis: 1 retrieved analog transferred to Qg2 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



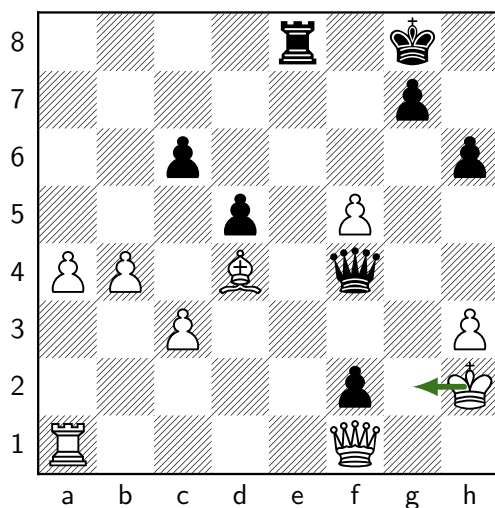
(a)



(b)

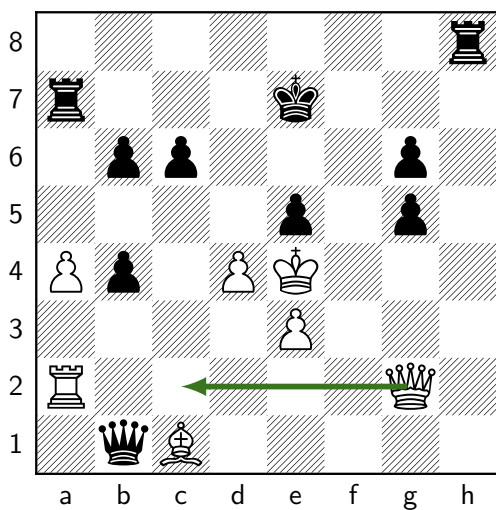


(c)

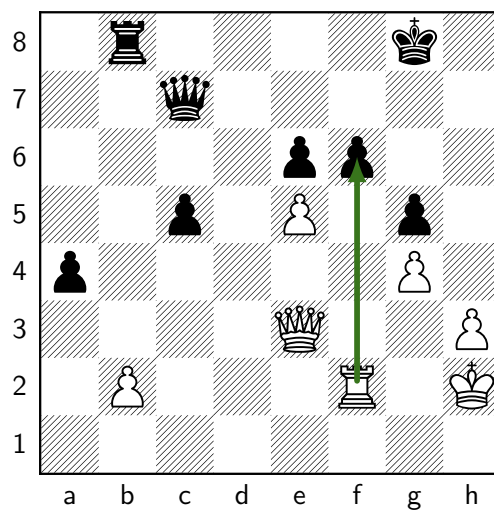


(d)

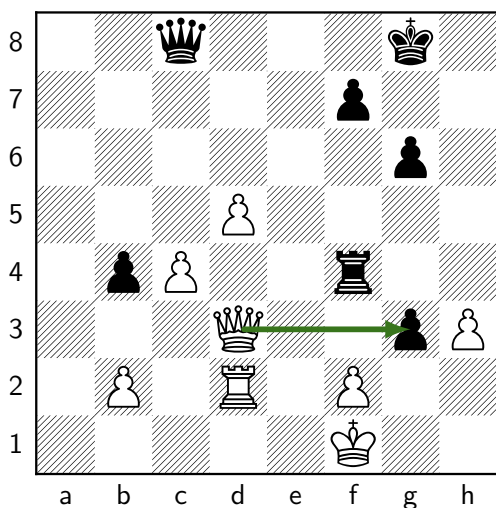
Figure 35: **Move 41, White: Ke4.** (a) Our position; Melanie plays Ke4 (d3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#2877` (ply 66) [structural distance  $d=69$ , QAP obj=50]: the master played Kh3; under  $\mu$  our Kd3 plays the role of that game’s Kg4; cross-type roles: our Qg2 as their Bb2, our pc6 as their Rd2; transferred move d3e4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#3055` (ply 95) [structural distance  $d=72$ , QAP obj=33]: the master played Kh7; under  $\mu$  our Kd3 plays the role of that game’s Kg1; cross-type roles: our ra7 as their Qe7, our pb6 as their Rd6; transferred move d3e4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#1586` (ply 66) [structural distance  $d=69$ , QAP obj=74]: the master played Kg2; under  $\mu$  our Kd3 plays the role of that game’s Kh2; cross-type roles: our pb4 as their Ra1, our ra7 as their Qf1; transferred move d3e2. Synthesis: 2 retrieved analogs transferred to Ke4 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



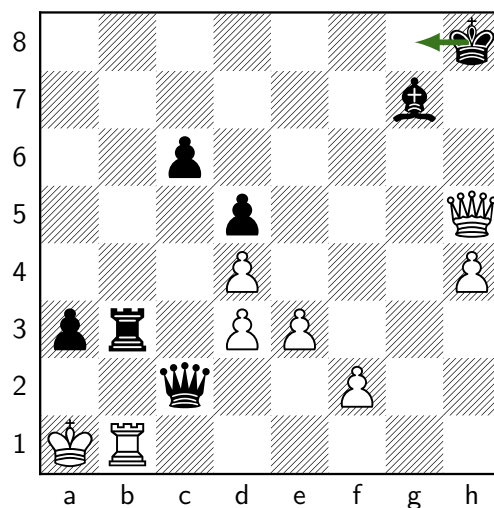
(a)



(b)



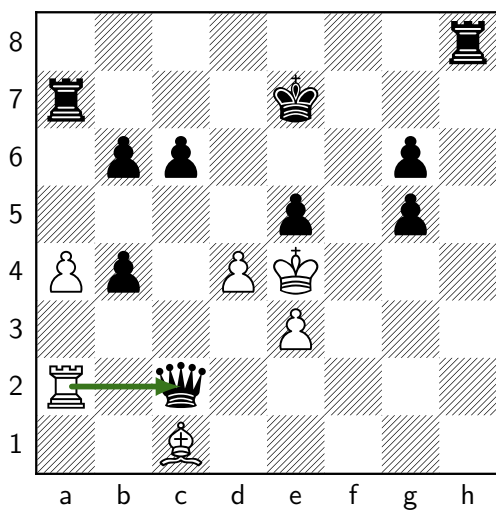
(c)



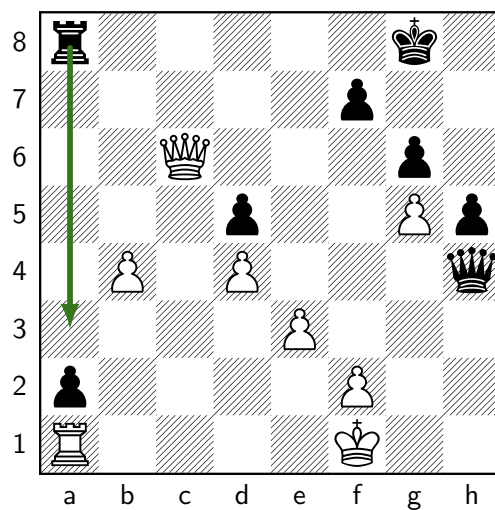
(d)

Figure 36: **Move 42, White: Qc2.** (a) Our position; Melanie plays Qc2 (g2c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#278` (ply 84) [structural distance  $d=55$ , QAP obj=19]: the master played Rxf6; under  $\mu$  our Qg2 plays the role of that game’s Rf2 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Pb2, our Ra2 as their Qe3; transferred move g2c2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#1108` (ply 84) [structural distance  $d=75$ , QAP obj=18]: the master played Qxg3; under  $\mu$  our Qg2 plays the role of that game’s Qd3; cross-type roles: our qb1 as their Ph3, our rh8 as their qc8; transferred move g2c2 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#2157` (ply 87) [structural distance  $d=79$ , QAP obj=87]: the master played Kg8; under  $\mu$  our Ke4 plays the role of that game’s Kh1; cross-type roles: our ra7 as their Bg7, our Bc1 as their ph4; transferred move e4f3. Synthesis: 2 retrieved analogs transferred to Qc2 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

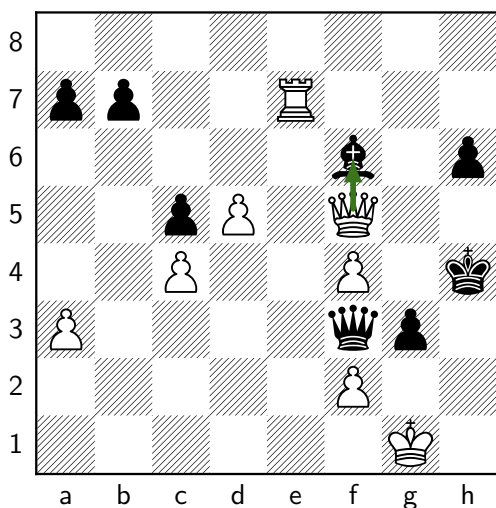




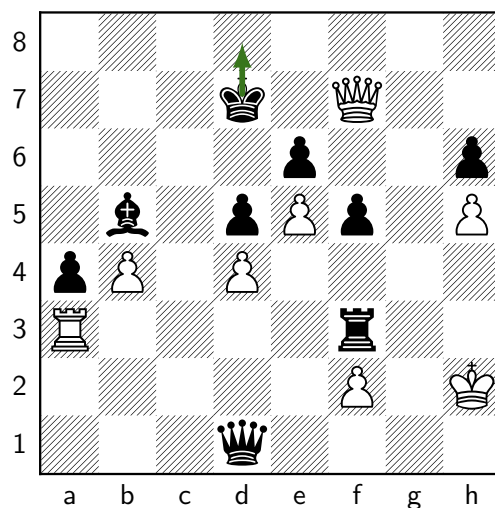
(a)



(b)

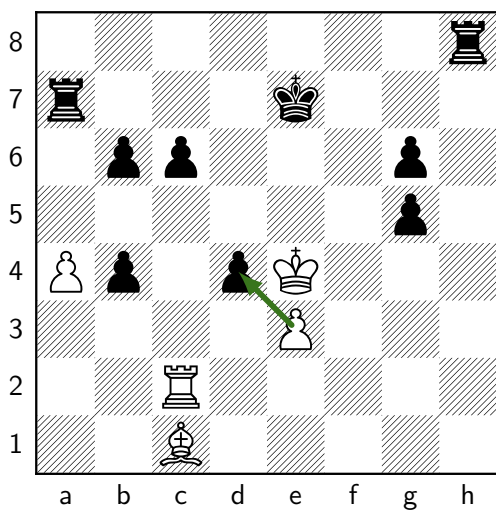


(c)

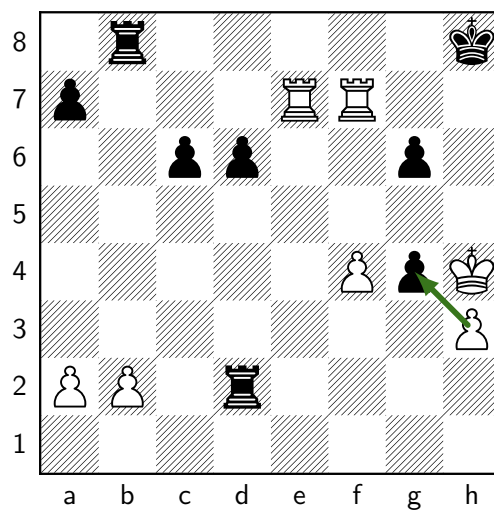


(d)

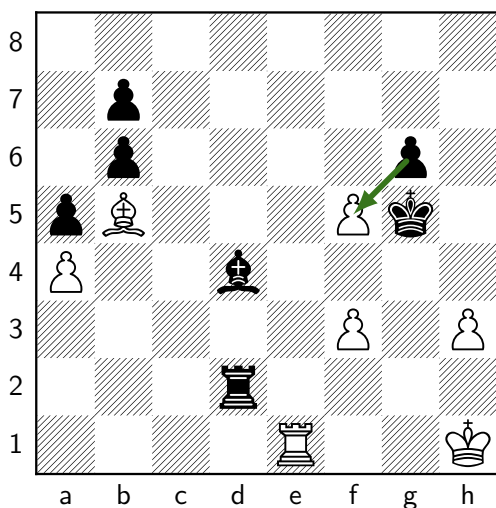
Figure 37: **Move 43, White: Rxc2.** (a) Our position; Melanie plays Rxc2 (a2c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#1182` (ply 81) [structural distance  $d=80$ , QAP obj=13]: the master played Ra3; under  $\mu$  our Ra2 plays the role of that game’s Ra1; cross-type roles: our rh8 as their Qh4, our Bc1 as their pe3; transferred move a2c2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#1793` (ply 88) [structural distance  $d=79$ , QAP obj=10]: the master played Qxf6+; under  $\mu$  our Ra2 plays the role of that game’s Qf5 (a **cross-type** correspondence); cross-type roles: our ra7 as their qf3, our rh8 as their Pc4; transferred move a2c2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0012#3292` (ply 91) [structural distance  $d=65$ , QAP obj=71]: the master played Kd8; under  $\mu$  our Ke4 plays the role of that game’s Kd2; cross-type roles: our Ra2 as their Ph6, our Pe3 as their Bb5; transferred move e4f3. Synthesis: 2 retrieved analogs transferred to Rxc2 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



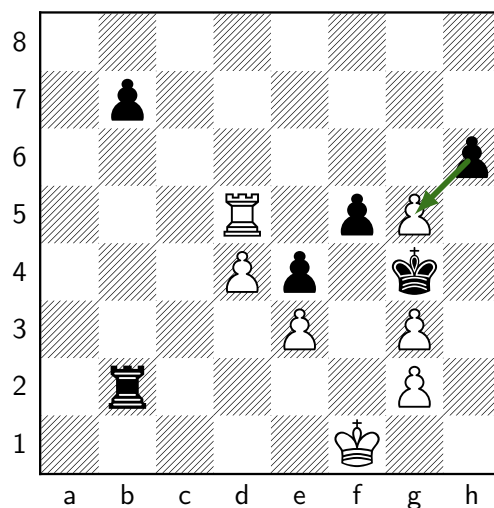
(a)



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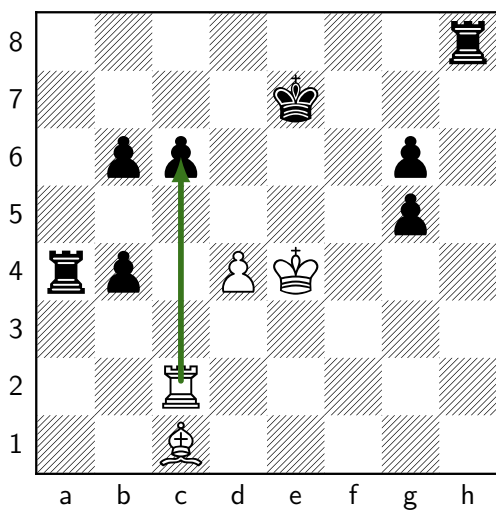


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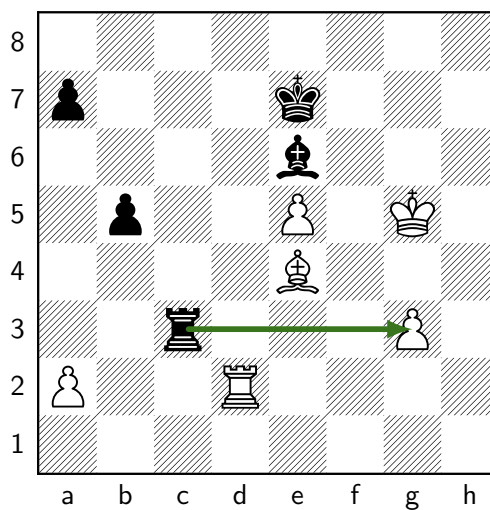


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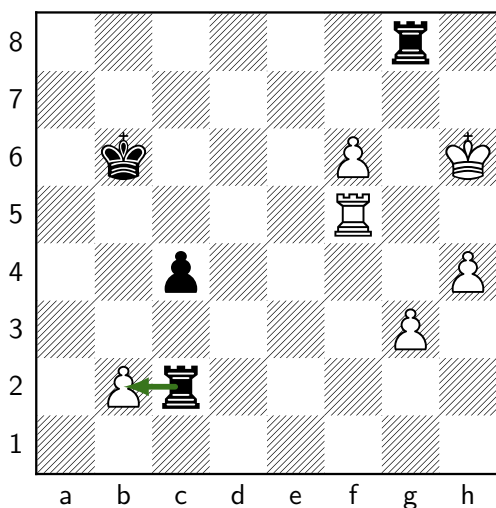
Figure 38: **Move 44, White: exd4**. (a) Our position; Melanie plays **exd4** (**e3d4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3989` (ply 70) [structural distance  $d=45$ , QAP obj=99]: the master played **hxg4**; under  $\mu$  our *Pe3* plays the role of that game’s *Ph3*; cross-type roles: our *rh8* as their *Pa2*, our *Bc1* as their *Pf4*; transferred move **e3d4** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#1540` (ply 73) [structural distance  $d=41$ , QAP obj=88]: the master played **gxf5**; under  $\mu$  our *Pe3* plays the role of that game’s *Pg3*; cross-type roles: our *rh8* as their *Pb7*, our *Bc1* as their *Pb6*; transferred move **e3d4** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#170` (ply 83) [structural distance  $d=44$ , QAP obj=87]: the master played **hxg5**; under  $\mu$  our *Pe3* plays the role of that game’s *Ph3*; cross-type roles: our *Rc2* as their *pd4*, our *Bc1* as their *Rb2*; transferred move **e3d4** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to **exd4** (summed QAP objective 422, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



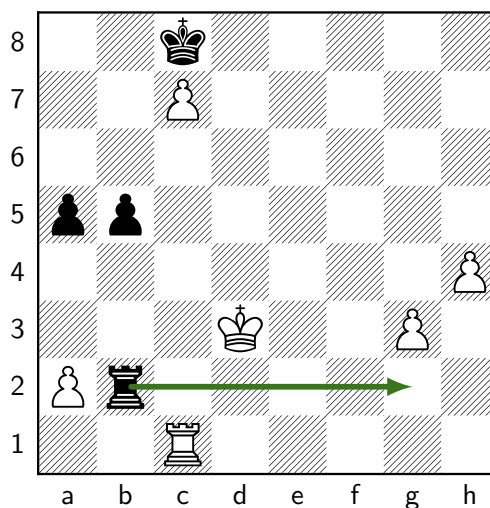
(a)



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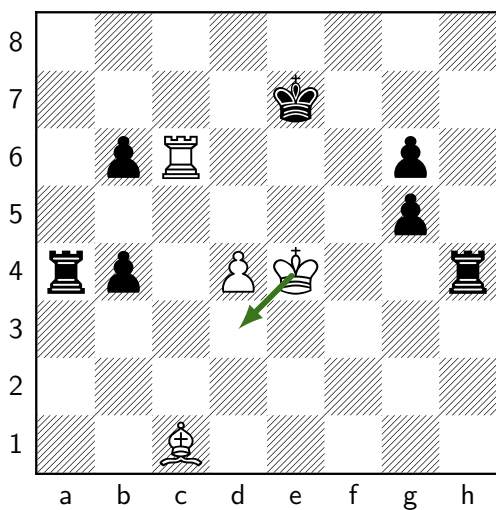


(c)

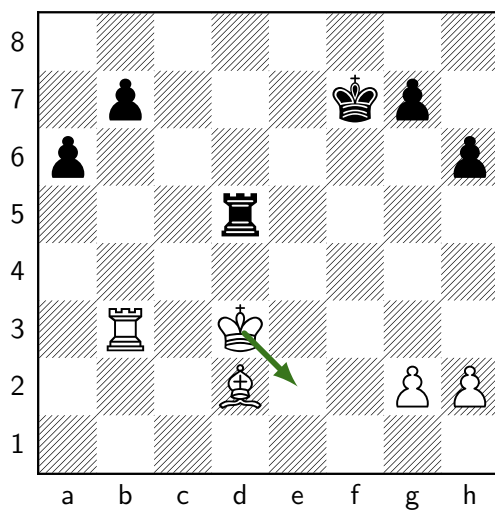


(d)

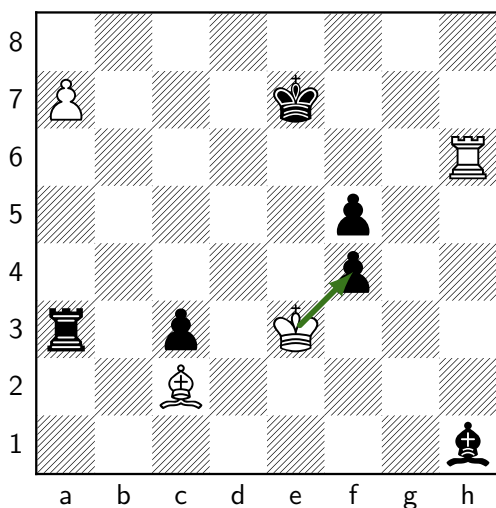
Figure 39: **Move 45, White: Rxc6**. (a) Our position; Melanie plays Rxc6 (c2c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#2291` (ply 91) [structural distance  $d=19$ , QAP obj=37]: the master played Rxc3+; under  $\mu$  our Rc2 plays the role of that game’s Rc6; cross-type roles: our ra4 as their Pb5, our pb6 as their be4; transferred move c2c6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#1561` (ply 91) [structural distance  $d=27$ , QAP obj=7]: the master played Rxb2; under  $\mu$  our Rc2 plays the role of that game’s Rc7; cross-type roles: our Bc1 as their Rg8; transferred move c2c6 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#1977` (ply 81) [structural distance  $d=27$ , QAP obj=6]: the master played Rg2; under  $\mu$  our Rc2 plays the role of that game’s Rb7; cross-type roles: our ra4 as their Pa5, our Bc1 as their Pb5; transferred move c2c6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rxc6 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



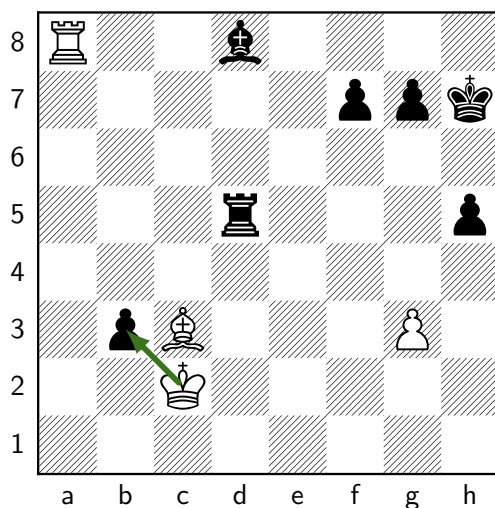
(a)



(b)

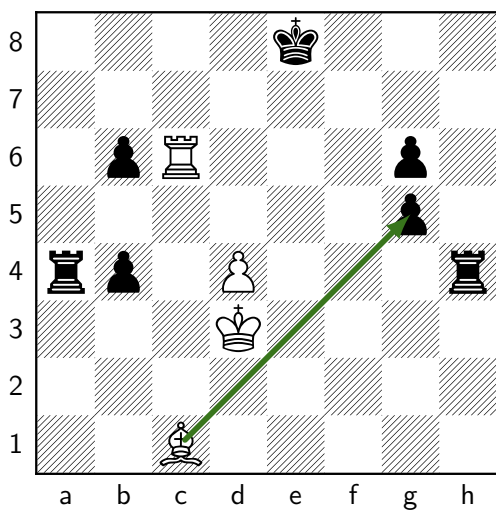


(c)

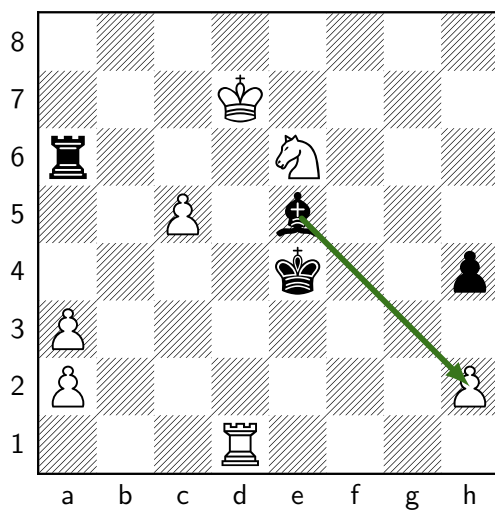


(d)

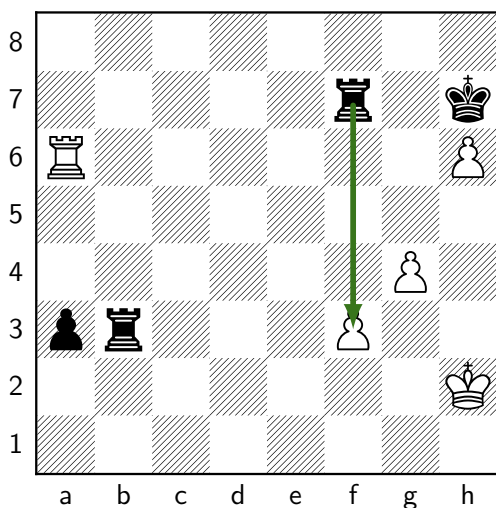
Figure 40: **Move 46, White: Kd3.** (a) Our position; Melanie plays Kd3 (e4d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#3641` (ply 98) [structural distance  $d=24$ , QAP obj=47]: the master played Ke2; under  $\mu$  our Ke4 plays the role of that game’s Kd3; cross-type roles: our pg5 as their Bd2, our ra4 as their Ph2; transferred move e4d3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#1245` (ply 90) [structural distance  $d=26$ , QAP obj=27]: the master played Kxf4; under  $\mu$  our Ke4 plays the role of that game’s Ke3; cross-type roles: our pg6 as their bh1, our rh4 as their pf4; transferred move e4d3 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#2000` (ply 80) [structural distance  $d=25$ , QAP obj=26]: the master played Kxb3; under  $\mu$  our Ke4 plays the role of that game’s Kc2; cross-type roles: our rh4 as their pb3, our Pd4 as their Bc3; transferred move e4d3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kd3 (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



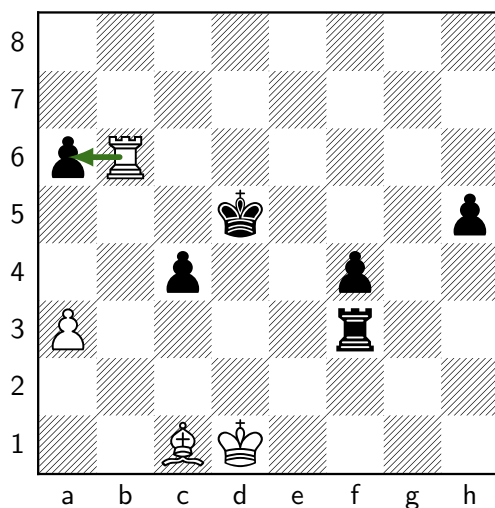
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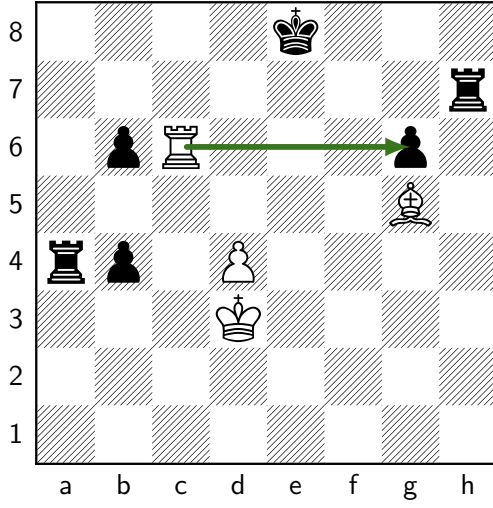


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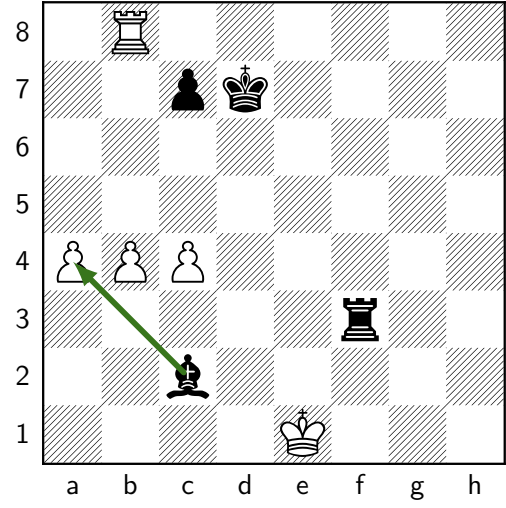


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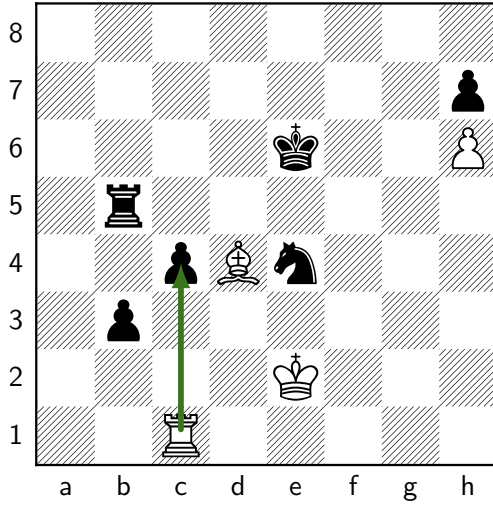
Figure 41: **Move 47, White: Bxg5.** (a) Our position; Melanie plays Bxg5 (c1g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#1844` (ply 103) [structural distance  $d=26$ , QAP obj=23]: the master played Bxh2; under  $\mu$  our Bc1 plays the role of that game’s Be4; cross-type roles: our pb6 as their ne6, our rh4 as their Ph4; transferred move c1g5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#293` (ply 101) [structural distance  $d=28$ , QAP obj=11]: the master played Rxf3; under  $\mu$  our Bc1 plays the role of that game’s Rf2 (a **cross-type** correspondence); cross-type roles: our rh4 as their pg4; transferred move c1g5 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#3976` (ply 106) [structural distance  $d=23$ , QAP obj=25]: the master played Rxa6; under  $\mu$  our Rc6 plays the role of that game’s Rb6; cross-type roles: our rh4 as their Pa3, our pg6 as their rf3; transferred move c6b6. Synthesis: 2 retrieved analogs transferred to Bxg5 (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



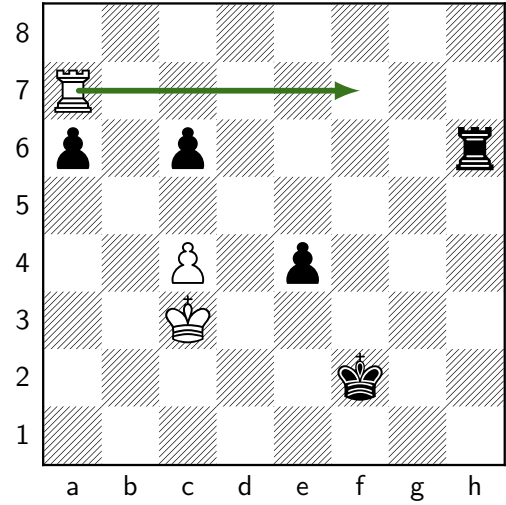
(a)



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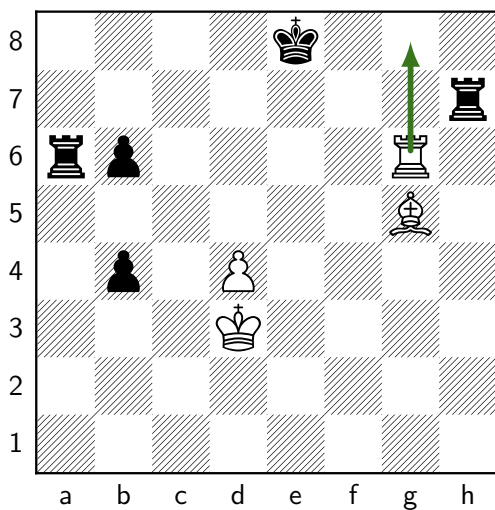


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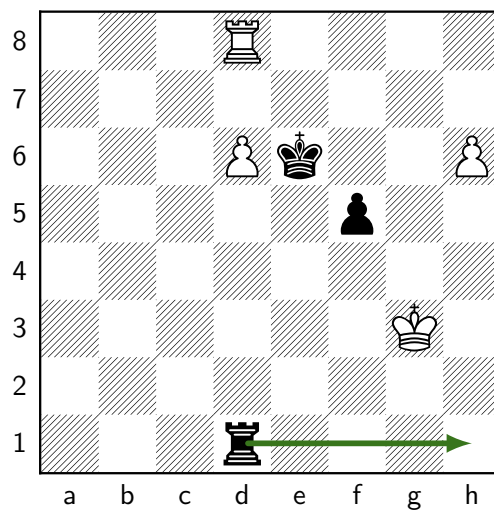


(d)

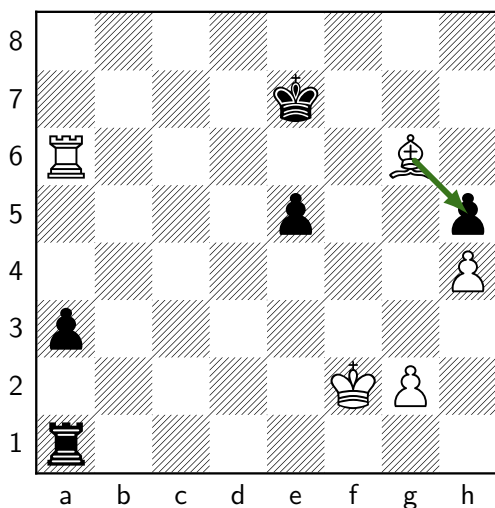
Figure 42: **Move 48, White: Rxc6**. (a) Our position; Melanie plays Rxc6 (c6g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#1915` (ply 111) [structural distance  $d=13$ , QAP obj=12]: the master played Bxa4; under  $\mu$  our Rc6 plays the role of that game’s Bc7 (a **cross-type** correspondence); cross-type roles: our pb4 as their rb8, our ra4 as their pb4; transferred move c6g6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#3368` (ply 96) [structural distance  $d=14$ , QAP obj=9]: the master played Rxc4; under  $\mu$  our Rc6 plays the role of that game’s Rc1; cross-type roles: our rh7 as their ne4; transferred move c6g6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0005#2777` (ply 96) [structural distance  $d=13$ , QAP obj=21]: the master played Rf7+; under  $\mu$  our Rc6 plays the role of that game’s Ra7; cross-type roles: our ra4 as their pc6, our Bg5 as their rh6; transferred move c6c8. Synthesis: 2 retrieved analogs transferred to Rxc6 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



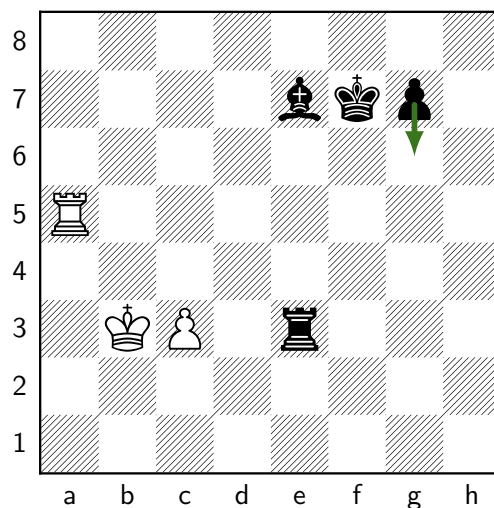
(a)



(b)



(c)



(d)

Figure 43: **Move 49, White: Rg8+.** (a) Our position; Melanie plays Rg8+ (g6g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#933` (ply 109) [structural distance  $d=15$ , QAP obj=24]: the master played Rh1; under  $\mu$  our Rg6 plays the role of that game’s Rd8; cross-type roles: our Bg5 as their Pf5; transferred move g6g8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#1616` (ply 92) [structural distance  $d=7$ , QAP obj=20]: the master played Bxh5; under  $\mu$  our Bg5 plays the role of that game’s Bg6; cross-type roles: our rh7 as their pe5; transferred move g5h4. (d) Retrieved from `lichess_elite_2024-09_p0026#3522` (ply 93) [structural distance  $d=15$ , QAP obj=20]: the master played g6; under  $\mu$  our Pd4 plays the role of that game’s Pg2; transferred move d4d5. Synthesis: 3 retrieved analogs transferred to Rg8+ (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

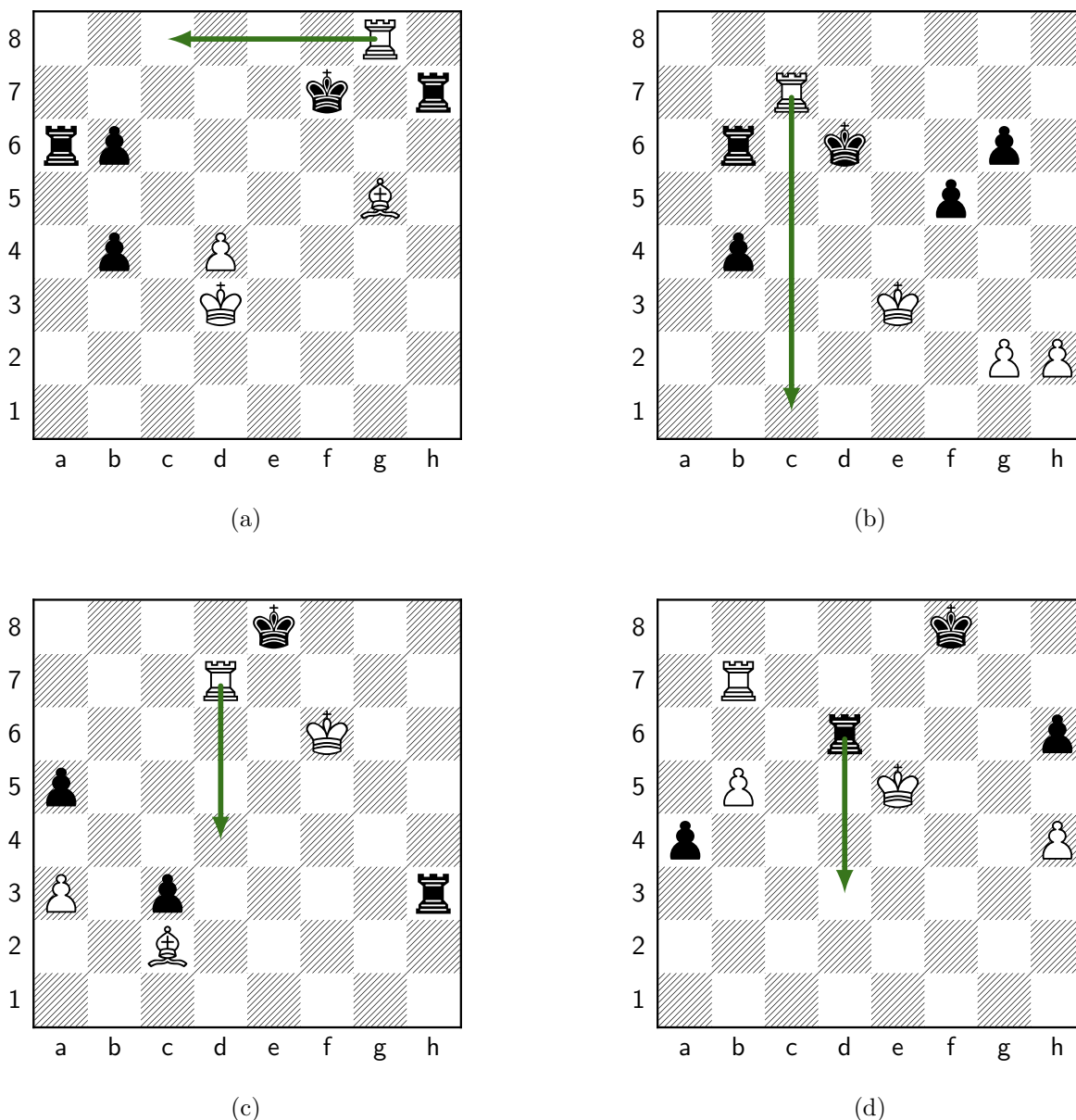
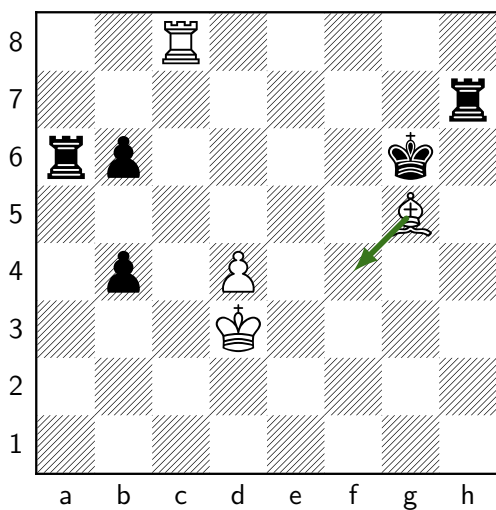
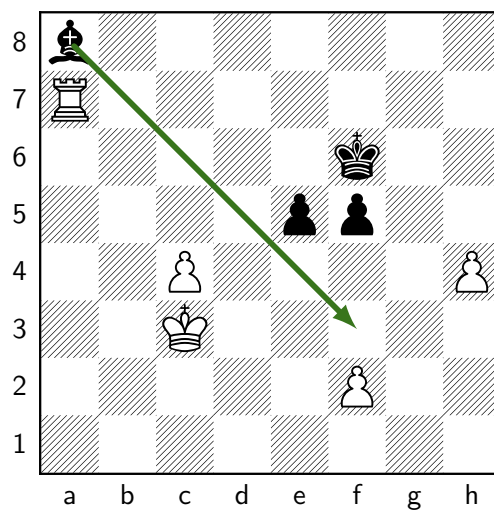


Figure 44: **Move 50, White: Rc8.** (a) Our position; Melanie plays Rc8 (g8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#2378` (ply 100) [structural distance  $d=16$ , QAP obj=23]: the master played Rc1; under  $\mu$  our Rg8 plays the role of that game’s Rc7; cross-type roles: our Bg5 as their Ph2, our ra6 as their pb4; transferred move g8c8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#3748` (ply 92) [structural distance  $d=16$ , QAP obj=19]: the master played Rd4; under  $\mu$  our Rg8 plays the role of that game’s Rd7; transferred move g8c8 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#1237` (ply 105) [structural distance  $d=15$ , QAP obj=19]: the master played Rd3; under  $\mu$  our Rg8 plays the role of that game’s Rd3; cross-type roles: our pb4 as their rb7, our Bg5 as their Ph6; transferred move g8c8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rc8 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

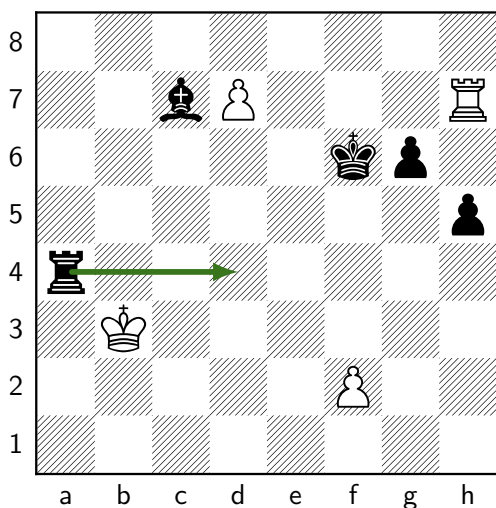




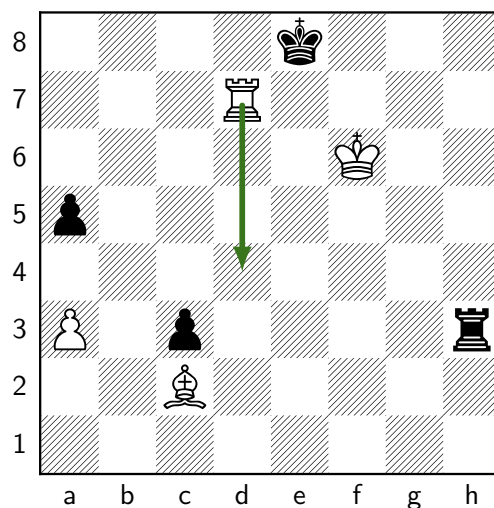
(a)



(b)

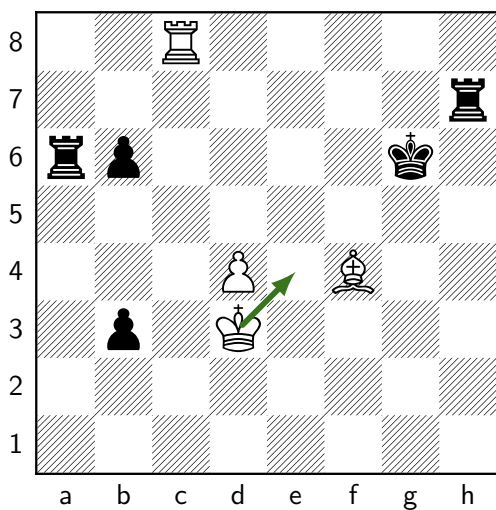


(c)

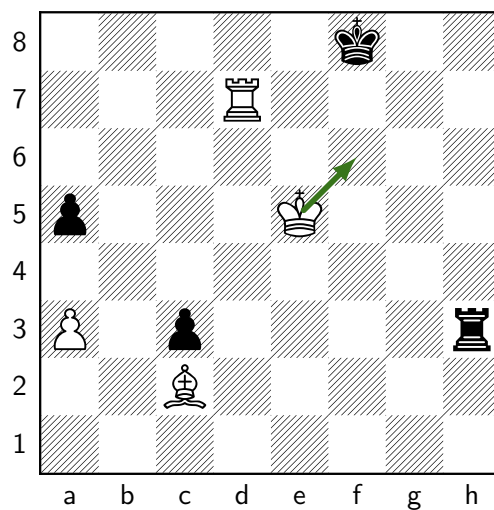


(d)

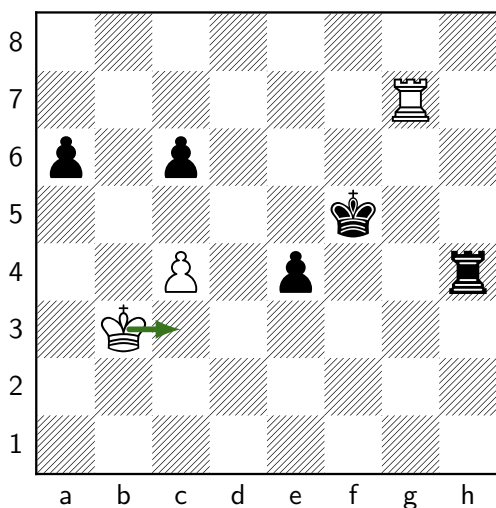
Figure 45: **Move 51, White: Bf4**. (a) Our position; Melanie plays Bf4 (g5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#2079` (ply 117) [structural distance  $d=16$ , QAP obj=15]: the master played Bf3; under  $\mu$  our Bg5 plays the role of that game’s Ba1; cross-type roles: our kg6 as their ra7, our rh7 as their pc4; transferred move g5f4 (**endorses this move**). (c) Retrieved from `twic1500#7756` (ply 91) [structural distance  $d=14$ , QAP obj=12]: the master played Rd4; under  $\mu$  our Bg5 plays the role of that game’s Ra5 (a **cross-type** correspondence); cross-type roles: our Rc8 as their Bc7, our rh7 as their pd7; transferred move g5f4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#3748` (ply 92) [structural distance  $d=13$ , QAP obj=12]: the master played Rd4; under  $\mu$  our Bg5 plays the role of that game’s Rd7 (a **cross-type** correspondence); cross-type roles: our Rc8 as their Bc2, our rh7 as their pc3; transferred move g5f4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bf4 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



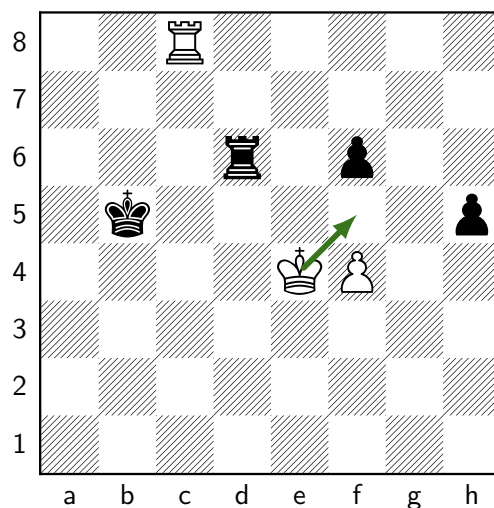
(a)



(b)

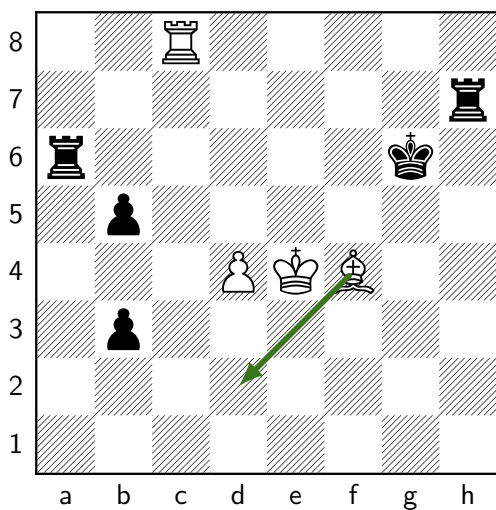


(c)

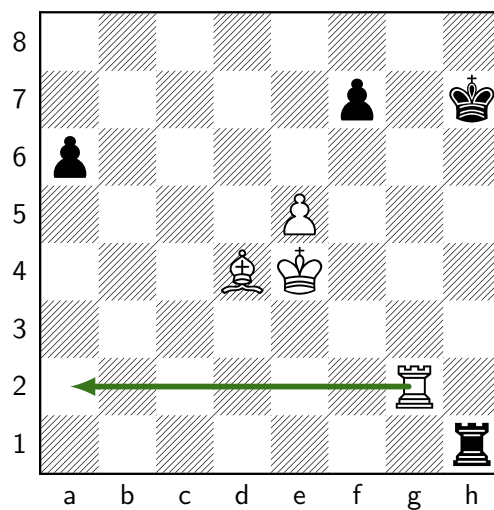


(d)

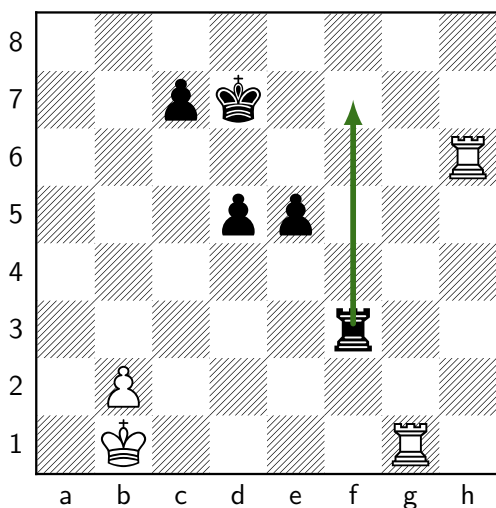
Figure 46: **Move 52, White: Ke4**. (a) Our position; Melanie plays Ke4 (d3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3748` (ply 90) [structural distance  $d=11$ , QAP obj=2]: the master played Kf6; under  $\mu$  our Kd3 plays the role of that game’s Ke5; cross-type roles: our Rc8 as their Bc2, our Bf4 as their Rd7; transferred move d3e4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#2777` (ply 88) [structural distance  $d=11$ , QAP obj=1]: the master played Kc3; under  $\mu$  our Kd3 plays the role of that game’s Kb3; cross-type roles: our Rc8 as their pc6, our Bf4 as their Rg7; transferred move d3e4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#2498` (ply 112) [structural distance  $d=12$ , QAP obj=1]: the master played Kf5; under  $\mu$  our Kd3 plays the role of that game’s Ke4; cross-type roles: our Bf4 as their Rc8; transferred move d3e4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ke4 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



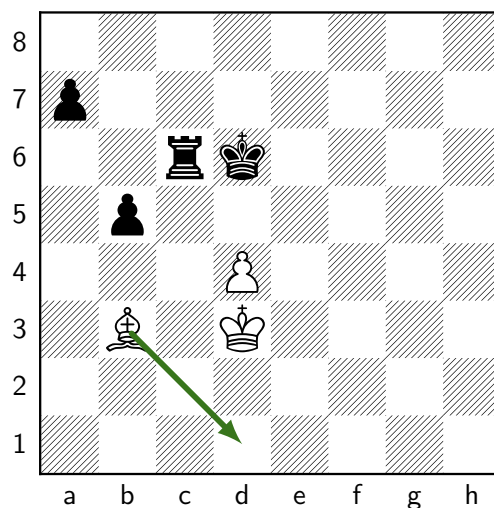
(a)



(b)

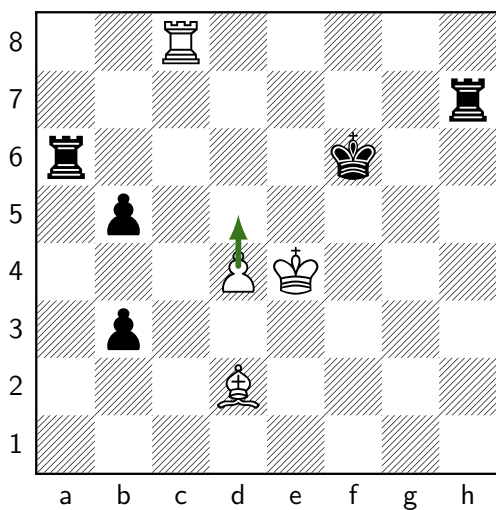


(c)

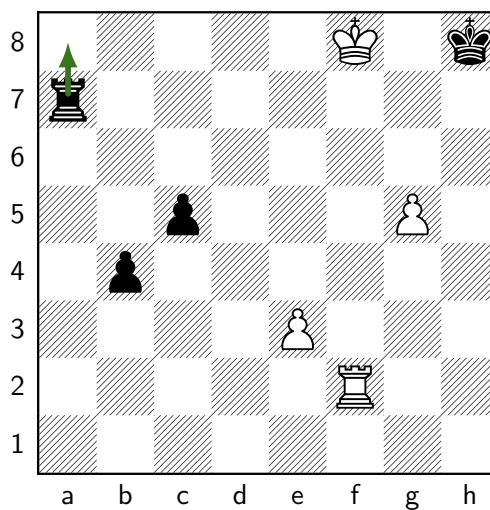


(d)

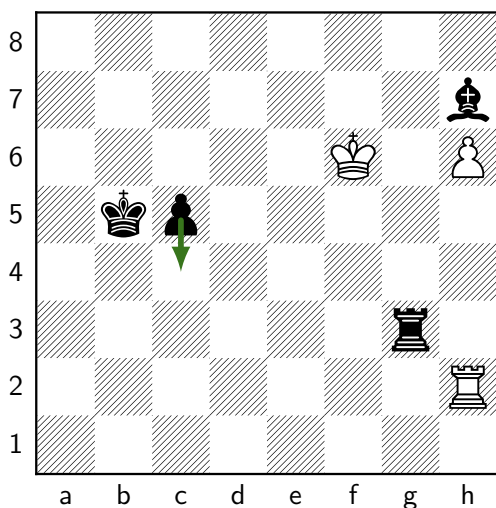
Figure 47: **Move 53, White: Bd2.** (a) Our position; Melanie plays Bd2 (f4d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2317` (ply 106) [structural distance  $d=11$ , QAP obj=3]: the master played Ra2; under  $\mu$  our Bf4 plays the role of that game’s Rg2 (a **cross-type** correspondence); cross-type roles: our Rc8 as their Bd4, our rh7 as their pf7; transferred move f4d2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#3894` (ply 93) [structural distance  $d=17$ , QAP obj=2]: the master played Rf7; under  $\mu$  our Bf4 plays the role of that game’s Rf6 (a **cross-type** correspondence); cross-type roles: our rh7 as their pb2; transferred move f4d2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0026#2093` (ply 108) [structural distance  $d=14$ , QAP obj=1]: the master played Bd1; under  $\mu$  our Bf4 plays the role of that game’s Bb3; cross-type roles: our rh7 as their pb5; transferred move f4d2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bd2 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



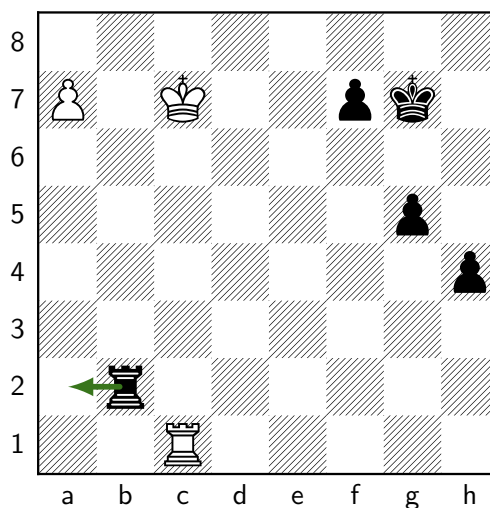
(a)



(b)

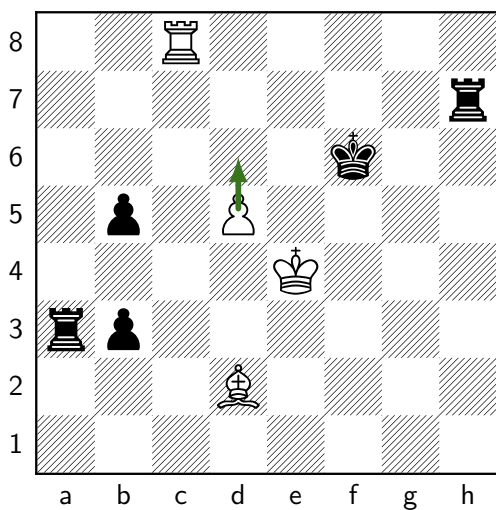


(c)

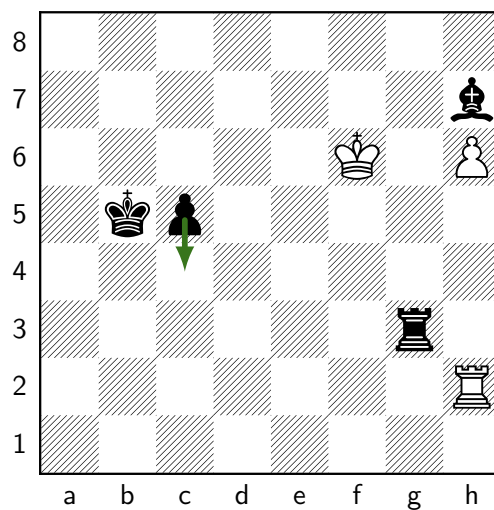


(d)

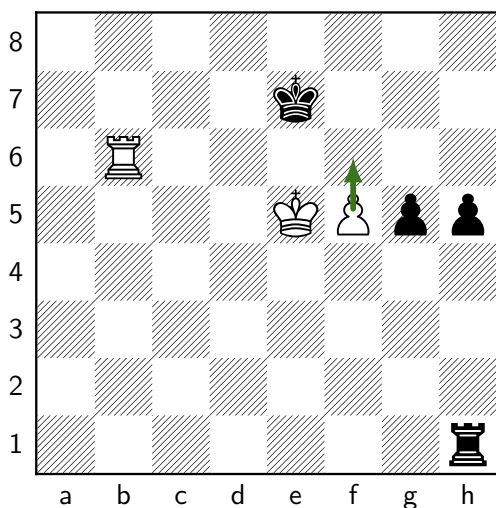
Figure 48: **Move 54, White: d5.** (a) Our position; Melanie plays d5 (d4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#1210` (ply 115) [structural distance  $d=10$ , QAP obj=3]: the master played  $Ra8+$ ; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Ra2$  (a **cross-type** correspondence); cross-type roles: our  $Rc8$  as their  $Pb4$ , our  $Bd2$  as their  $pe3$ ; transferred move d4d5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#249` (ply 95) [structural distance  $d=11$ , QAP obj=1]: the master played c4; under  $\mu$  our  $Pd4$  plays the role of that game’s  $Pc4$ ; transferred move d4d5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#3569` (ply 97) [structural distance  $d=12$ , QAP obj=3]: the master played  $Ra2$ ; under  $\mu$  our  $Bd2$  plays the role of that game’s  $Rb7$  (a **cross-type** correspondence); cross-type roles: our  $Rc8$  as their  $Ph4$ ; transferred move d2b4. Synthesis: 3 retrieved analogs transferred to d5 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



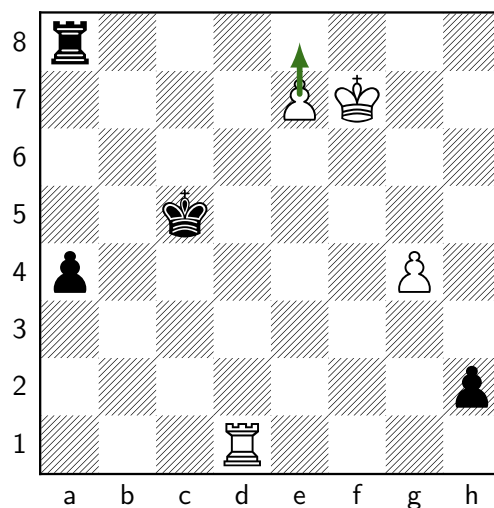
(a)



(b)

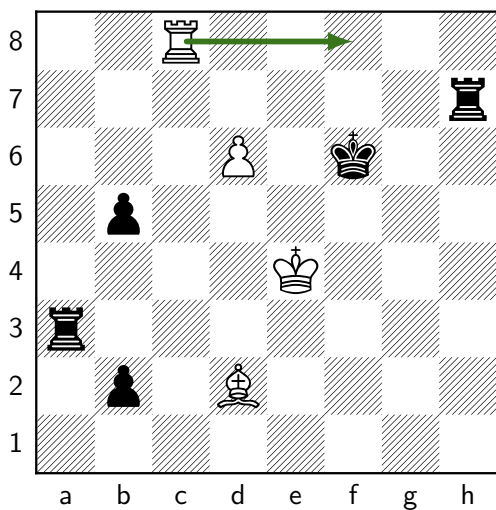


(c)

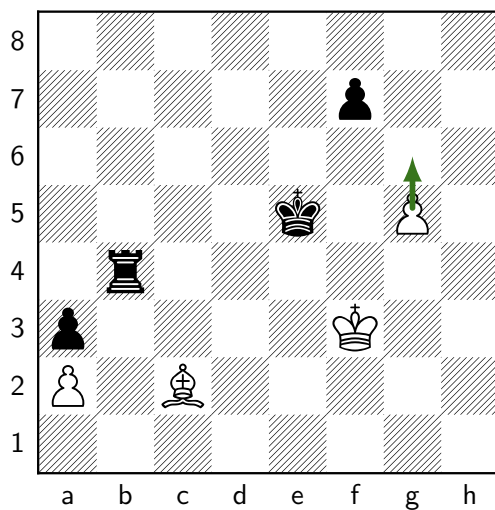


(d)

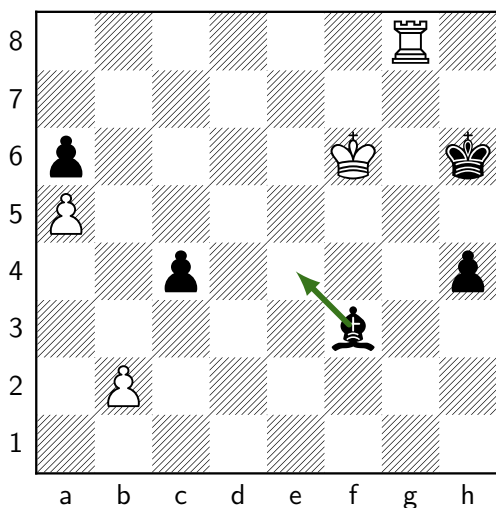
Figure 49: **Move 55, White: d6**. (a) Our position; Melanie plays d6 (d5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#249` (ply 95) [structural distance  $d=7$ , QAP obj=3]: the master played c4; under  $\mu$  our  $Pd5$  plays the role of that game’s  $Pc4$ ; transferred move d5d6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#1455` (ply 118) [structural distance  $d=9$ , QAP obj=3]: the master played f6+; under  $\mu$  our  $Pd5$  plays the role of that game’s  $Pf5$ ; cross-type roles: our  $Bd2$  as their  $pg5$ ; transferred move d5d6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#840` (ply 116) [structural distance  $d=7$ , QAP obj=3]: the master played e8=Q; under  $\mu$  our  $Pd5$  plays the role of that game’s  $Pe7$ ; cross-type roles: our  $Bd2$  as their  $Rd1$ , our  $rh7$  as their  $Pg4$ ; transferred move d5d6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to d6 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



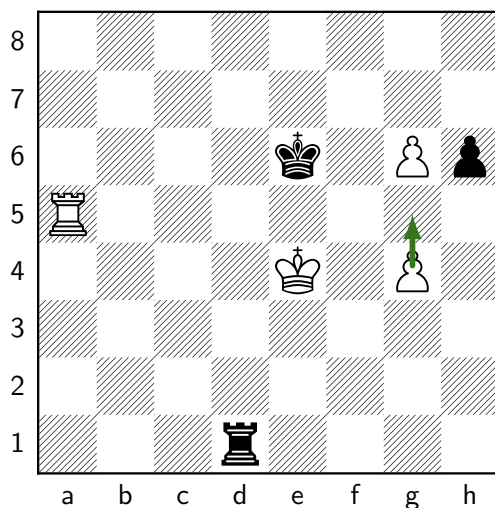
(a)



(b)



(c)



(d)

Figure 50: **Move 56, White: Rf8+.** (a) Our position; Melanie plays Rf8+ (c8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#3523` (ply 98) [structural distance  $d=7$ , QAP obj=1]: the master played g6; under  $\mu$  our Rc8 plays the role of that game’s Pg5 (a **cross-type** correspondence); cross-type roles: our rh7 as their pa3; transferred move c8c4. (c) Retrieved from `lichess_elite_2024-09_p0003#2823` (ply 99) [structural distance  $d=10$ , QAP obj=1]: the master played Be4; under  $\mu$  our Bd2 plays the role of that game’s Bf6; cross-type roles: our rh7 as their Pa6, our Rc8 as their pb2; transferred move d2a5. (d) Retrieved from `lichess_elite_2024-09_p0019#3561` (ply 100) [structural distance  $d=14$ , QAP obj=1]: the master played g5; under  $\mu$  our Bd2 plays the role of that game’s Pg4 (a **cross-type** correspondence); transferred move d2a5. Synthesis: no retrieved analog transferred to Rf8+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

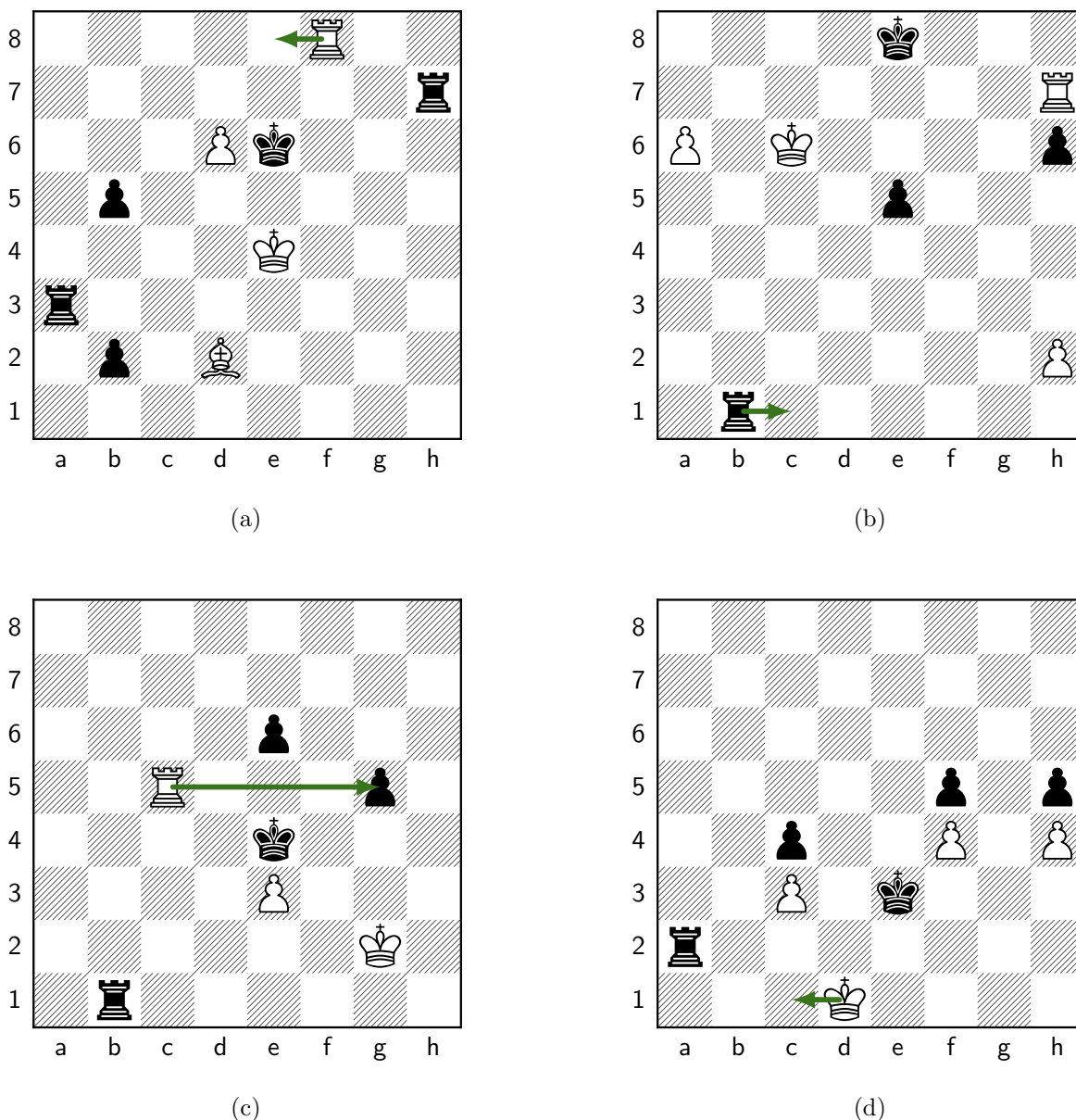
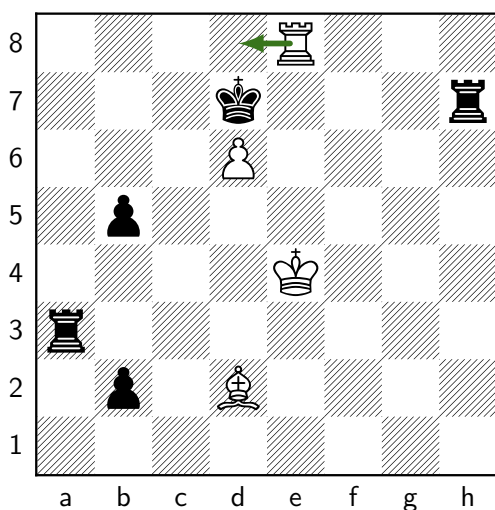
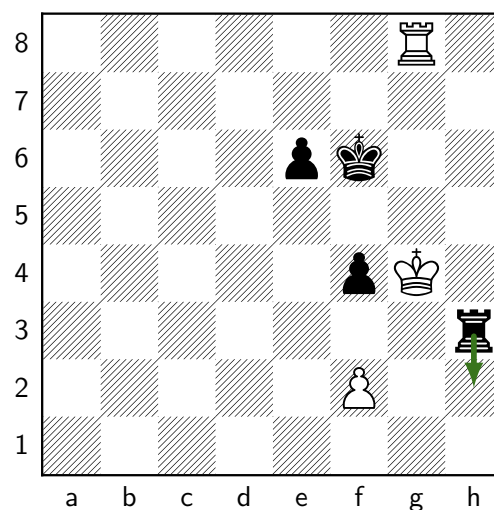


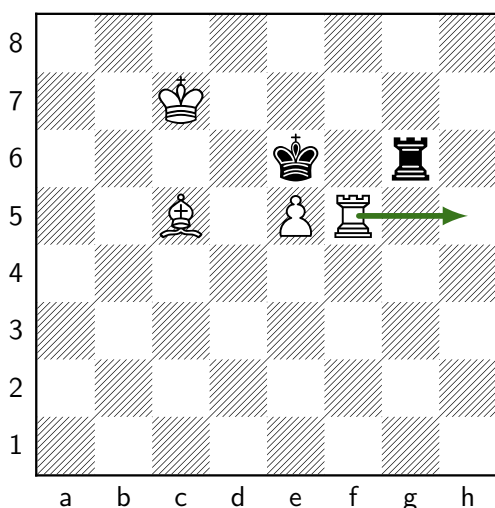
Figure 51: **Move 57, White: Re8+.** (a) Our position; Melanie plays Re8+ (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#536` (ply 105) [structural distance  $d=14$ , QAP obj=14]: the master played Rc1+; under  $\mu$  our Rf8 plays the role of that game’s Rb8; cross-type roles: our ke6 as their rh7, our Bd2 as their Pe5; transferred move f8e8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#3221` (ply 118) [structural distance  $d=15$ , QAP obj=19]: the master played Rxg5; under  $\mu$  our Rf8 plays the role of that game’s Rc5; cross-type roles: our rh7 as their pg5; transferred move f8c8. (d) Retrieved from `lichess_elite_2024-09_p0016#1314` (ply 102) [structural distance  $d=16$ , QAP obj=19]: the master played Kc1; under  $\mu$  our Ke4 plays the role of that game’s Kd1; cross-type roles: our Rf8 as their Pc3, our Bd2 as their Ph4; transferred move e4d4. Synthesis: 1 retrieved analog transferred to Re8+ (summed QAP objective 14, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



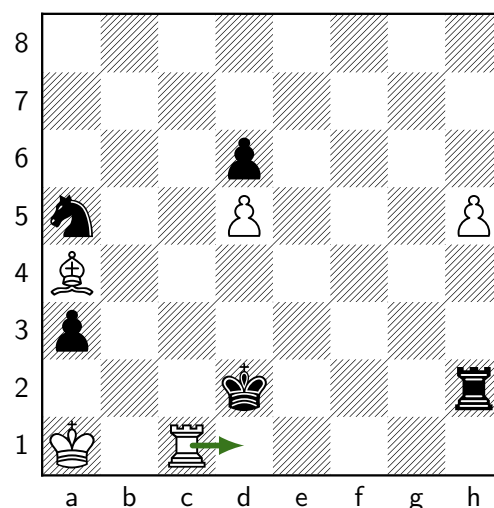
(a)



(b)



(c)



(d)

Figure 52: **Move 58, White: Rd8+.** (a) Our position; Melanie plays Rd8+ (e8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#1652` (ply 121) [structural distance  $d=22$ , QAP obj=26]: the master played Rh2; under  $\mu$  our *Re8* plays the role of that game’s *Rh6*; cross-type roles: our *Bd2* as their *Pe6*; transferred move e8b8. (c) Retrieved from `lichess_elite_2024-09_p0022#2317` (ply 130) [structural distance  $d=20$ , QAP obj=25]: the master played Rh5; under  $\mu$  our *Re8* plays the role of that game’s *Rf5*; transferred move e8g8. (d) Retrieved from `lichess_elite_2024-09_p0023#1480` (ply 118) [structural distance  $d=16$ , QAP obj=24]: the master played Rd1+; under  $\mu$  our *Re8* plays the role of that game’s *Rc1*; cross-type roles: our *ra3* as their *pa3*, our *Pd6* as their *Ba4*; transferred move e8e6. Synthesis: no retrieved analog transferred to Rd8+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



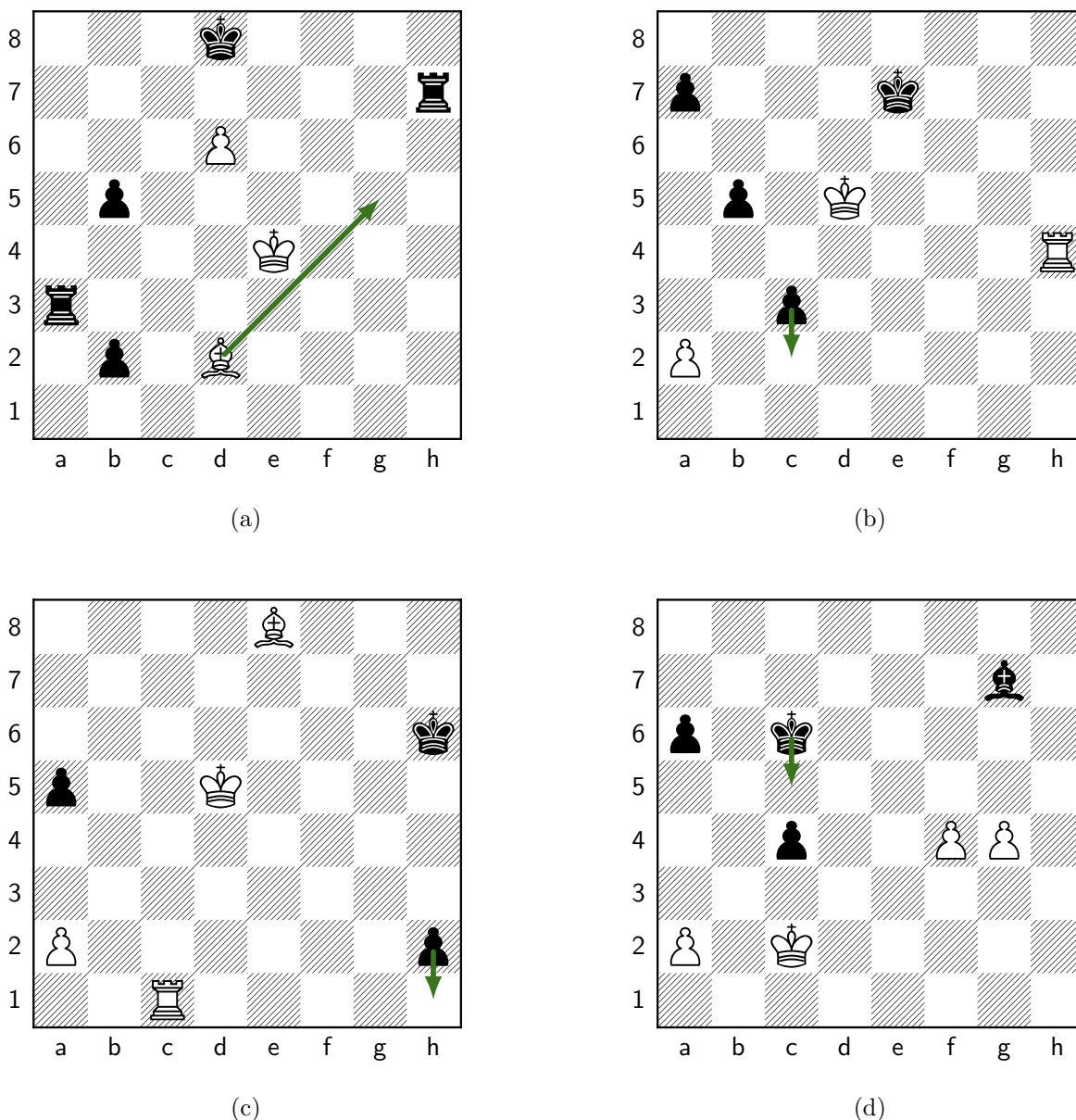


Figure 53: **Move 59, White: Bg5+.** (a) Our position; Melanie plays Bg5+ (d2g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#2075` (ply 105) [structural distance  $d=11$ , QAP obj=1]: the master played c2; under  $\mu$  our Bd2 plays the role of that game’s Pc6 (a **cross-type** correspondence); cross-type roles: our ra3 as their pa2; transferred move d2g5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#2587` (ply 105) [structural distance  $d=11$ , QAP obj=1]: the master played h1=Q+; under  $\mu$  our Pd6 plays the role of that game’s Ph7; cross-type roles: our ra3 as their be8, our Bd2 as their Pa5; transferred move d6d7. (d) Retrieved from `lichess_elite_2024-09_p0026#2302` (ply 113) [structural distance  $d=11$ , QAP obj=1]: the master played Kc5; under  $\mu$  our Ke4 plays the role of that game’s Kc3; cross-type roles: our ra3 as their pg4, our rh7 as their pa2; transferred move e4e5. Synthesis: 1 retrieved analog transferred to Bg5+ (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

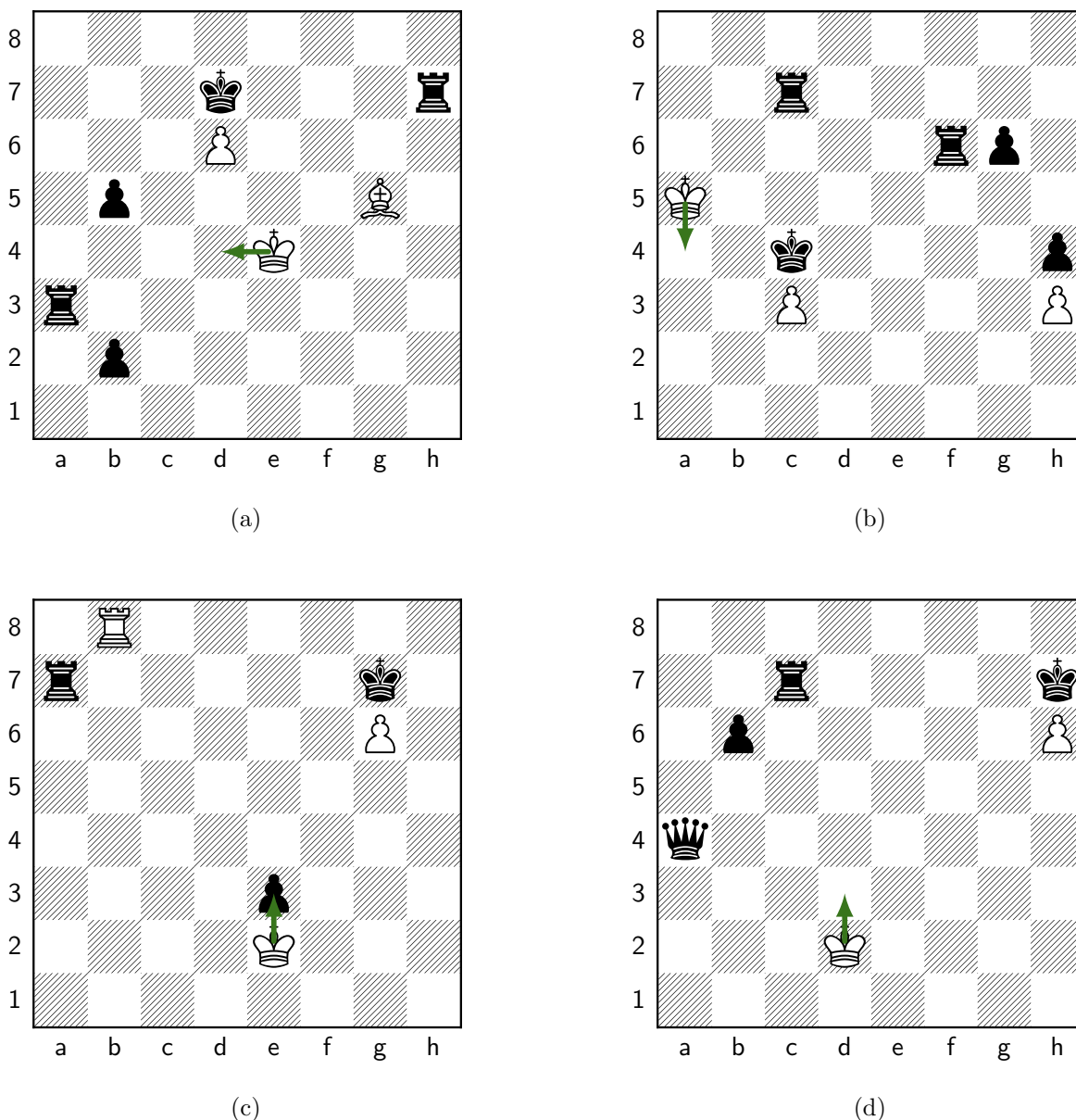
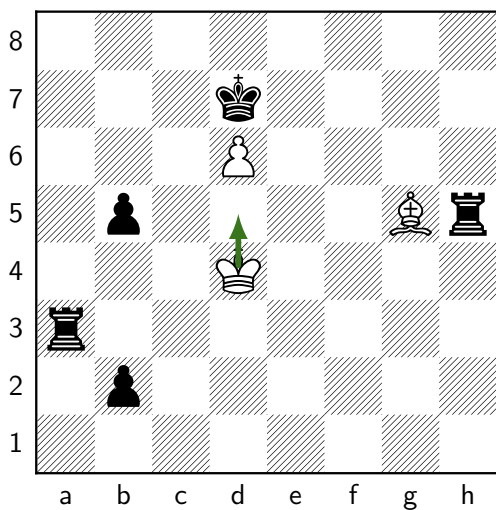
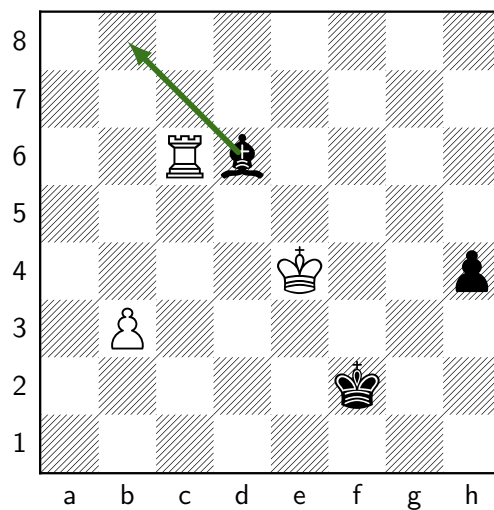


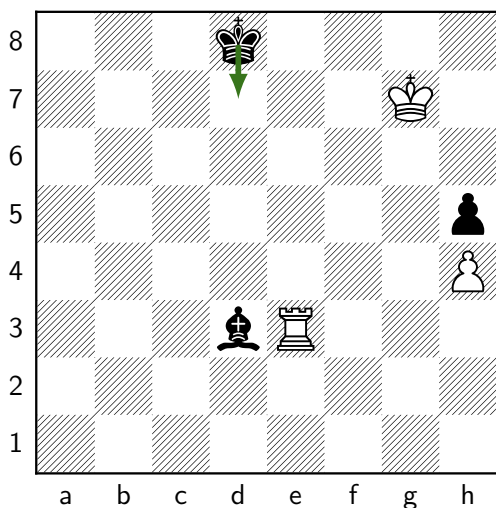
Figure 54: **Move 60, White: Kd4.** (a) Our position; Melanie plays Kd4 (e4d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#1152` (ply 122) [structural distance  $d=9$ , QAP obj=22]: the master played Ka4; under  $\mu$  our Ke4 plays the role of that game’s Ka5; cross-type roles: our Bg5 as their ph4; transferred move e4d4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#61` (ply 124) [structural distance  $d=14$ , QAP obj=22]: the master played Kxe3; under  $\mu$  our Ke4 plays the role of that game’s Ke2; cross-type roles: our ra3 as their pe3, our Bg5 as their Rb8; transferred move e4d4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#3105` (ply 104) [structural distance  $d=10$ , QAP obj=22]: the master played Kd3; under  $\mu$  our Ke4 plays the role of that game’s Kd2; cross-type roles: our ra3 as their qa4; transferred move e4d4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kd4 (summed QAP objective 88, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



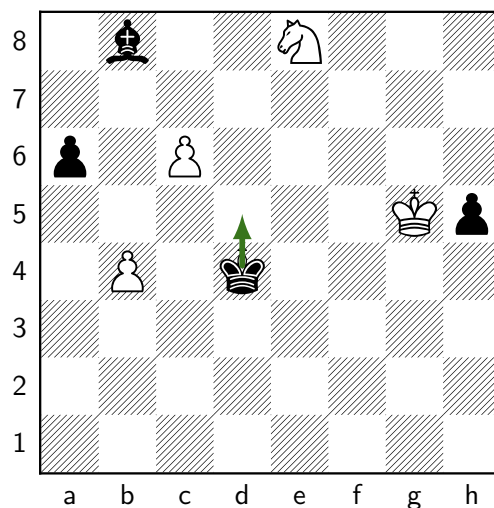
(a)



(b)



(c)



(d)

Figure 55: **Move 61, White: Kd5.** (a) Our position; Melanie plays Kd5 (d4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#1867` (ply 131) [structural distance  $d=12$ , QAP obj=18]: the master played Bb8; under  $\mu$  our Bg5 plays the role of that game’s Bd3; cross-type roles: our ra3 as their Ph4; transferred move g5d2. (c) Retrieved from `lichess_elite_2024-09_p0022#3230` (ply 133) [structural distance  $d=11$ , QAP obj=18]: the master played Kd7; under  $\mu$  our Kd4 plays the role of that game’s Kd1; cross-type roles: our ra3 as their ph4; transferred move d4e4. (d) Retrieved from `lichess_elite_2024-09_p0024#2268` (ply 109) [structural distance  $d=11$ , QAP obj=7]: the master played Kd5; under  $\mu$  our Kd4 plays the role of that game’s Kd5; cross-type roles: our rh5 as their ne8, our ra3 as their Pa6; transferred move d4e4. Synthesis: no retrieved analog transferred to Kd5 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

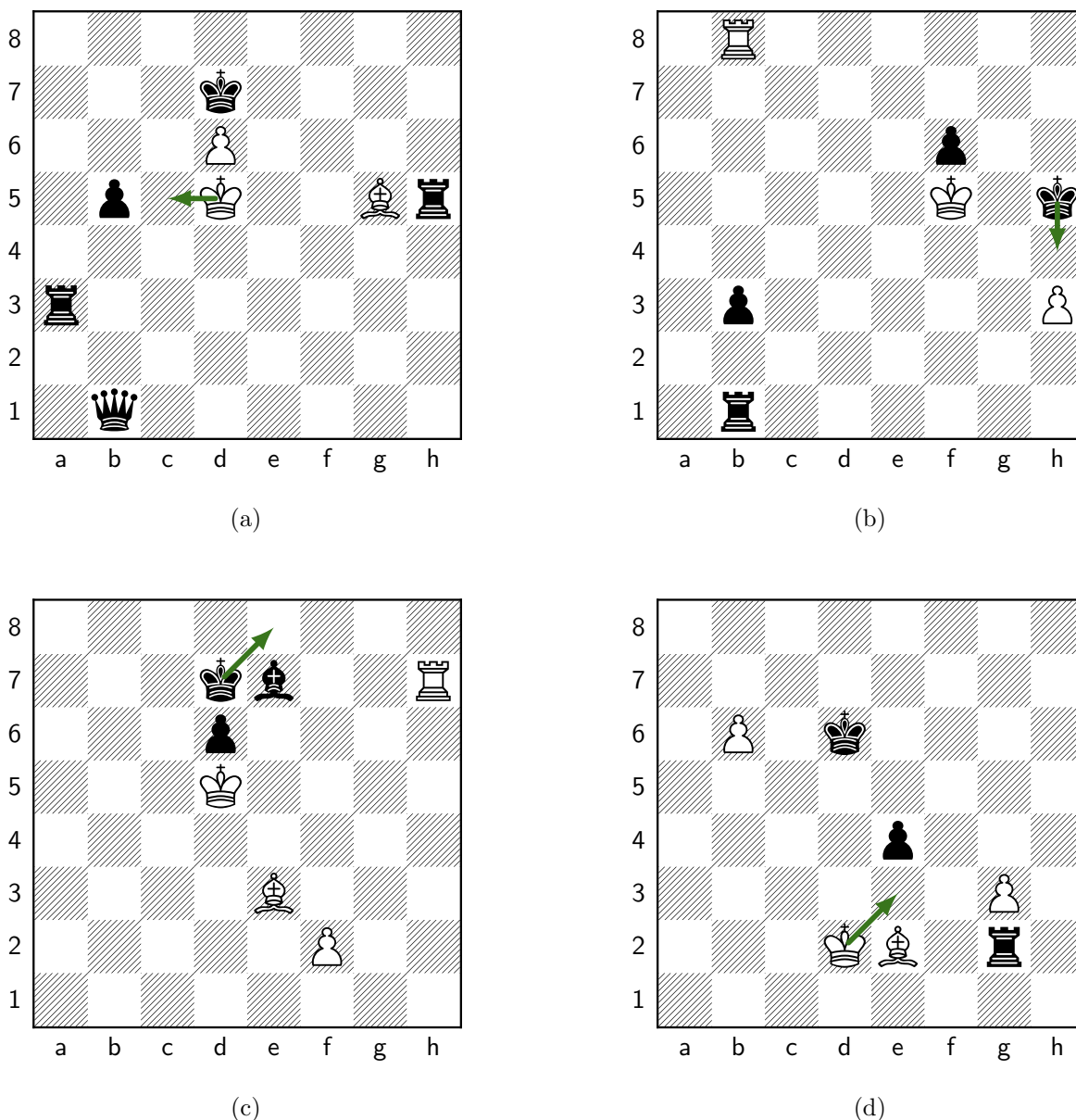
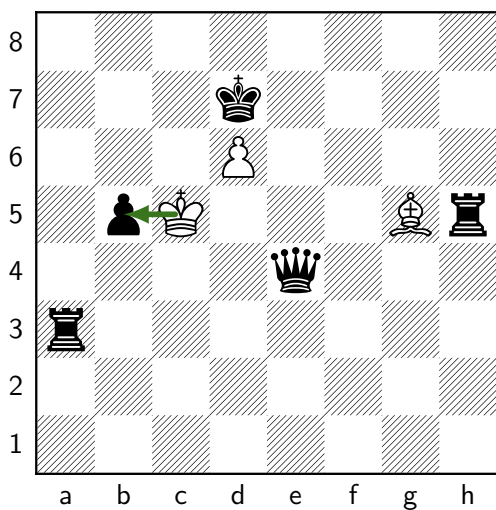
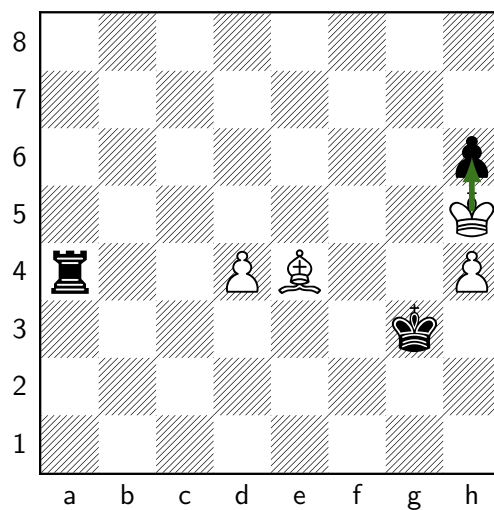


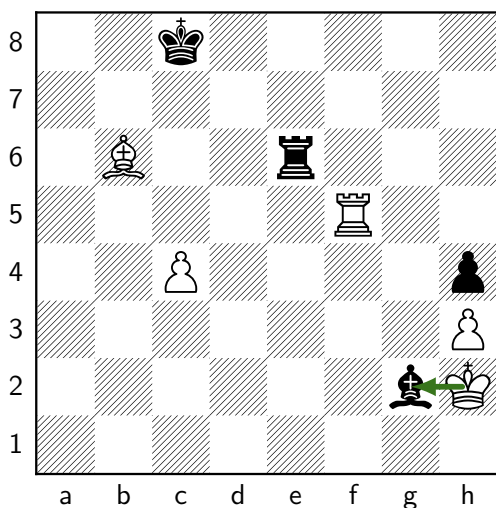
Figure 56: **Move 62, White: Kc5.** (a) Our position; Melanie plays Kc5 (d5c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#3761` (ply 111) [structural distance  $d=20$ , QAP obj=8]: the master played Kh4; under  $\mu$  our Kd5 plays the role of that game’s Kh4; cross-type roles: our Bg5 as their rb8, our rh5 as their ph3; transferred move d5c5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#801` (ply 119) [structural distance  $d=17$ , QAP obj=22]: the master played Ke8; under  $\mu$  our Kd5 plays the role of that game’s Kd2; cross-type roles: our qb1 as their be3; transferred move d5d4. (d) Retrieved from `lichess_elite_2024-09_p0020#822` (ply 108) [structural distance  $d=17$ , QAP obj=21]: the master played Ke3; under  $\mu$  our Kd5 plays the role of that game’s Kd2; cross-type roles: our Bg5 as their Pg3; transferred move d5d4. Synthesis: 1 retrieved analog transferred to Kc5 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



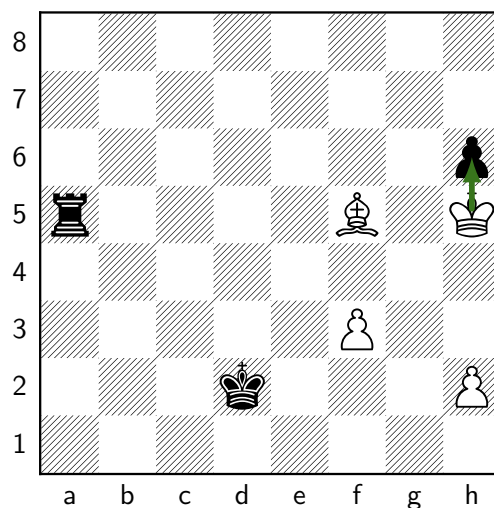
(a)



(b)

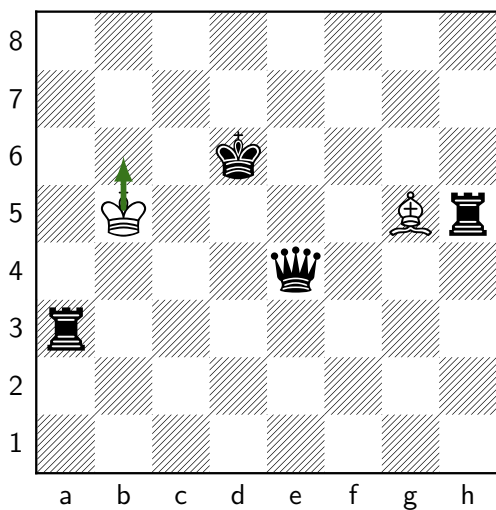


(c)

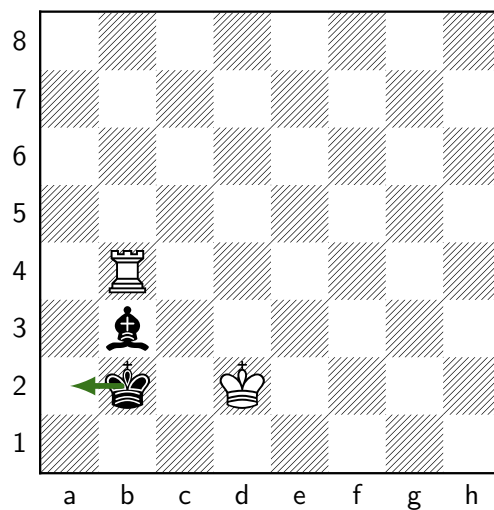


(d)

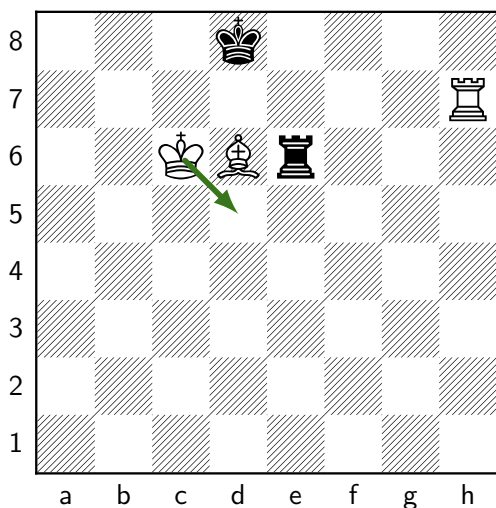
Figure 57: **Move 63, White: Kxb5.** (a) Our position; Melanie plays Kxb5 (c5b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#604` (ply 114) [structural distance  $d=15$ , QAP obj=16]: the master played Kxh6; under  $\mu$  our Kc5 plays the role of that game’s Kh5; cross-type roles: our Bg5 as their Pd4, our qe4 as their Be4; transferred move c5b5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#2837` (ply 112) [structural distance  $d=20$ , QAP obj=14]: the master played Kxg2; under  $\mu$  our Kc5 plays the role of that game’s Kh2; cross-type roles: our pb5 as their bg2, our ra3 as their Pc4; transferred move c5b5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0012#3543` (ply 114) [structural distance  $d=17$ , QAP obj=8]: the master played Kxh6; under  $\mu$  our Kc5 plays the role of that game’s Kh5; cross-type roles: our ra3 as their Ph2, our qe4 as their Pf3; transferred move c5b5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kxb5 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



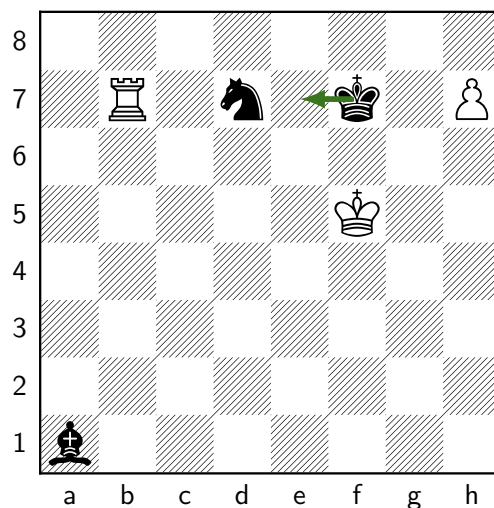
(a)



(b)



(c)



(d)

Figure 58: **Move 64, White: Kb6.** (a) Our position; Melanie plays Kb6 (b5b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#2129` (ply 127) [structural distance  $d=10$ , QAP obj=39]: the master played Ka2; under  $\mu$  our Kb5 plays the role of that game’s Kb7; transferred move b5b6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#2016` (ply 130) [structural distance  $d=10$ , QAP obj=39]: the master played Kd5; under  $\mu$  our Kb5 plays the role of that game’s Kc6; transferred move b5b6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0023#1889` (ply 115) [structural distance  $d=12$ , QAP obj=21]: the master played Ke7; under  $\mu$  our Kb5 plays the role of that game’s Kf2; cross-type roles: our Bg5 as their Nd7, our ra3 as their ph7; transferred move b5b6 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kb6 (summed QAP objective 165, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.